SURGERY

Workbook

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MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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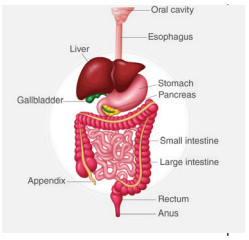


PAEDIATRIC SURGERY

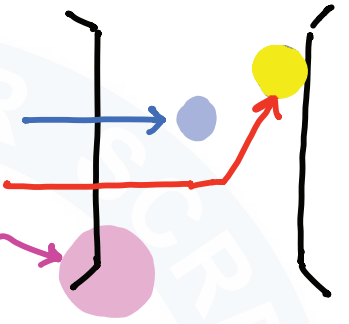
Face - Cleft Lip, Cleft Palate

Neck - Swellings/Fistula -

Thyroglossal Cyst, fistula

Branchial Cyst, "

Cystic hygroma



Thorax - Esophageal Atresia

Abdomen -

Duo. Atresia

Annular Pancreas

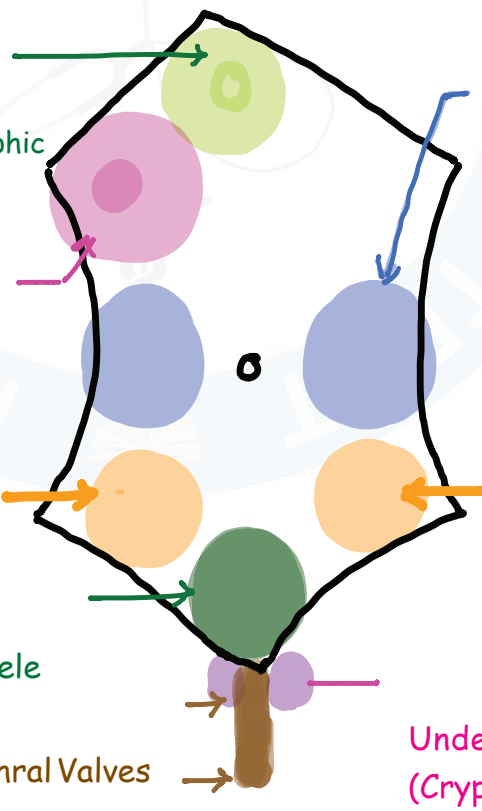
Infantile hypertrophic

Pyloric Stenosis

Biliary Atresia

Choledochal Cyst

Intussusception



Ectopic"

Polycystic "

Horseshoe "

Meckel's div

Hirschsprung's

Duplication of
Ureter, Ureterocele

Postr Urethral Valves

Undescended Testis
(Cryptorchidism)

Hypospadias

- Imperforate Anus

Active space

FACE - CLEFT LIP AND PALATE

MC Cong anomaly of oro facial structures

Combined CL + MC

(CL = Cleft Lip
CP = Cleft Palate)

- RF (Risk Factors) :
- Genetic -
- Environmental -
  - Maternal Epilepsy
  - Drugs taken during Preg-
    (Phenytoin, Diazepam, Steroids)
- Associated with congenital heart diseases
- Asso c̄ Syndromes → Van der Woude / Stickler / Di George / Down's / Apert's / Treacher Collin's / Pierre Robin / Patau down's syndrome **Q July 24**

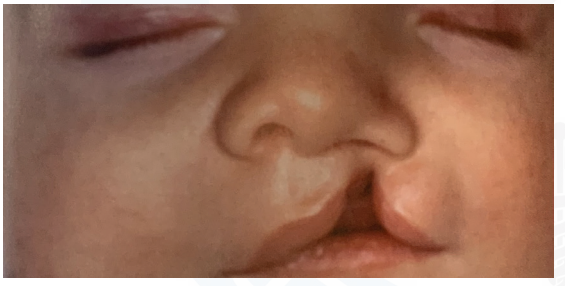


FIGURE- 1.1 Unilateral Partial Cleft Lip- Left Side

FIGURE- 1.2 Bilateral Partial Cleft Lip

- Role of Folic acid supplementation during Pregnancy is equivocal

FIGURE- 1.3 Unilateral Complete Cleft Lip- Right Side and Bilateral Complete Cleft Lip

Cleft Palate →

- Incomplete - If Palate attached c̄ Vomer / Nasal Septum
- Complete - If P not attached c̄ Vomer / Nasal septum

Active space

Reason for CL & CP → Failure of fusion of facial processes

a) If Median Nasal process fail to unite → Cleft of primary palate (Philtrum/Alveolus)

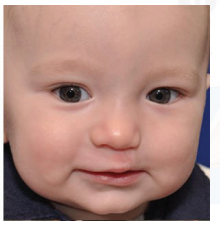
b) If Median Nasal process fail to unite c̄ maxillary process→Cleft of Nose/Lip

C) If Maxillary process fail to unite, ie.. 2 palatine shelves fail to unite → Cleft Palate

- Problems : Airway, Breathing, Swallowing, Dentition, Speech
- CL Can be diagnosed on Antenatal USG after 18 weeks of Pregnancy
- Rx-Plastic Sx reconstruction - Timing of Sx

Q July 24

1-2months of age	3-6months	6-12months	1-7yrs.	7-12yrs.
QQ • No Sx • Feeding Expressed breast mild by mdified feeding bottle / teats • Squeeze & Suck	Sx Clef of Nose Lip, Hard palate * M u s c l e Recons*	Sx Clef of soft Palate ↓ Intravelar veloplasty Levator sling recons	Feeding Dentition Speech Acsthetics	Alveolar bone grafting



Operations -

- Millard's
- Wardill's

NECK SWELLINGS- THYROGLOSSAL CYST AND FISTULA

Thyroglossal Cyst

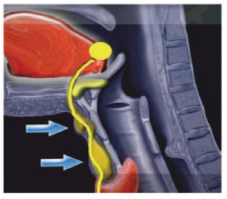
- → Accumulation of secretions. in a persistent thyroglosssal tract

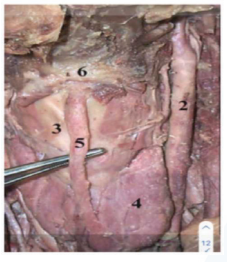
MIST

- Always in .....; (except in the region of thyroid cartilage)
- Children
- MC - Beneath hyoid bone (subhyoid)

Active space

- MC - Swelling - move c̄ deglutition
- Charac → Also move c̄ .....





It can be mistaken as Ectopic thyroid tissue

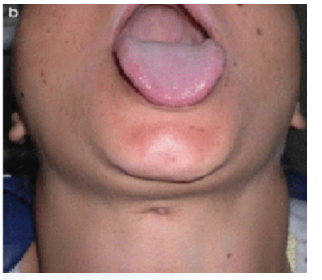
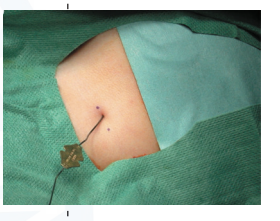


FIGURE- 1.1 Thyroglossal Cyst

- Complicn
  - Infection → Abscess → Rupture → Thyroglossal ..... Rarely fistula (Never Congenital)
  - ..... Thyroid Cancer

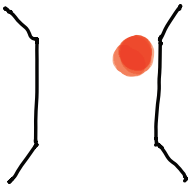


Q July 24
Probing

- Rx - ..... Operation (Excision of cyst+Thyroglossal tract upto tongue including central portion of hyoid bone)

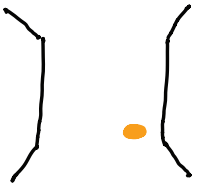
NECK SWELLINGS- BRANCHIAL CYST AND FISTULA

Branchial Cyst



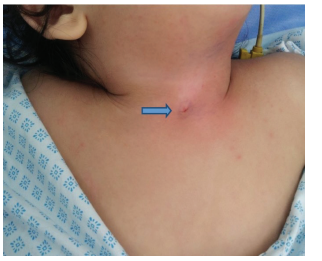
- Remnant of ..... branchial cleft
- Antr border of Sternomastoid muscle, at the junction of upper and middle 1/3 of neck
- Lined by Squamous epithelium
- Rx - Excision

Branchial Fistula



- Persistent of ..... branchial cleft
- Antr border of sternomastoid muscle - Lower neck

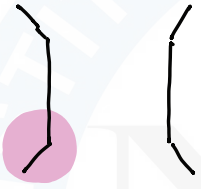
- Internal opening - Antr aspect of Postr faucial pillar behind Tonsil
- Internal opening of type III /IV is at Pyriform sinus
- Lined by ciliated columnar epithelium
- Rx - Complete Excision



MIST

NECK SWELLINGS- CYSTIC HYGROMA

Cystic Hygroma



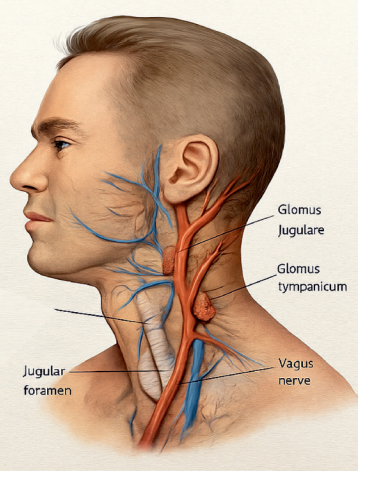
- Postr Δ of neck
- Lined by Endothelium
- 50% Neonate - can obstruct labour
- Clear Lymphatic fluid inside
- Most brilliantly .....
- Rx - Complete Excision
  - Sclerotherapy - Picibanil (OK 432)

Other swellings c
⊕ Transillumⁿ
1. Ranula
2. Hydrocele
3. Meningocele
4. Rarely Branchial Cyst

FIGURE- 1.2 Cystic Hygroma

CHEMODECTOMA (PARAGANGLIOMA)

Benign but highly vascular neuro- endocrine tumour arising from parasympathetic paraganglia that respond to hypoxia. Derived from neural crest.



Active space

Common Sites :

Carotid Body	Carotid bfurcation
Glomus Jugulare	Jugular bulb

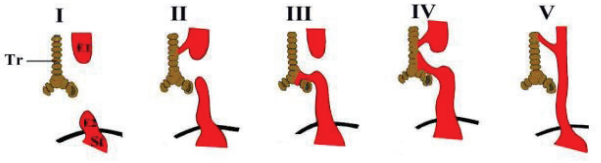
- Pulsatile, lateral neck mass on angle of the mandible
- 'Salt and pepper' appearance
- Lyre sign on angiography
- Biopsy contraindicated due to high vascularity

Rx - Surgical excision after embolization

THORAX- ESOPAGEAL ATRESIA AND DYSPHAGIA LUSORIA

Esophageal Atresia

- Absence of some part of esophaqus
- Due to Postr deviation of Tracheo-esophageal Septum
- Types :



Tr - trachea E1 - proximal esophagus E2 - distal esophagus St - stomach

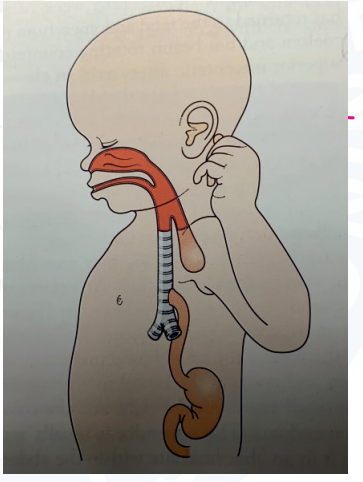


FIGURE- 1.3 Esophageal atresia with Tracheo-esophageal fistula

- MC - Type C → Upper eso is .....
  Lower eso communicate .....
  - H Type - Isolated TE Fistula

Active space

- Asso. Cong anomalies - 'VACTERL' -

  (Vertebral, Anorectal, MC VSD PDA TOT Cardiac, TE fistula, Renal, Limb defects)

- Pregnancy - Polyhydramnios
- C/F :-
- MC - At birth, continuous pouring of saliva from baby's mouth
- Risk - Regurgitation, Aspiration, Pulmonary complications
- Initial Bed Side Test-Failure to pass Nasogastric tube beyond 10 cm
- Initial Invⁿ - CXR- 1. Coiling of NG Tube in upper esophagus

  2. Distended stomach / intestine c̄ gas suggest Type ..... (connection of lower esophagus c̄ trachea)

QQ

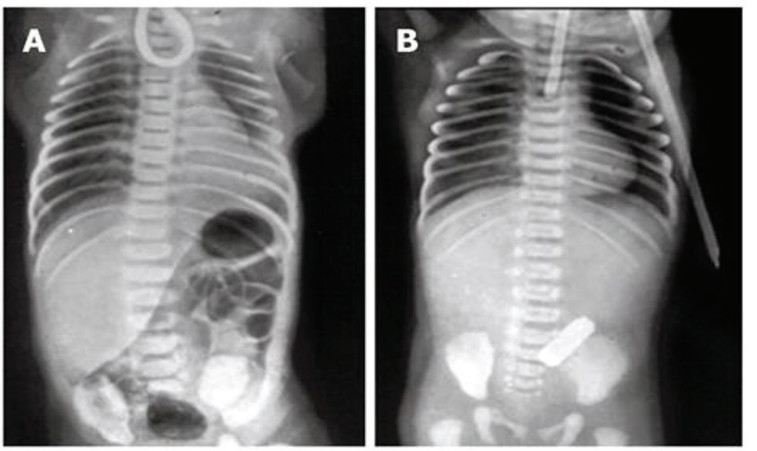


FIGURE- 1.4 Coiling of Nasogastric tube in upper esophagus, distended stomach with gas in Eso Atresia

- Best Invⁿ - IOC - ..... MIST
- Rx : Sx correction on Day 1 or 2 after birth, Thoracotomy→ Division of fistula

+

.....

Active space

→ If 2 ends of eso are far, then Wait for .....
(Feeding Jejunostomy) or Feeding Gastrostomy

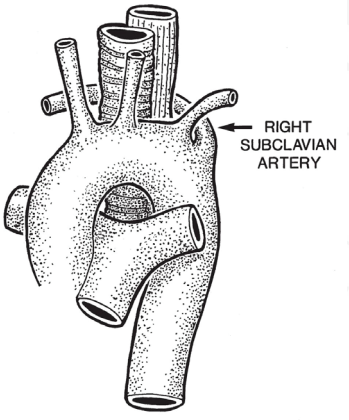
Reassess 2 ends of eso after 4-6 weeks

close Far

Eso-Eso-stomy Eso Reconstruction (Colon/Jejunum/Stomach)

Dysphagia Lusoria:

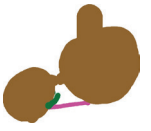
- Compression of esophagus by Aberrant .....



- Double Aortic arch
- Ⓡ Aortic arch

ABDOMEN EPIGASTRIUM- DUODENAL ATRESIA AND ANNULAR PANCREAS

Duodenal Atresia



- Absence of some part of duodenum
- d/t failure of fusion of Fore gut c̄ Midgut
- c̄ in few hrs after birth with .....

- - X-Ray - Double Bubble Sign
- Rx - .....

  .....
- Alternative Rx - .....

  .....

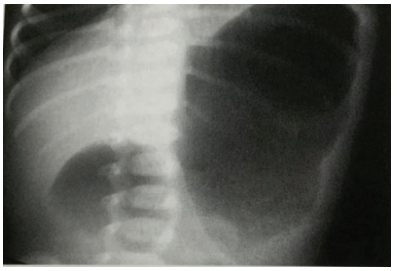


FIGURE- 1.5 Double Bubble Sign in Duodenal atresia and Annular Pancreas

TRIPLE BUBBLE SIGN QQ

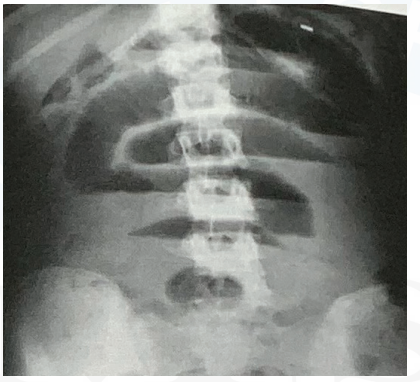


FIGURE- 1.6 Multiple air fluid levels in central part of abdomen suggest Small Intestinal Obstruction

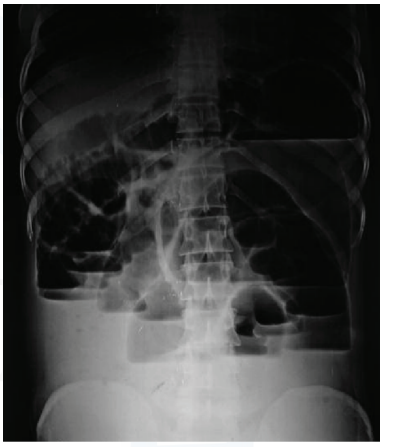


FIGURE- 1.7 Multiple air fluid levels in peripheral part of abdomen suggest Large Intestinal Obstruction

Q July 24

1 SBS - Pyloric obstruction (Infantile hypertrophic Pyloric Stenosis)

2 DBS - Duodenal obsⁿ (Duodenal atresia/Annular Pancreas)

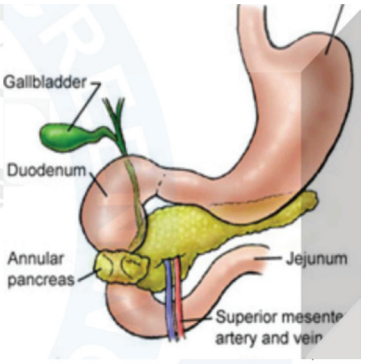
3 TBS - Jejunal obsⁿ (Jejunal atresia) QQ

4 MBS - Central - Small Intestinal obsⁿ

- Peripheral - Large Intestinal obsⁿ

Annular Pancreas

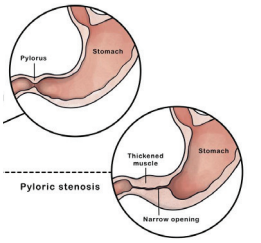
- Ring of Pancreatic tissue surrounds duodenum
- X- Ray - DBS
- Rx - D - D
- Alternative Rx - DJ..



Active space

EPIGASTRIUM- (INFANTILE) HYPERTROPHIC PYLORIC STENOSIS

Idiopathic Q



- hypertrophy of circular muscle fibres of Pyloric musculature
- Usually 1st born male child

- Presents during 3rd week of life (2nd - 6th week)

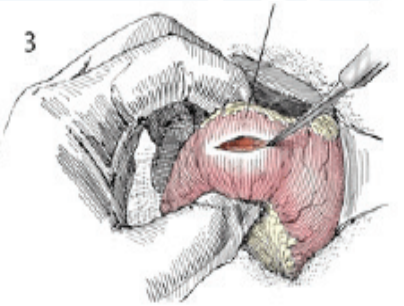
C/F : **Q July 25**

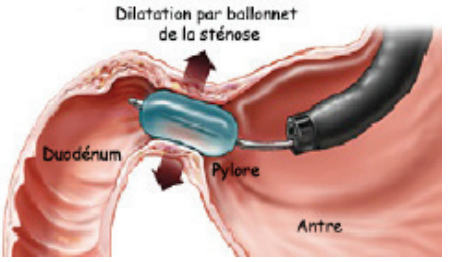
- Olive Lump in RHC / Epigastrium - Thickened Pyloric ms
- Peristalisis visible from (L) → (R) after Test meal
- Projectile..... vomiting
- Metabolic ..... QQ
- ..... chloremia, Kaliuresis, ... kalemia **Q July 24**
- Paradoxical aciduria QQ
- Cl. app : Emergency → 1st → Rehydration & Electrolyte correction

QQ ↓

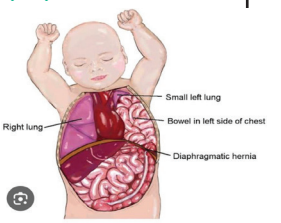
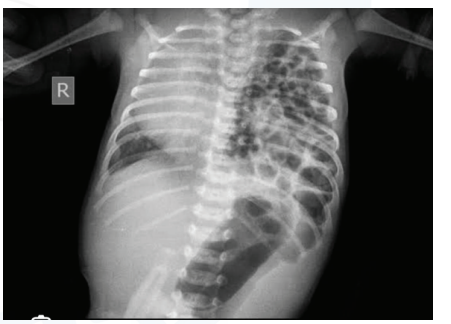
- 1st Invn (IOC) US abd → Thick Pyloric ms (> 3 - 4 mm)
  Lengthening of Pyloric canal (> 16 - 17 mm)
- Rx - Initially - Pneumatic balloon dilatation
- Rx OC - Ramstedt's .....

45-day old baby was brought to the hospital with vomiting, failure to thrive and a mass in the epigastrium. Which of the following is the most likely diagnosis in this baby?
a. Duodenal atresia
b. Hypertrophic pyloric stenosis
c. Esophagitis
d. Intussusception





Active space

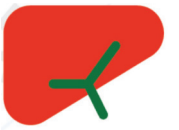
Morgagni H.	Bochdalek H.
d/t failure of fusion of septum transversum with sternum & ribs Mc Site - Anteromedial Mc Organ – Large bowel C/F – Asymptomatic (defect is small) Dx - Chest X- Ray Air filled large bowel loops in anterior mediastinal shift mediastinum Incidental finding Rx:- Transabdominal mesh repair	d/t failure of fusion of septum transversum with pleuroperitoneal memb. Mc site - Posterolateral at left Mc Organ – Small bowel/ stomach enters in thoracic cavity. C/F – ARDS d/t pulmo. Hypoplasia Barrel shape chest Scaphoid abdomen Suncken abdomen  Dx – Chest X-ray Air filled bowel loops in chest Absence of lung shadow.  Rx:- NGT + Tracheal intubation Thoracotomy & laproscopic repair defect. Ambu bag & Pleurocentesis - Contraindicated

Active space

RIGHT HYPOCHONDRIUM- BILIARY ATRESIA

- Absence of biliary channels - Exact cause X
- d/t Inflammatory fibrosis of biliary channels starting in 1st week of life
- Types

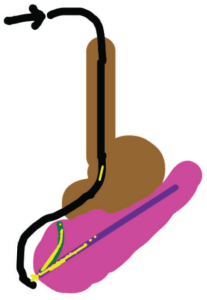
I - (-) CBD (Common Bile duct)



II - (-) CHD, CBD (Common Hepatic duct)

III - (-) R/LHD, CHD, CBD (Right & Left Hepatic ducts)

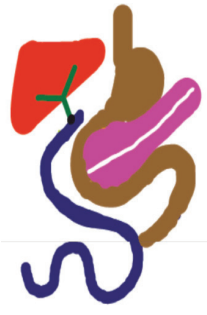
- MC - III .....



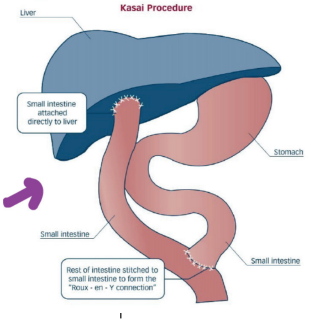
- C/F: Jaundice
  Steatorrhoea
  Hepatomegaly
  } 1st week /2nd week of life
- Cl. app : Jaundice - 1st → LFT (Liver Function Test)
  - 2nd → US abd

Invᴺ for Biliary Chonnels

- MRCP - Magnetic Resonance Cholangio Pancreaticography
- ERCP - Endoscopic Retrograde Cholangio Pancreaticography
- HIDA Scan - Hepatobilary Imino Diacetic Acid (Isotope)



- IOC - HIDA Scan .....
- Rx :
  - I - .....
  - II, III - ..... (Hepatico Porto enterostomy)



- Eventually Liver Transplant

RIGHT HYPOCHONDRIUM- CHOLEDOCHAL CYST

Normal

- Cystic dilatations of biliary channels
- Exact cause X
- d/t Anomalous Pancreaticobiliary duct Junction
- Types- Todani classi - V types

  I, II, III - involve .....

  IV, V - involve .....
- MC - I - Diffuse dilatation of CBD

  II - Eccentric dil^{n} of CBD above Pancreas

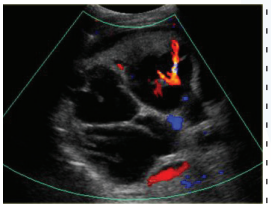
  III - Dil^{n} of CBD c̄ in Pancreas → Choledochocele

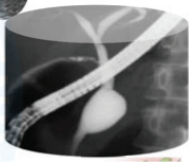
  IV_a - both Intra t Extra hep. ducts

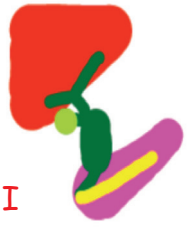
  IV_b - only Extrahep ducts

  V - only Intrahep ducts - Caroli's disease

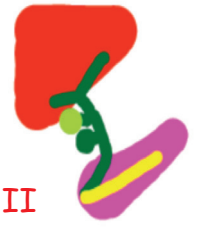
  (Central dot sign which corresponds to Portal vein)



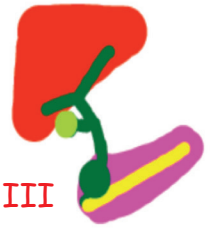




I

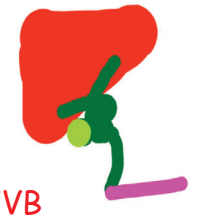


II



III

IVA

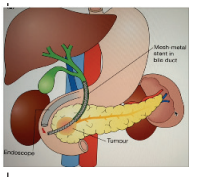


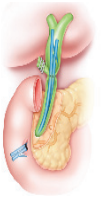
IVB

V

Active space

OPD	Emergency
• Painless Intermittent Jaundice in infancy • Mass in RHC ↓ LFT US abd ↓ IOC - ..... ↓ Rx - Cyst Excision + Roux en YHJ (Hepatico Jejunostomy)	• Cholangitis - Sudden, Severe pain in RHC, high grade fever, chills rigors, Jauridce PFJ ⟶ ↓ P, F, J, + Septicemia (hypotension and Mental Confusion) - Reynold's Pentad ↓ Basic Resuscitation IVF, IV Antibiotics ↓ LFT, US abd ↓ Endobiliary Stenting .....





➪ Complicⁿ:

- Cholangitis
- Choledocholithiasis (Chole=bile ; docho = duct ; Litho = Stone, Osis = disease) (Cholelithiasis = Stones in GB)
- Cholangiocarcinoma

  MC Site of Cholangioca - Confluence of (R) & (L) hepatic ducts

  Klatskin's Tumor (Hilar Cholangiocarcinoma)

Active space

RIGHT HYPOCHONDRIUM- INTUSSUSCEPTION

- Invagination (Telescoping) of one part of intestine into adjacent part

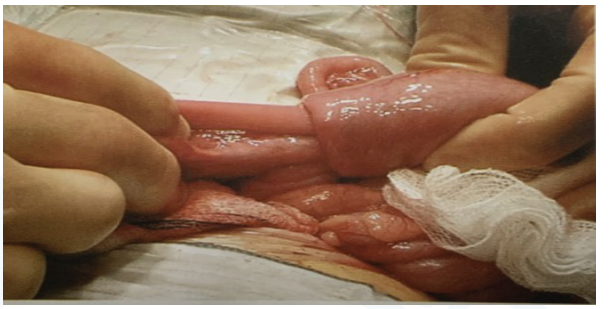


FIGURE- 1.8 Ileocolic Intussusception- Appendix is visible for identification

- MC - ..... Intussusception
- MC - ..... Months (5- 10 months)
- MCC - Idiopathic (hypertrophy of Peyer's patches)

Rota Virus Vaccine / Infection / Meckel's diverticulum

MC in Adults - Colo-colic, (Polyps in colon)

C/F :

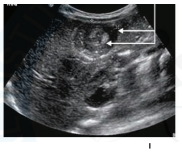
- Saucage lump - RHC / (L) Paraumbilical region concavity of lump towards umbilicus

- Blood + Mucus in stool - Red currant Jelly stools

FIGURE- 1.9 Red Currant Jelly Stools in Intussusception

- **Empty RIF** - Sign of Dance
- DRE - Examiner's finger is stained with blood (Digital Rectal Examination)

1st Invn US abd - Doughnut Sign (Target Sign)



- X - Ray - Crescent Sign (Meniscus Sign)
- **Ba Enema** - ..... Sign / Coiled Spring Sign
- Best Invn - .....

FIGURE- 1.10 Ba Enema- Claw Sign in Intussusception

- Rx :RxOC - Hydrostatic Reduction - .....enema
  Saline enema
  Barium enema MIST

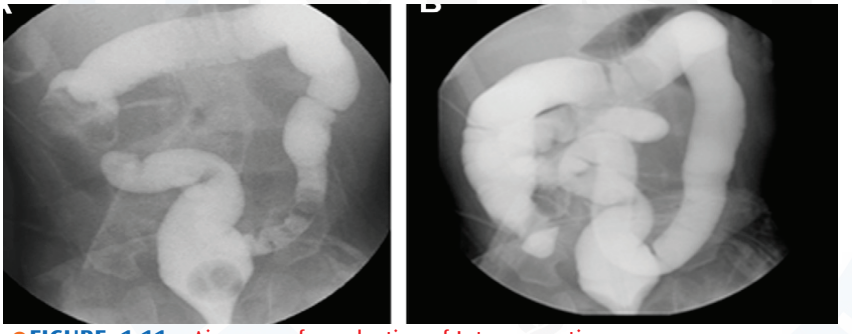


FIGURE- 1.11 Air enema for reduction of Intussusception

If fails → Operative Reduction
-Never Pull, Always Push
(Gentle milking of outer tube to reduce the inner tube)

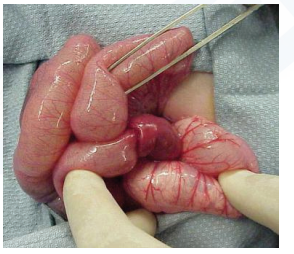
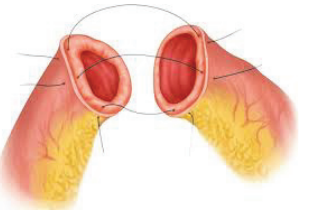


FIGURE- 1.11 Operative Rx

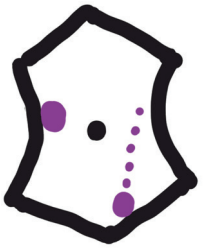
- If gangrenous-Resection & end to end anastomosis



QQ
Q. July 24

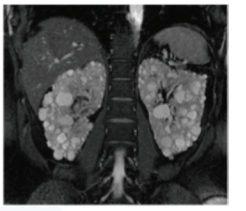
Active space

FLANKS- ECTOPIC, HORSE SHOE, POLYCYSTIC KIDNEY DISEASE



Polycystic Kidney

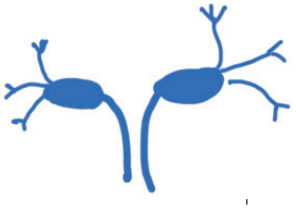
- congenital (Adult) PC KD - Aut. dominant
- Infantile PCKD - Aut. recessive,
  Rare ↑ mortality

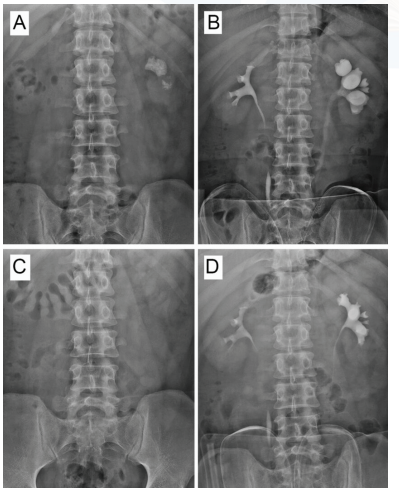


⇨ Congenital (Adult) PCKD -

- Bilateral
- Asso c̄ cysts in other organ
  (MC - Liver Spleen, Pancreas, Ovaries)
- Asso c̄ Berry Aneurysm (Circle of Willis)
- Mitral valve Prolapse

- Clinically manifest > 30 y
- MC - Hypertension - 75%
- Swelling, Pain in Bilateral Lumbar region
  Hematuria, Uremia
- Clinical approach :
  Ist Invⁿ - US abd
  ..... - IOC
  IVU (IVP) - Spider Leg appearance





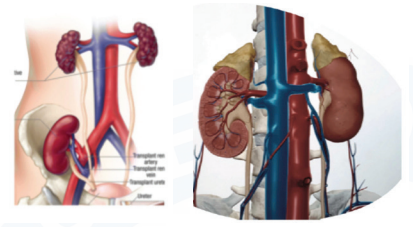
Q July 25

Active space

Rx : Initially - Deroofing of cyst

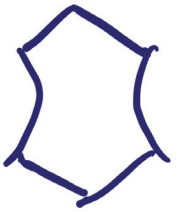
Eventually - Renal Transplant

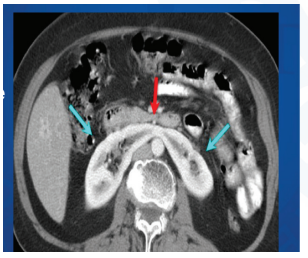
(Transplanted Kidney placed in Iliac fossa, Usually (L) Kidney preferred from donor d/t Long Left Renal vein)

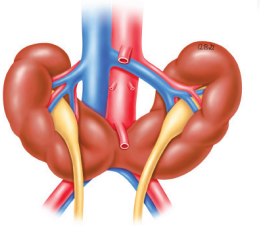


Orthotopic Transplant - Same Anatomical Location - Liver
Heterotopic Transplant - Different Anatomical Location Kidney

- Horse shoe Kidney : 1 in 900
  - Pair of Ectopic Kidney
  - Fusion of Medial subdivision of Low poles of both Kidneys
  - Ascent is impeded d/t Infr mesenteric artery
  - Isthumus - .....







- Usually Asymp
- Rarely Swelling, Pain - Lower abdomen
- Complicⁿ - UTI, Urinary stones
- 1st Invⁿ - US abd
- Best .....
- IVP - Flower Vase ureters
  (Hand Joining Sign)

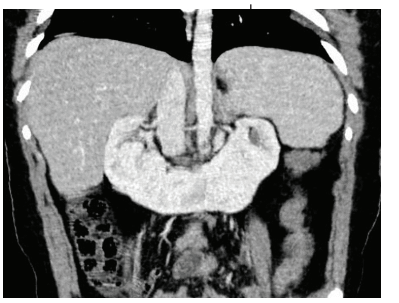
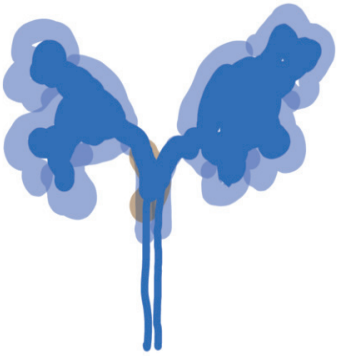


FIGURE- 1.12
CT Scan image of Horse shoe kidney

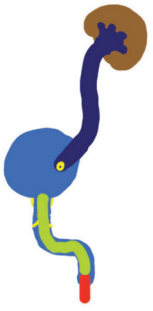


- Rx : UTI - Antibiotics
  Stones - Removal
  (Division of Isthumus is not the Rx OC)

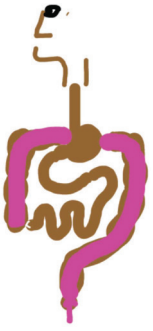
MCC of Abd mass in Newborn - Hydronephrosis (PUJO)
2nd MCC of Abd mass in Newborn - Multicystic Kidney disease

Contrast Studies of Urinary Tract :

⇨ Upp Uri Tract - IVU (IVP)
Retrograde Pyelography
Antegrade Pyelography



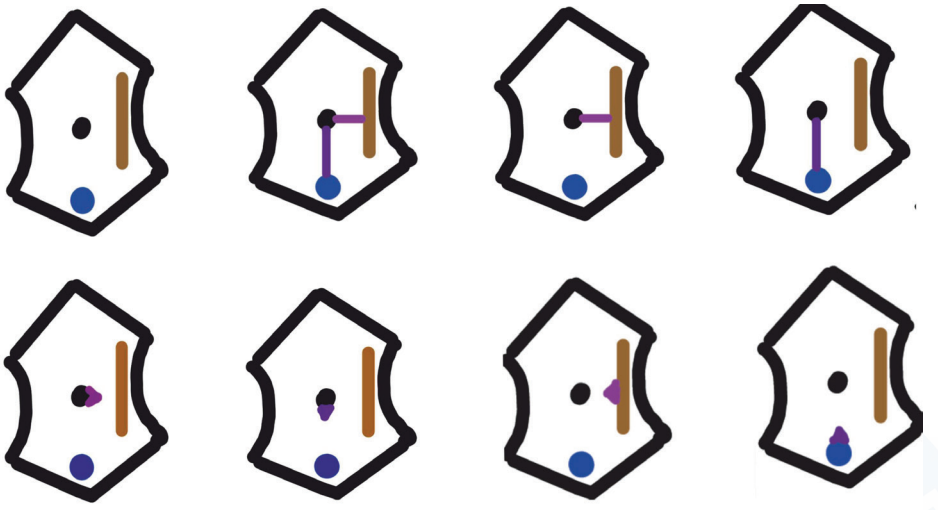
⇨ Low Uri Tract - Cystogram (Bladder)
RGU - Retrograde Urethrogram (Urethra)
MCU - Micturating Cystourethrogram
Voiding Cystogram
For 1. Postr Urethral Valve
2. Vesico ureteric reflux



Contrast Studies of GIT

1. Esophagus = Swallow
2. Stomach = Meal
3. Duo/Jej/Ileum=Meal Follow through
4. Colon = Enema

ILIAC FOSSA- MECKEL'S DIVERTICULUM



* Umb. Faecal Fistula - Failure of Involution of Vitello intestinal duct (Omphalo - mesenteric duct)

  Intestinal Contents coming from umbilicus in a newborn baby

* Umb. Urinary Fistula - Failure of Involution of Allantois urine

  watery discharge from umbilicus in a new born baby

* Umb. sinus - Patent Umb end of VID / Allantois

  Mucus discharge from umbilicus

* Meckel's div - Patent Int end of VID
* Cong UB div - Patent vesical end of Allantois

Cong. div → ....., contain all layers of organ

Meckel's, Cong UB div, Since .....

Acquired div → ....., do not contain all layers

Consist of only mucosa or sup.

Submucosa, ...

Sigmoid colon div, Zenker's div, Acquired UB div

Active space

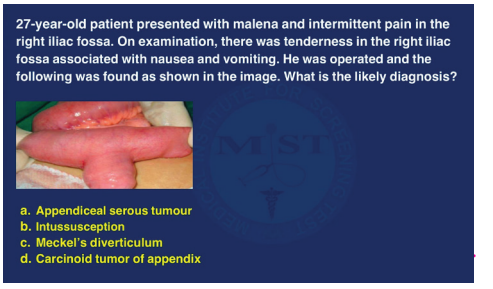
Meckel's Diverticulum

Q. Jan 25

FIGURE- 1.13 Meckel's diverticulum is present on the antimesenteric border of intestine

- Persistence of Intestinal end of vitello intestinal duct (Omphalo - Mesenteric duct)
- Antimesenteric border
- Cong., True, all Layers of intestine
- Has its own blood supply
- Rule of 2 : 2% Population
- 2 cm wide
- 2 inches long (3 - 5 cm)
- 2 feet (60 cm) proximal to Ileocaecal junction
- Presents at around 2yrs. of age
- 2% of patients are symptomatic QQ
- Consist of heterotopic epithelium - MC - ..... epithelium
  Pancreatic, colonic epithelium
- If Meckel's found in hernia sac → Littre's hernia
- Usu Asymp
- Rarely Symp - mimick Acute Appendicitis

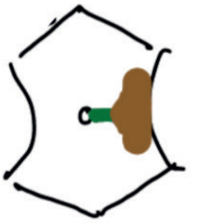
- Complicⁿ :
  1. Bleeding MC
  2. Diverticulitis, Div. Perforation
  3. Peptic ulceration



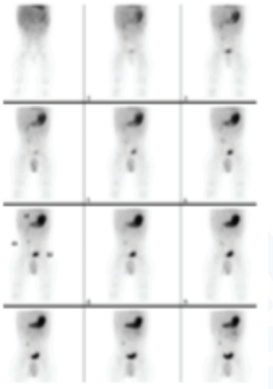
Active space

4. Intussusception

5. Int. obstruction - d/t band from Umbilicus to apex of diverticula



- IOC - meckel's scan - Radioisotope scan
  - Technetium pertechnetate scan



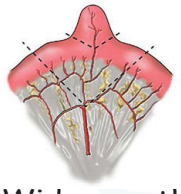
Barium meal follow through
Video capsule endoscopy
CECT

- Rx :

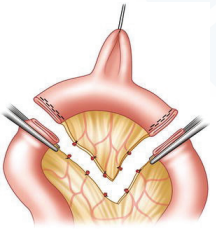
⇨ If Asymp → No Rx

⇨ If Symp or Complicⁿ

a) Narrow mouth of Meckel's → Diverticulectomy



b) Wide mouth of Meckel's → Resection of segment of intestine bearing Meckels, c̄ end to end anastomasis



QQ

Staplers preferred

Linear Stapler

Note : If found incidentally at operation for some other disease →

a) Narrow mouth → Diverticlectomy

b) Wide mouth → Leave as such

Active space

ILIAC FOSSA- HIRSCHSPRUNG'S DISEASE

- Congenital Megacolon / Congenital Aganglionosis
- Absence of Ganglion cells in nerve plexus
- d/t failure of migration of neural crest cells
- MC - .....
  Rectum Always involved

- Short Segment Aganglionic - OPD
  → Childhood → Constipation
- 1st Invn / IOC for Extent of disease - Ba Enem
- Narrow contracted Part - Pathological
  Proximal dilated part - Normal
- Ano Rectal Manometry - For severity of obstruction
  IOC for diagnosing → Suction Rectal Biopsy

FIGURE- 1.14 Hirschsprung's disease- short aganglionic segment (Contracted), dilated proximal segment

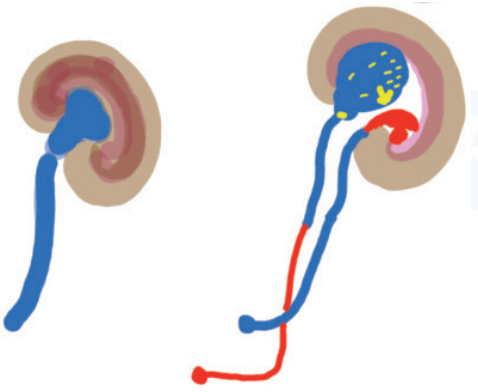
FIGURE- 1.15 Hirschsprung's disease- long aganglionic segment (Contracted), dilated proximal segment

Rx - Duhamel / Swenson's opn /
Soave colo anal anastomosis

- Emergency :
  Long Segment is Aganglionic - Emergency : →
  Neonate → Int. obsn →
  Proximal Colostomy

HYPOGASTRIUM URINARY TRACT- DUPLICATION OF RENAL PELVIS AND URETER

- MC Congenital anomaly of Upper Urinary Tract is Duplication of Renal Pelvis and ureter



- Duplication of Renal Pelvis - ..... Sign
  (Water Lily Sign - Hydatid Cyst)
- Duplication of Ureter - Meyer Weigert Law

  Usually - If there are 2 ureters on 1 side they join with each other, so that only 1 ureteric orifice is seen in bladder

  Rarely - 2 ureters on 1 side do not join in such case, upper ureter passes distally & open more medially than its counter part which is called as - Ectopic Ureter

FIGURE- 1.16 Drooping Lily sign in Duplication of Renal pelvis

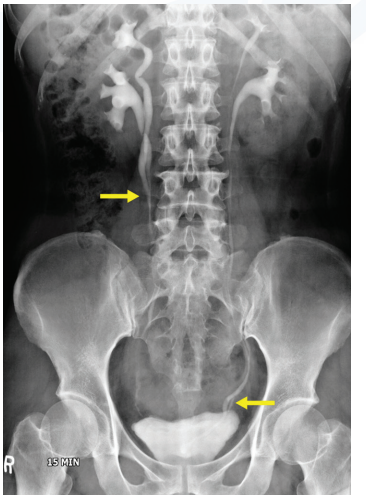
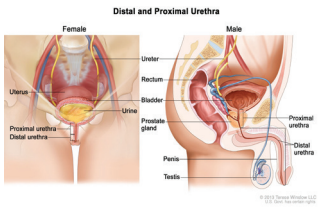


FIGURE- 1.17 MEYER WEIGERT

Active space



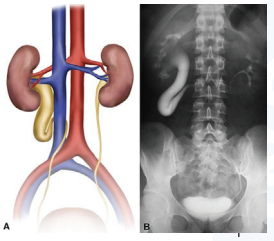
- Ectopic Ureter -

  Males - Apex of Trigone (MC), Seminal vesicle. posterior urethra, Ejaculatory duct
  - Recurrent UTI

  Females - Distal to distal Sphincter
  - Dribbling urinary incontinence

- IOC - .....
- Rx for Ectopic Ureter - Ureteric Reimplantation

HYPOGASTRIUM- URINARY TRACT- RETROCAVAL URETER



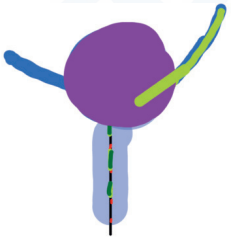
Fish hook Sign

Normally (R) Ureter is on (R) side of IVC, but if right ureter is lying behind the IVC, then it is called retrocaval ureter

- Usually Asymp.
- Rarely obstruction of ureter by IVC at L_3 - L_4 → Proximal dilatation → Unilateral right sided Hydronephrosis → UTI → Stones
- Reverse J Sign on contrast Study **Q. July 24**
- Rx - If Symptomatic Uretero - ureterostomy or Pyeloplasty

HYPOGASTRIUM- URINARY TRACT- URETEROCELE

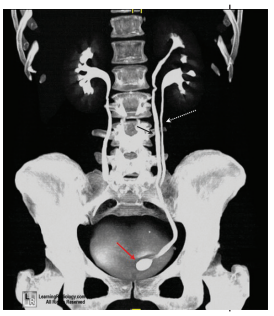
QQ
26 July 25



- Partial atresia of ureteric orifice
- ♀ • Childhood
- UTI
- IVU = ...

IBQ

or Address head appearance

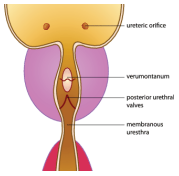


- Rx - Surgical excision and reimplantation of ureter

  Rarely - Endoscopic Longitudinal Incision of ureteric orifice

  (Complication - Vesico ureteric reflux - IOC - MCU)

URETHRA AND PENIS- POSTERIOR URETHRA VALVES



- MC - Prostatic urethra → Just distal to Verumontanum.
- Only in .....
- MCC of obstructive urethral lesion in boys

- Incomplete valves → OPD → Childhood → Recurrent UTI

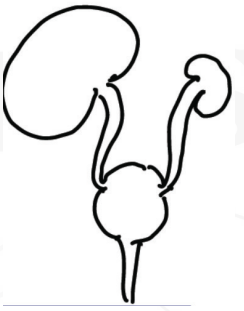
  IOC - .....

  Rx - TransurethralFulguration QQ of valves QQ

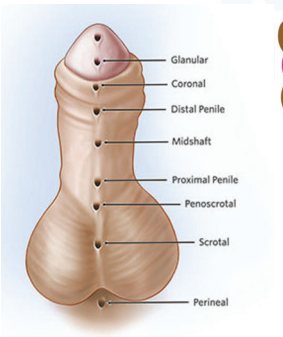
- Complete valves → Emergency → Neonate → Retention of urine → Suprapubic Cystostomy (Supracath)

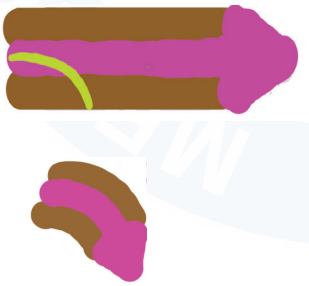
Note :

Whitaker Test- in Hydronephrosis to compare pressures in kidney and bladder in urinary tract obstrution



URETHRA AND PENIS- HYPOSPADIAS

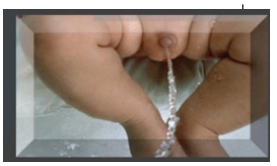




- MC Cong anomaly of Male Urethra
- .....

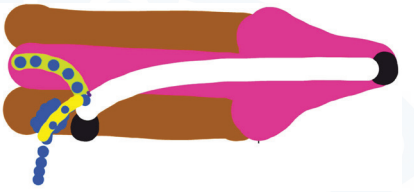
FIGURE- 1.18 Hypospadias- Urethral opening on the undersurface of penis

Active space



Perineal - Most Severe, Least Common

- Sex is difficult to differentiate in Perineal Hypospadias
- Use of Diethyl stilboesterol during Pregnancy is asso c̄ hypospadias
- Asso c̄ Chordee (downward curvature of Penis), Meatal stenosis
- Problems - Urination, Erection, Sperm deposition
- Rx -



Sx - Chordee correction + Urethroplasty

Age of opn - months (6-18 months)

MC - Prepuce is used for Urethroplasty

Circumcision is C.I.. in hypospadias

Complicn of Sx - Urethrocutaneous fistula

SCROTUM- UNDESCENDED TESTIS

- Undescended Testis - Cryptorchidism

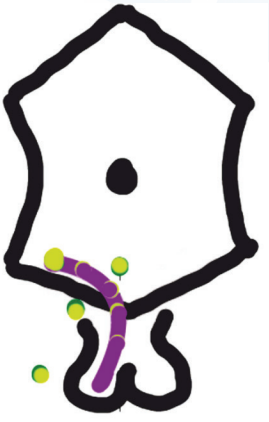


FIGURE- 1.19 Undescended Testis (Cryptorchidism)- Left

Active space

- MC Cong anomaly of Male genitalia
- 3% in Full Term Baby at Birth
- 30% in Pre Term baby
- (R) = 50%; (L) = 30%; (B/L) = 20%
- 75% of UDT, descend into bottom of scrotum c̄ in 3 months of Life
- UDT = (T) arrested in normal path of descent
- Ectopic = (T) deviate from normal path of descent
- Vanishing Testis → During intrauterine life, testis develop, but disappear before birth
- Retractile = (T) ascend upwards, but can be brought down to bottom of scrotum No Rx
- UDT - MC Site - High Scrotal QQ 2nd MC Site- Inguinal canal
- Ectopic T-Site - Superficial Inguinal Pouch (Denys Brown Pouch)QQ
  - Femoral Triangle
  - Root of Penis
  - Perineum

Note : Histological changes in UDT Start at 1 year of age

- **Complicn of UDT : 'A T E S T I S'**

  Atrophy, Trauma, Epididymo orchitis, Sterility (B/L)

  Torsion, Inguinal hernia (MC), Seminoma
- Cl. examn - Inspection, Palpation

  1st Inv - US abd & scrotum

  Diagnostic Laparoscopy - Gold Standard for non Palpable Testis QQ
- Rx : Orchidopexy

  Age of opn - ..... months (months)

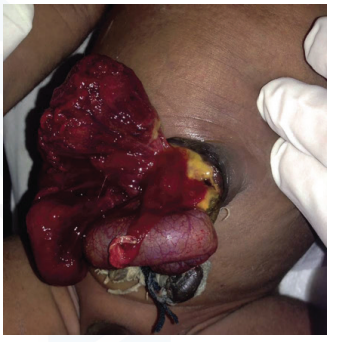
Active

CONGENITAL ABDO. WALL DEFECT

Gastroschisis (1:3000)

Protrusion of Abd organs through
a defect in the abdominal wall
d/t failure of closure of Layers of ant abd wall.

Always Right to umbilicus
Not..... covered by peritonum hernial sac.



Associated c – Intestinal atresia
RF – young mother, smoking, Alcohol, Aspirin

Rx - Primary closure < 24 hrs

If no atresia, then good Prognosis

Omphalocele (1:5000)

Herniation of Abd organs through the umbilical ring
d/t ..... midgut in abd. cavity on 10th wk
Always through umbilicus
Hernial sac formed by peritoneum & amnion.

Associated c – Trisomy 13/18/21
Beckwith Weidman Syndrome

Rx

Antenatal

C-section to Prev Rupture of Sac

Postnatal

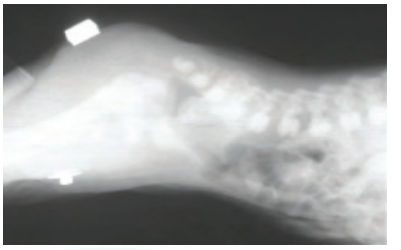
Small defect → 1-ry closure < 24 hs

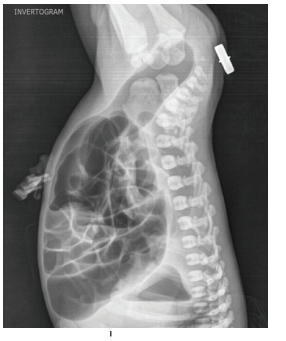
Large defect → Sterile Plastic Wrapping 1-ry closure

..... Prognosis -

Active space

IMPERFORATE ANUS -





- IOC - X-Ray in lateral Prone (Invertogram)

  By 24 hours - Distance between marker and air indicate low or high anomaly

- Rx - Low → Anoplasty

  - High → Plastic Sx - PSARP Postr Sagiltal Anorectoplasty

Art of Solving Q :

1. Simple Single Liner Q → Avoid silly mistakes not true / True / Except / Least Likely
2. Lengthy Q → 1st - Last line

   2nd - Option

   3rd - Stem → Key words
3. Image based Q → (IBQ) →

   1st - Image

   2nd - Option

   3rd - Last line of Q

   4th - Stem

Avoid Frequent changing of answers

Active space

Workbook Questions

1. Which of the following is true about Cleft lip and palate?

   A. In the initial 1 to 2 months, no active intervention is preferred

   B. Muscle reconstruction is a key factor in the initial 1-2 months of life.

   C. Alveolar bone grafting should be done in 1-2 years of age.

   D. Intravelar veloplasty should be done in 1-2 months of life.

2. Which of the following is not true about infantile hypertrophic pyloric stenosis?

   A. There is hypertrophy of longitudinal muscle fibres

   B. US of the abdomen is 100% sensitive modality to detect

   C. Presents with projectile non bilious vomiting

   D. Baby is dehydrated with metabolic alkalosis

3. One and a half year old child is having problem in urination for the last 3 months. You have introduced a contrast dye from the urethra to visualize and understand the problem as shown in the image below. What is this investigation most likely?

Active space

A. Micturating cystourethrogram

B. Retrograde urethrogram

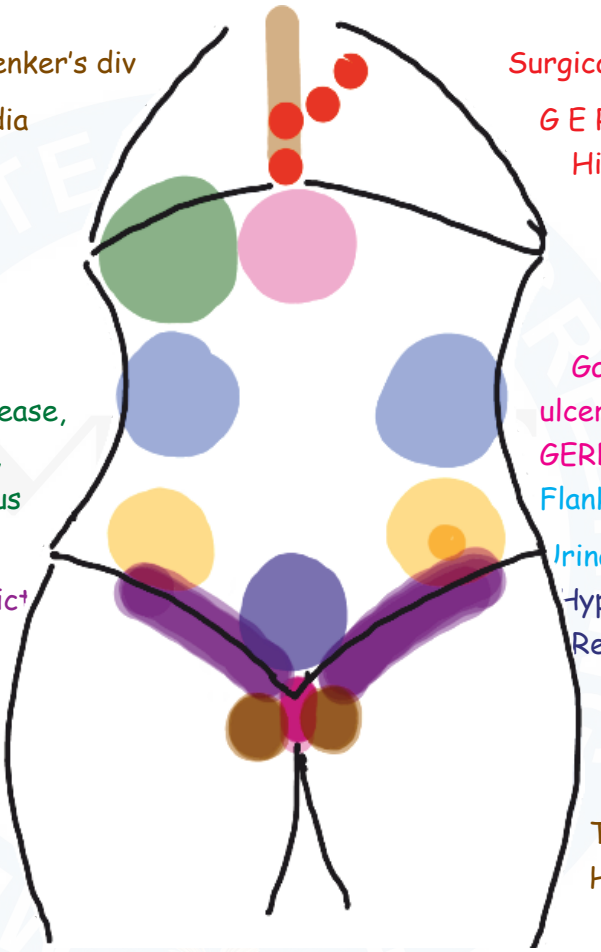
C. Antegrade pyelography

D. Retrograde pyelography

Answers: 1. (A), 2. (A), 3. (B)

Active space

ADULT SURGICAL PROBLEMS



Dysphagia - Zenker's div
Achalasia Cardia

Surgical Chest Pain
G E R D
Hiatus hernia
Boerhave Syn

Epigastrium -
Gastritis, Peptic ulcer, Pancreatitis, GERD, Cholecynostitis

Flank -
Irinary Stones, TB

Right hypoch.
Gall Stone Disease,
Liver Abscess,
Sigmoid Volvuus

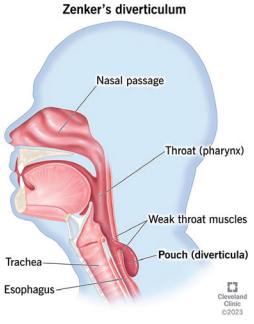
Iliac Fossa
Acute Appendict
9BD,
Div of colon

Hypogastrium -
Retention of urine
Trauma, BPH

Penis -
Phimosis,
Peyronie's,
Priapism

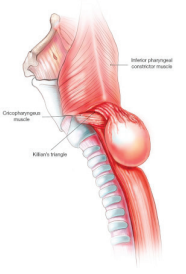
Groin - Hernia
Scrotum - Testis - Torsion, Varicocele, Hydrocele
Rectum, Anal Canal - Hemorrhoids, Abscess, Fistula, Fissure, Sinus

DYSPHAGIA- ZENKER'S DIVERTICULUM



Zenker's div - Pharyngeal Pouch - Cricopharyngeal Achalasia

- Out pouching of ..... pharyngeal wall Postrly
- Above CPS, Through Killian's dehiscence
- Inferior Constrictor muscle (fibres of thyropharyngeus and Cricopharyngeus muscle)



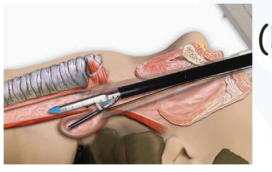
- Pulsion diverticulum
- Acquired, False Usually mucosa/Superficial Submucosa
- Elderly
- 50% - associated with cervical webs
- MC - C/F .....
- Regurgitation
- Halitosis (foul Smelling breath)
- Complication - Aspiration → Pulmonary abscess
  Postr neck swelling to the left of midine
- Boyce Sign - Gurgling sound heard on reducing the swelling

Bryce Sign = Laryngocele

IOC - .....

- Rx - If < 2 cm - C P myotomy
  2 - 4 cm - Diverticulopexy
  > 4 Cm - Dohlman's Option

QQ



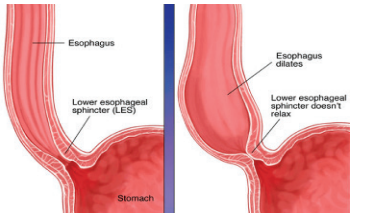
(Diverticulo-eso- Stomy using Linear Stapler)

FIGURE- 2.1 Zenker's diverticulum on Barium swallow

DYSPHAGIA- ACHALASIA CARDIA

Tone of LES is regulated by Acetyl Choline

- Failure of relaxation of LES (Lower Eso Sphincter) due to unopposed action of ACH
- There is Absence of Ganglion cells in Auerbach plexus in Eso
- Esophageal Peristalsis is Absent - Pathognomic
- Primary - Exact cause X
- Secondary - Secondary to disease
  - Chaga's disease
  - DM
  - Scleroderma



Active space

- Pseudo Achalasia - Tumours around esophagus
  (Mediastinal tumours / Tumours of cardia of stomach)
- C/F : 25-40 yrs of age
- MC Dysphagia - Initially for ..... gradually for both < Liquids / Solids
- Regurgitation, Aspiration, Pulmonary Complicⁿ

Esophagitis - Middle 1/3 of Eso

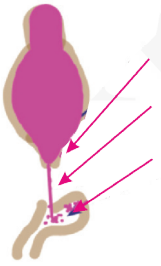
Squamous cell Ca - Middle 1/3 of Eso

FIGURE- 2.2 Bird beak/ Pencil tip sign in Achalasia cardia on Barium swallow

- Cl. app :

IBQ QQ

- 1st Invⁿ - Ba Swallow



..... - Pencil Tip

Rat Tail appearance

Absence of Gastric Air bubble

QQ

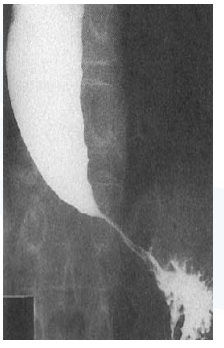


FIGURE- 2.3 Bird beak/ Pencil tip sign in Achalasia cardia on Barium swallow

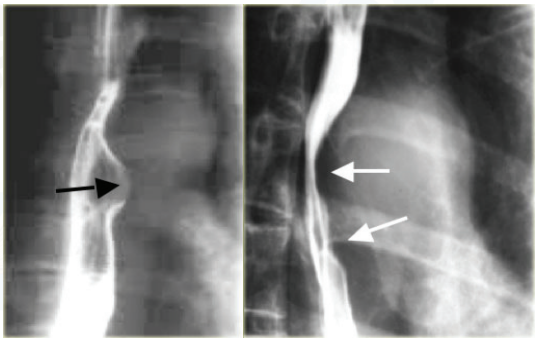


FIGURE- 2.4 Filling defect and Rat tail appearance in Cancer esophagus on Barium swallow

Q Jan 25

A 40-year-old lady presenting with dysphagia, initially for liquids and then for solids, with immediate regurgitation of liquids. Which of these signs would be seen in this lady on imaging?

a Irregular tapering at the lower end
b. Rat tail sign
c. Nutcracker esophagus
d. Irregular filling defect

Active space

- Standard Invⁿ - .....
- (N) IEP = 10 - 25 mm of Hg
- Achalasia Cardia = 25 - 100 of Hg
- Nutcracker Eso = 180 - 200 of Hg
- Diffuse Eso spasm = > 300 - 400 " "

FIGURE- 2.5 Cork screw esophagus in Diffuse esophageal spasm on Barium swallow

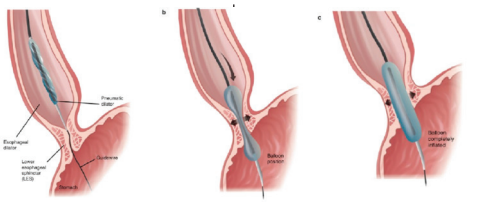
QQ

- DES - On Ba Swallow = Cork Screw appearance QQ

  Diff. ESO Spasm

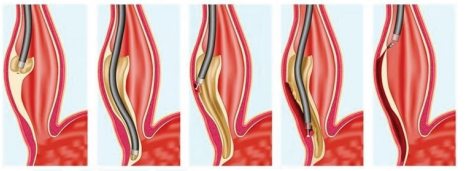
- Rx of Achalasia
  - Drugs - CCB's Nitrates
  - Pneumatic balloon dilatation

    - Risk - Eso rupture
  - Rx OC - Sx - Myotomy



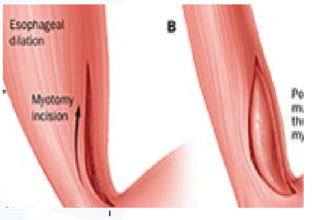
- Heller's Option - open Sx/Laparoscopic Sx
- .....Per Oral Endoscopic Myotomy

  (To avoid reflux - Fundoplication) Floppy Nissen's



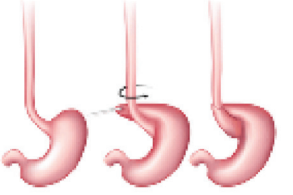
DYSPHAGIA- SCHATZKI'S RING

- Circumferential mucosal, submucosal thickening at Squamo columnar junction
- common in GERD patients
- sometimes can lead to Intermittent dysphagia
- Rx - Single Endoscopic dilatation



Lye Stricture

- d/t ingestion of Caustics (alkali)



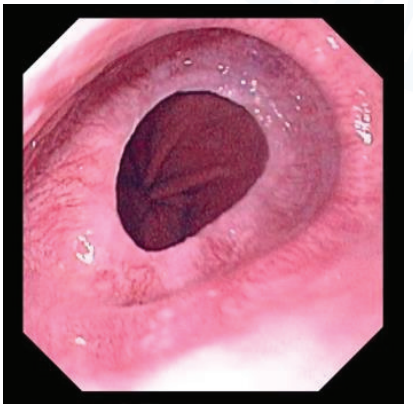


FIGURE- 2.6 Schatzki's Ring- at Squamo-columnar junction- Endoscopic view

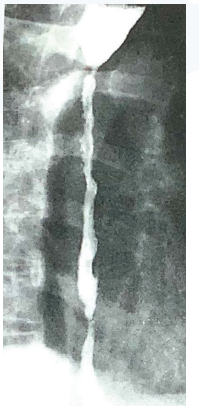


FIGURE- 2.7 Lye stricture of esophagus after ingestion of caustics on Barium swallow

Floppy Nissen's Fundoplieⁿ

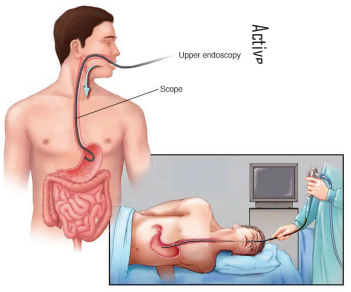
	Achalasia	DES	Nutcraker
Pathogen-esis	LES Fail to relax Eso Peristalsis reduced or absent	Uncoordinated Eso contractions (> 120mm Hg Multipeaked contraction)	Coordinated high amplitude contractions (> 180mm Hg)
Onset	Gradual	Sudden	Intermittent
Ba Swallow	Bird Beak	Cork Screw	Slow Clearance
Dysphagia	Liquids	Liquids	(-)
Rx	POEM	CCB'S Nitrates	CCB'S Nitrates
IEP	25-100	> 300 - 400	> 180 - 200

SURGICAL CHEST PAIN- GERD

- GERD - Gastro Eso Reflux Disease
- Protective factors : Integrity of LES / Good Gastric Emptying / Good Eso Peristalsis
- TLOSR'S - Transient LES relaxation can occur for 3-4% of times in 24 hours in normal individuals
  MC in ..... position (Supine)

**

- RF : Most Imp - Smoking, Alcohol, Tea, Coffee, Hot spicy food, hot Liquids, Obesity, Progesterone, CCB's - Nitrates
- C/F - MC - Retrosternal Pain or discomfort (Heartburn)
  Epigastric Pain
  Regurgitation
- Cl. app : Lifestyle modification
  ↓ stress, X smoking, Alc, Tea, coffee, hot spicy food
  ↓ wt



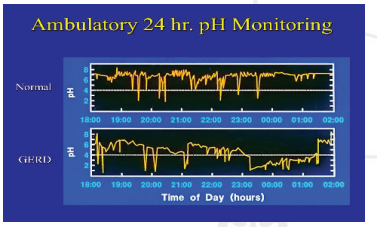
- Do not sleep for 3 - 4 hrs after meals
- Elevate head end of bed by 5 - 8 inches
- If fails → ..... - Rx oc for GERD 90 - 95%
- If PPI fails Successful

↓

Ist Invn - UGIE ..... (IOC - GERD)

Upper GI Endoscopy (OGD-Oesophago Gastro duodenoscopy)

Low Eso	Stomach	Duodenal
	Ulcer	Ulcer
↓	↓	↓
QQ 24 hr Ambulatory pH monitoring	Biopsy from • Ulcer -to rule out Ca • Prepyloric - for H. Pylori	Biopsy from • Prepyloric for H. Pylori
↓		
Drop in pH < 4 for > 4% times in 24 hrs → GERD		
Demeester Score		
Gold Standard		
QQ Invⁿ for GERD		



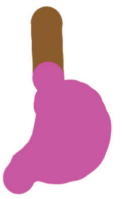
- Sx Rx for GERD - Floppy Nissen's Fundoplication
  (Loose 360° wrappping of fundus of stomach around Low eso)
- Complications of GERD :
- MC " " " - MC-Esophagitis (Lower 1/3 ESO)
  - Stricture Eso
  - Shortening of Eso
  - ..... Eso
  - Sliding Hiatus Hernia

Active space

Barret's Esophagus

Q July 25 QQ

- Intestinal columnar metaplasia of squamous mucosa of lower eso
- Hallmark - Goblet cells
- SC Jⁿ move upwards
- ↑↑ Risk of Adenoca
- Rx : Endoscopic mucosal resection of dysplatic areas followed by RFA + PPI's



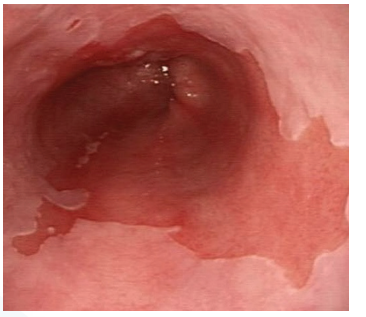
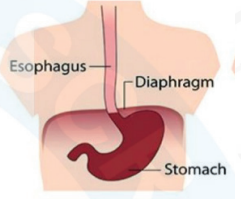
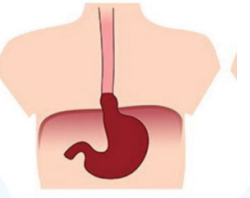
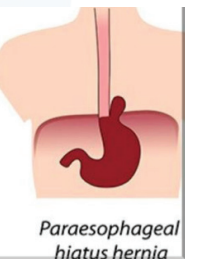


FIGURE- 2.8 Barret's esophagus- intestinal columnar metaplasia of squamous mucosa of lower esophagus in GERD on Endoscopy

SURGICAL CHEST PAIN- HIATUS HERNIA

	Sliding HH	Paraeso HH Rolling or .....HH
	 Normal	 Sliding hiatus hernia	 Paraesophageal hiatus hernia
• GE Jⁿ	Thorax	Abdomen
• Incidence	Common	
• Asso c̄ Reflux	✓	×
• Risk/Complication	↓	↑ (Ischemia, Strangulation)
• C/F	Chest Pain Dyspnoea	Chest Pain Dyspnoea
• 1st Invⁿ IOC	Chest X - Ray ..... C ..... CE CT ..... 2ⁿ → Barium meal	Chest X - Ray
• Rx	..... Fundoplication	

Active space

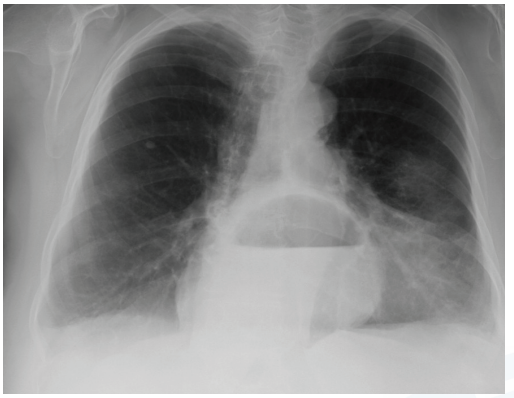


FIGURE- 2.9 Hiatus Hernia- Stomach bubble in Thorax on Chest X-Ray

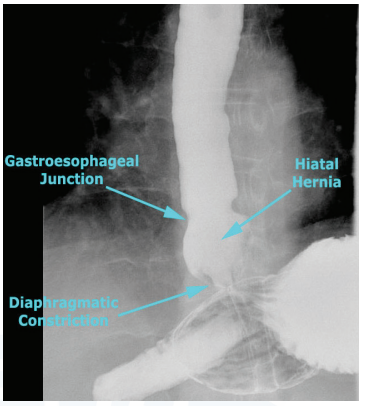


FIGURE- 2.10 GE Junction above diaphragm suggest Sliding Hiatus Hernia on Barium meal

Los Angeles classification of Reflux Esophagitis -

1 or more mucosal breaks

Grade A - Does not extend between 2 mucosal folds
≤ 5mm

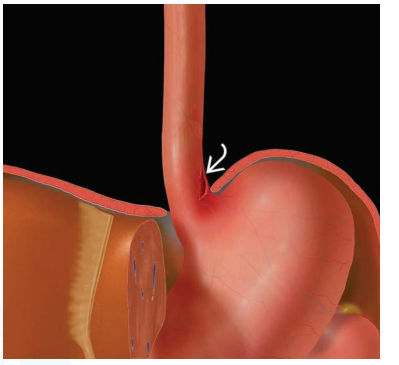
Grade B - Does not extend between 2 mucosal folds
> 5mm

Grade C - Extend between 2 or mucosalfolds
< 75% of Eso circumference

Grade D - Extend between 2 or mucosalfolds
≥ 75% of Eso circumference

Active space

SURGICAL CHEST PAIN- BOERHAAVE SYNDROME (BAROTRAUMA)



- Spontaneous Eso Perforation
- Transmural (All layers involved)
- Elderly - after heavy meals/binge QQ drinking, ↑ Intragastric Pressure → Forceful Vomiting → Eso ruptures at the weakest point

Q July 24

- MC Site of Eso rupture - Left, Low eso, Posterolateral wall
- Severe Mediastinitis, Risk of Septicemia
- C/F : Sudden, severe, (L) Chest pain radiating to (L) Shoulder, (L) arm (Mimicks Myocardial Infarction)

(Air benath skin) Subcutaneous Emphysema - Pathognomic

- Mackler's Triad - Vomiting, chest pain and subcutaneous emphysema
- Cl; app : Emergency - Basic Resuscitation

Rule out Cardiac event (ECG, Cardiac enzymes)

↓

After Stabilization

↓

- Auscultation - Foot steps on the snow (Hamman's Sign / Hamman's Crunch / Mediastinal Crunch)

Homan's and Moses sign are seen in DVT

Barium is hypertonic → Not Used in GI Perforation

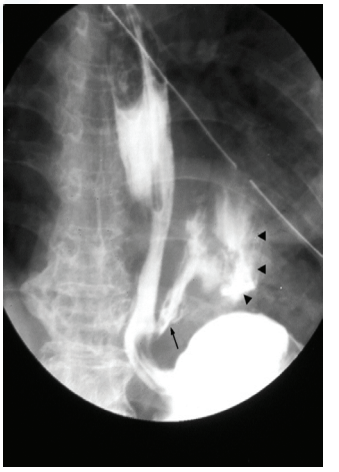


FIGURE- 2.11 Boerhaave syndrome (Barotrauma)- leakage of contrast in perforation of lower esophagus on Contrast swallow

Active space

- 1st Invn : CXR - Pneumomediastinum
- IOC - Best : ..... - If oral contrast has to be used ➔ Iohexol - Preferred

  Water soluble - Less injurious
- Rx : Thoracotomy + Mediastinal lavage ➔ Life saving + closure / non closure of perforation depends on amount of contamination

  a) < 4 - 6 hrs - Minimal contamination ➔ Clase the perforation

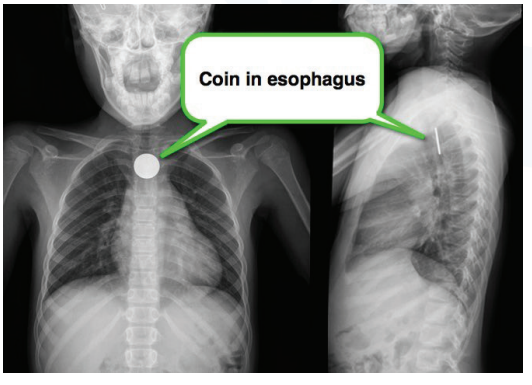
  b) > 4 - 6 hrs - Gross contamn ➔ Leave the drain

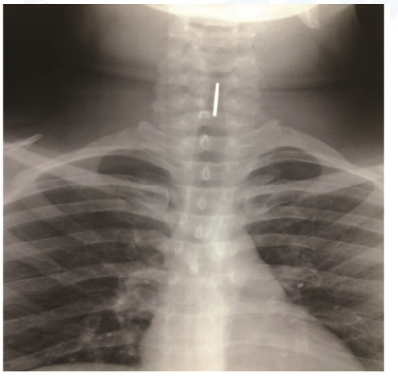
SURGICAL CHEST PAIN- IATROGENIC ESOPHAGICAL PERFORATION

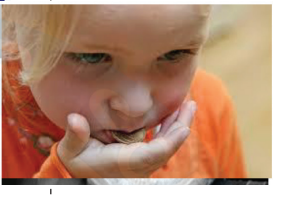
- Iatrogenic Eso Pervorn ››› Boerhaave syn
- MCC of Eso Perforation ➔ UGIE (Upper GI Endoscopy) Rigid Endo ›› Flexible Endo
- MC Site - .....
- Rx - Non operative

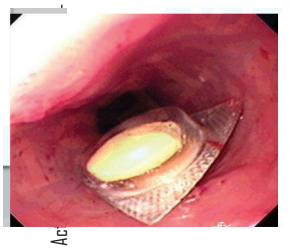
  - Conservative Rx Leave a NG Tube for 5 -7 days

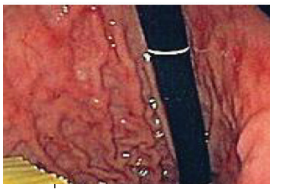
SURGICAL CHEST PAIN- FOREIGN BODIES IN ESOPHAGUS











MC FB in ESO :

- Child - Coin

Q Jan 25 - ENT - Trachea

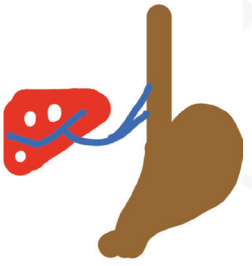
- Hemlich's

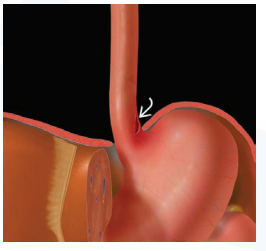
- Adult - Food bolus (Fish bone)
- Elderly - Loose teeth / Dentures
- If impacted at cricopharynx or symptoms → Endo removal
  (Esophagoscopic removal)
- If food bolus gets inpacted in eso → Give fizzy drinks

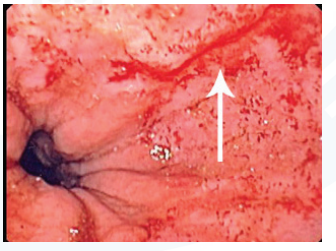
SURGICAL EPIGASTRIC PAIN- GASTRITIS- ACUTE, CHRONIC AND SPECIAL GASTRITIS

Acute Gastritis

- - NSAID's, Smoking, H. Pylori, Alcohol
- - Hemetemesis
- - MCC of Hemetemesis → Peptic ulcer (Duodenal ulcer)
  → Gastric erosions
  → Mallory Weiss Tear
  → Eso varices - (Portal hypertension/ Alcoholic Cirrhosis)







Boerhaave Syndrome	Mallory Weiss Tear	Esophageal Varices
Binge drinking	Binge drinking	Chronic alcoholic / Chronic Liver disease
Esophagus	GEJ^{N}	Esophagus
Transmural	Mucosal	Mucosal / Submucosa
Chest Pain	Hemetemcsis	Hemetemesis

Active space

- Cl. App. in hemetemesis :
- IVF ✓ Crystalloids - Normal Saline
- IF > 30% blood loss → BT ; Trenexaemic acid

(BT - Leukodepleted blood)

- → 1st Invn → UGIE

↓

Bleeding vessel

↓

- If identified → Endo Sclerotherapy / " Adrenaline / " Diathermy / " Clips

If not identified → 2nd Invn - Angiography → Fogarty Catheter in vascular Sx

Angioembolisation (Therapeutic Embolisation MC - Coil)

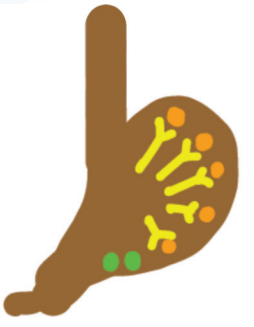
Steps for bleeding Eso varices :
1. ABC — Q July 25
2. EVL (Endoscopic variceal Ligation) Rubber bands
3. Definitive Rx - Later

Chronic Gastritis

★ Type A - Atrophic Gastritis /Autoimmune - Ab against Parietal cells

↓ Hcl → ↓ IF

- Hypochlorhydria → ↓
- ↑ Gastrin → Pernicious
- Hypergastrinemia → anemia



(MCC of Hypergastrinemia → .....)

- MC Site of Cancer in Pernicious anemia - Fundus of stomach

Active space

★ Type B Gastritis - H. Pylori

- Toxins- cag A, vac A, Proteases, Phospholipases

↓

damage Gastric acid mucus barriere

↓

Gastritis, Ga ulcer, Ga Cancer

- Enzyme - Urease

Urea ⟶ NH3 (Alkaline)

↑↑ Hcl

Duo ulcer

- Cl app in Epigastric Pain :

↓

1st - Lifestyle modification

↓

If fails → PPI 'S

↓

If fails → UGIE (IOC)

Biopsy from Prepyloric region

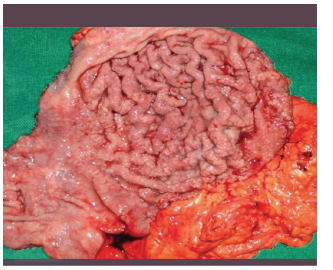
- Stains - Giemsa, Ethin Starrey Silver Stains
- Rapid urease test - CLO Test - Campylobacter Like Organism Test
- Urea Breath Test - Radiolabelled Carbon ($^{13}C/^{14}C$)
- If H. Pylori → ⊕ → Eradication therapy

2Ab + PPI

Q July 25

Active space

Special Gastritis



1. Menetrier's disease - gross hypertropy of gastric mucosal folds, Premalignant

2. Reflux G - Alkaline Gastritis

- After Gastric Sx (Damage to Pyloric Sphincter)

Reflux of Alkali from Jejunum into stomach

→ can cause Intestinal metaplasia

→ Adeno Carcinoma

SURGICAL EPIGASTRIC PAIN- PEPTIC ULCER

- MC Site - D1 (1st Part of duodenum)
- 2nd MC " - Gastric (Lesser curvature of stomach)
- Low eso, Meckel's div, Stoma of GJ
- RF :
- MCC - .....

NSAID'S

Smoking

Spirits

Spices

Stress

Steroids

Syndrome - Zollinger, Ellison Syn-Gastrinoma

Active space

- Peptic ulcer in Head Injury → Cushing's Ulcer
- Peptic ulcer in Burn → Curling's Ulcer

(Skin ulcer in Burn - Marjolin's ulcer)

Features	Duo ulcer	Ga ulcer
Incidence	Common	Less Common
Age	Young	Elderly
Socioeconomic status	Rich	Poor
Size of ulcer	Small	Large
Pathogenesis	↑↑ HCL	Disruption of Gastric acid mucus barrier
Hcl secretion	....↑↑.....	(N) / ↓ .....
C/F	Epigastric Pain	Epigastric Pain
Relation of Pain to meals	Pain occurs 2-3 hrs after meals	Pain occurs immediately after meals
Pain relieved by	Food	Vomiting
Appetite, wt	....↑.....	.....↓.....
Blood group asso.	'O'	'A'

- Cl. app. of Epigastric Pain

LM (Lifestyle modification)

↓

If fails - ($R_x$oc)

In Gastric Ulcer → Add Sucralfate

Active space

If fails → 1st Invn - UGIE

Low Eso	Stomach	Duo.
✓	✓	✓

H. Pylori ⊕
↓
Eradication therapy
↓
If Symp. recur
↓
Reinfection c̄ H. Pylori
↓
Reeradication therapy
↓
If fails
↓
Sx

- Sx Rx for DU : Aim → ↓Hcl Secretion
- Rx OC
  - Highly Selective Vagotomy or Parietal cell Vagotomy (No drainage opn required)

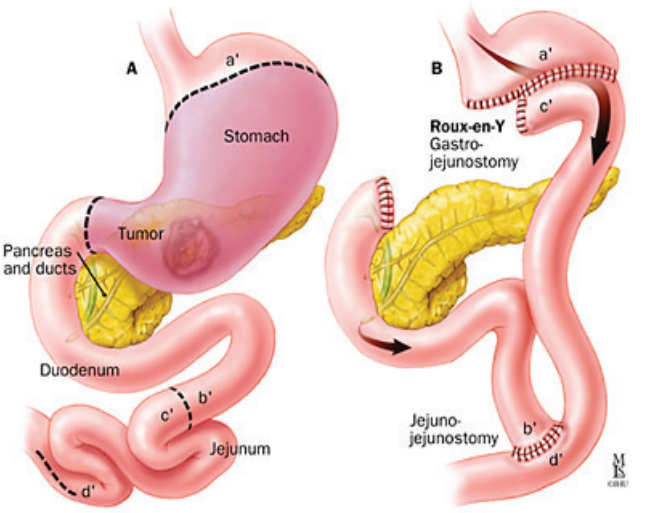
Risk of recurrence is high after Highly Selective Vagotomy

- Selective Vagotomy
- Truncal "

} + drainage opn- Pyloroplasty or GJ

- Sx Rx For GU : - Aim - Excision of ulcer to remove Cancer risk
- Rx OC - Roux enY GJ (Modification of Bilroth II to prevent Bile / Panc. Juice reflux into residual stomach)
- Bilroth I - GD
  II - GJ

Active space



SURGICAL EPIGASTRIC PAIN - COMPLICATIONS OF SURGERY FOR PEPTIC ULCER

1. Duodenal Stump blow out on 4th Postop day
2. Dumping Syndrome (Post cibal Syndrome)

QQ Early Dumping:

→ c̄ in 5-10 mins after meals

→ Bloating Epigastric Pain

Tachycardia, Cold Peripheries

- d/t ↑ osmotic load in intestine

Late Dumping:

→ in 2nd hr after meals

→ Faintness, Tremors

- d/t Reactive hypoglycemia (d/t Insulin excess)

Rx - Small frequent meals MIST

3. Post vagotomy diarrhoea

QQ 4. Nutritional def. - Iron, Vit B_{12}, Ca, Phosphorus Protein

5. Gall Stones QQ

6. Ga Cancer

Active space

SURGICAL EPIGASTRIC PAIN- COMPLICATIONS OF PEPTIC ULCER

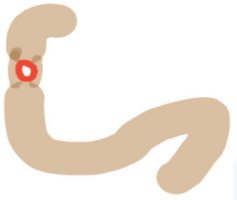
1. Bleeding - MC - Duo ulcer

   MC - Posterior surface of D1

   MC - Gastroduodenal artery Q July 25

   GU bleed is rare, (L) Gastric artery

2. Perforation - Antr. surface ulcer, Peritonitis



Diffuse abd pain, Nausea, vomiting / Shock Like

Basic Resuscitation - IVF, IV Ab 1st

Cl. examⁿ - Cardboard Like rigidity in Peritonitis

(C X R) Pneumoperitoneum

In Supine - (L) Lateral decubitus Neet PG 25

Prefer seeing gas below (R) dome of diaphragm Hollow viscus Perforation QQ

Gas under diaphragm

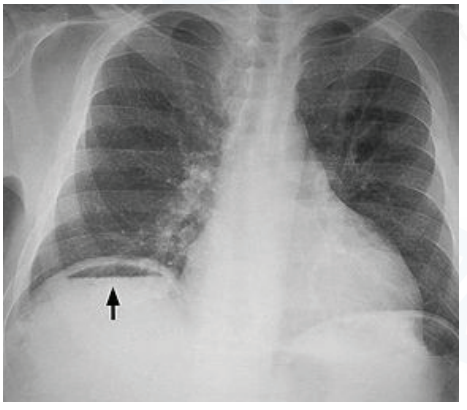


FIGURE- 2.12 Pneumoperitoneum- Gas under Right dome of diaphragm on Chest X-Ray

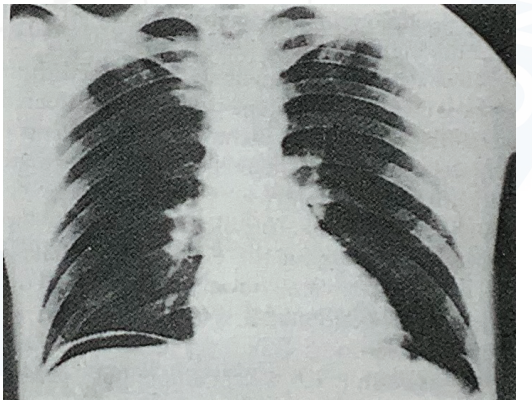


FIGURE- 2.13 Pneumoperitoneum- Gas under Right dome of diaphragm on Chest X-Ray

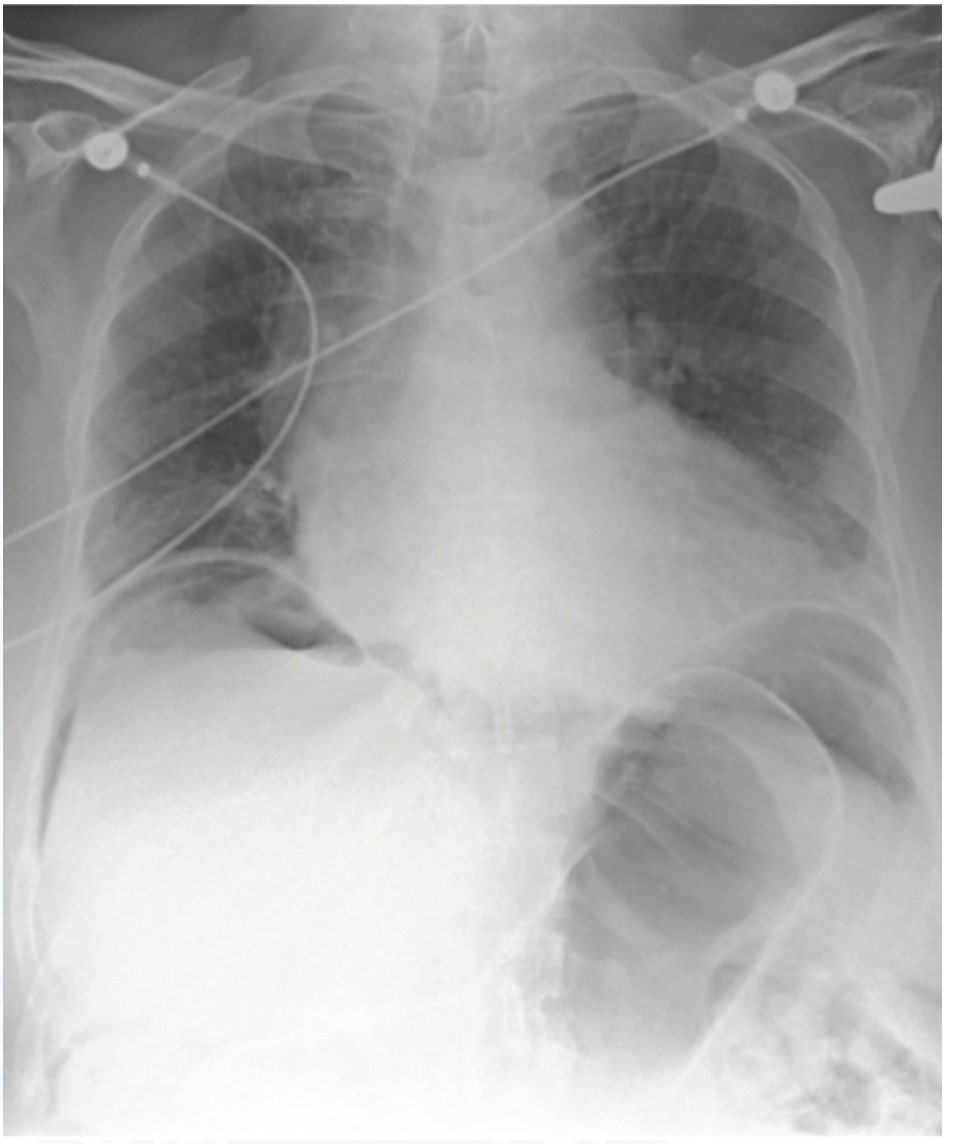
- Rx : Laparotomy + Peritoneal lavage (Life Saving) MIST

  +

  a) < 24 - 48 hrs - Minimal contamination - Closure of perforation → Omental patch - Graham's patch

  b) > 24 - 48 hrs - Gross contamination - Leave the drain

Active space



Gas under diaphragm :

1st Make stable - IVF, IV Ab

Nasogastric tube, Urinary catheter

Analgesia

- Laparotomy Lavage
- Drainage Repair Exteriorization

Active space

SURGICAL EPIGASTRIC PAIN- BEZOARS

- Trichobezoar : hairs in stomach
  Young
  Psychiatric ♀

QQ
IBQ

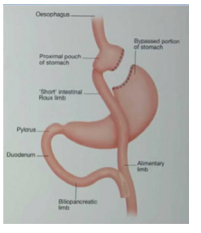
FIGURE- 2.14 Trichobezoar

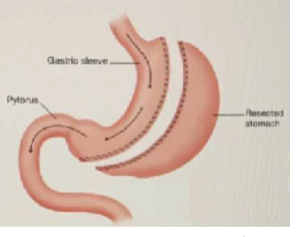
- Phytobezoar : Vegetable matter
- Lactatobezoar : Milk & Milk products

} d/t Gastric Stasis
Prolonged Immobilisation
and
Previous Gastric Surgery

Bariatric Surgery

- - Sx for Morbid Obesity
  Metabolic Sx
- Indication - 1. BMI> 40
  2. BMI > 35 + Co-morbidities (DM, HTN, Osteoarthritis, Sleep apnoea Syndrome, CAD, CVA)
- Most effective - Roux en Y Gastric by pass
- LAGB - Laparoscopic Adjustable Gastric banding
- Vertical banded Gastroplasty
- Sleeve Gastrectomy .....





RIGHT HYPOCHONDRIUM- AETIOPATHOGENESIS, CLINICAL FEATURES, MANAGEMENT OF GALL BLADDER STONE DISEASE

- Gall stone disease (Cholelithiasis)

1. Cholesterol - Pale Yellow
2. Pigment -

1. Cholesterol: Fat, Fertile, Female, Forty

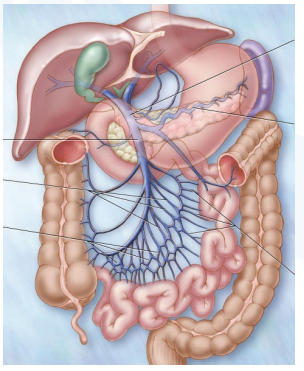
- ↑ cholesterol
- ↓ Bile salts, acids

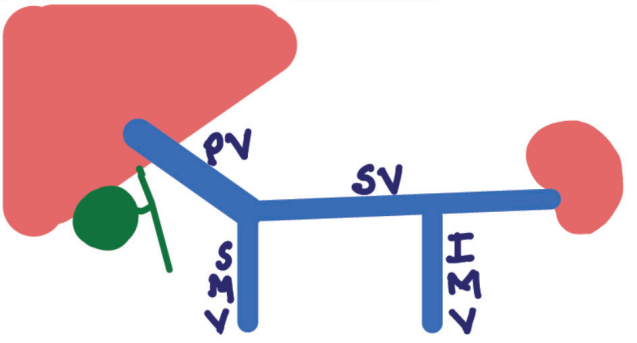
} Lithogenic bile or Supersaturated bile

→ - Abnormal Enterohepatic Circulation -

Diseases of Terminal Ileum - Tuberculosis of intestine

Crohn's disease





2. Pigment stones - hemolytic disorders →

HS (Hereditary Spherocytosis), SCA (Sickle cell anemia)

Q Jan 25 Patho

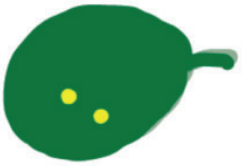
Parasitic infestation - Ascaris, Clonorchis Sinensis

Bacterial Infecⁿ - E. Coli

Cirrhosis of Liver

Active space

- C/F :

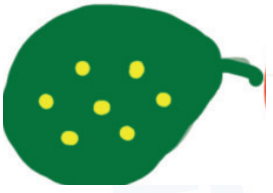


- Asymp > 80%

↓

No Rx but Cholecystectomy if-

a) Single Stone > 3 cm, b) Multiple Stones, c) Elderly
d) Immunocompromised, e) Pregnancy f) Typhoid carrier
g) Porcelain GB h) Adenomatous Polyps > 10 mm
i) Hemolytic disorders j) Bariatric Sx (Gastric Sx)



- OPD

↓

Mild Pain, off & on ..... in right upper abd.

↓

IOC - ..... Gold standard Inv Stone-hyperechoic

Postr acoustic shadow

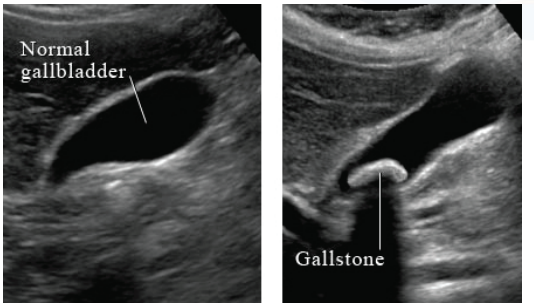


FIGURE- 2.15 Ultrasound comparing a normal Gall bladder with a stone inside the Gall bladder. Note the stone casting its posterior acoustic shadow

Active space

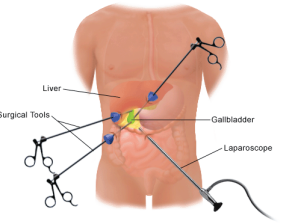
Radiolucent Stones - Not Visible on X-Ray [PG LUX] Parotid duct, Gall bladder, Lysine Uric acid Xanthine

Elective Rx : Laparoscopic Cholecystectomy

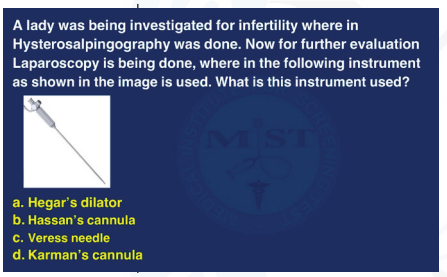
Verres Heedle

Q Jan 25 / Q July 23

Q Jan 25 / Q July 24



4 Ports



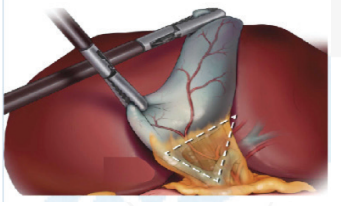
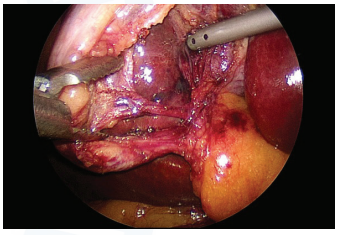


FIGURE- 2.16 Triangle



Verres needle, CO_2 gas (Carbon dioxide) Q July 23

Hepatobiliary/ Hepatocystic Δ :

Sup - Infr surface of Liver

Laterally - Body of GB, Cystic duct

Medially - ..... CHD

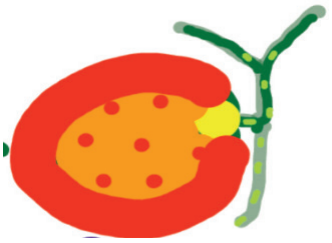
Content - Cystic LN of Lund and Cystic artery

Calot's Δ :

- Suprly Cystic artery
- Laterally - Same (HB Δ)
- Medially - Same (HB Δ)

Content - Can be Cystic LN of Lund

Active space



Pregnancy - 2nd Trimester - Operate if moderate or severe

1st & 3rd Trimester - Conservative

- Emergency Large stone obstructing mouth of GB / Cystic duct

↓

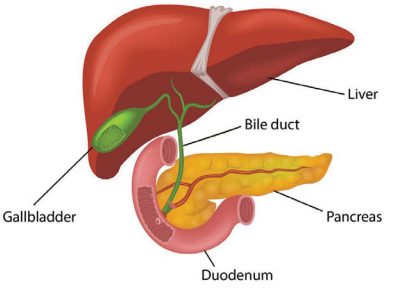
Sudden, severe RHC pain radiating to Infr angle of Ⓡ scapula/tip of Ⓡ shoulder

Anorexia Nausea, Vomiting, Fever

↓

Cl. Examn -

.....sign Acute Cholecystitis



↓

US abd IOC

↓

Rx - Conservative IVF, IV Ab 5 - 7 days

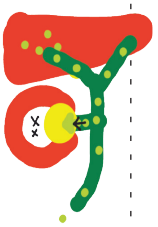
↓

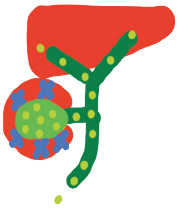
Discharge advice

Interval cholecystectomy after.....weeks

Active space

HIDA Scan IOC to differentiate Acute Calculus Cholecystitis from Acalculus Cholecystopathy





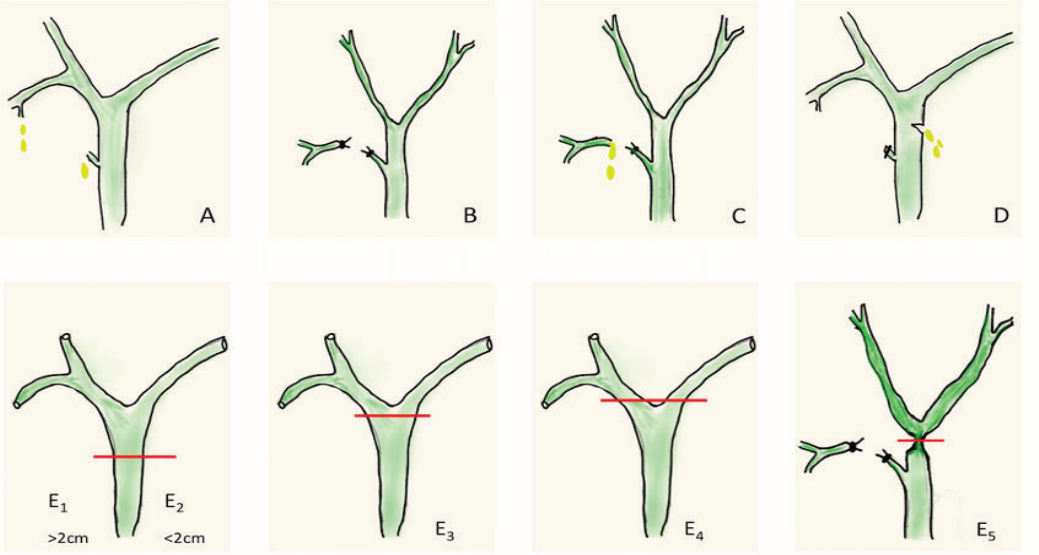
Acute Calculus cholecystitis	Acalculus cholecystopathy
↓	↓
Non visualisation of GB	Persistence of isotope c̄ in GB despite washout of isotope from remainder of biliary tree.

- Complications of Lap Chole :

1. Bile leak - MC from Cystic duct stump-Resolve c̄ Endobiliay Stenting
   - Collection in Subhepatic area c̄ symp - Require US guided Catheter drainage

Q July 25 ⟶

2. Bile obstruction - D/t Stone in CBD - Require ERCP
3. Biliary stricture



Active space

Bile duct injury	Bismuth	Strasberg
Cystic duct leak or leaks from small ducts in liver bed	–	A
Occlusion of an aberrant right hepatic duct	–	B
Leak from an aberrant right hepatic duct	–	C
Lateral injury to the common bile duct (<50% of the circumference)	–	D
Common hepatic duct stricture, stump >2 cm	Type I	E1
Common hepatic duct stricture, stump <2 cm	Type II	E2
Hilar stricture with preserved biliary confluence	Type III	E3
Hilar stricture with involvement of confluence	Type IV	E4
Stricture to an aberrant right hepatic duct and to common hepatic duct	Type V	E5

RIGHT HYPOCHONDRIUM- GALL BLADDER PATHOLOGIES OTHER THAN STONE

1. Mucocele

- Mucus distension of GB

2. Empyema GB

- Pus inside GB- Tube cholecystostomy
- (US guided percutaneous catheter drainage)

3. Cholesterosis

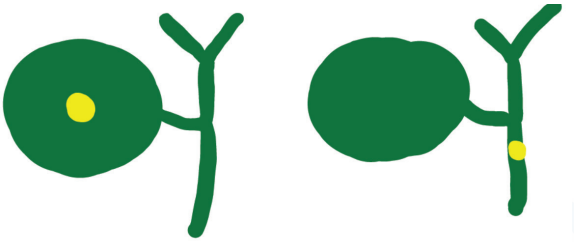
- - Strawberry GB - Yellow flakes on the mucosal surface of GB

Active space

4. Porcelain GB

- Calcium deposition in GB/ ↑ Risk of Cancer

(1) Jaundice (2)



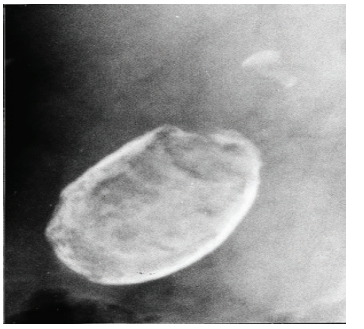


FIGURE- 2.17 Porcelain GB or Limey Bile- Calcium carbonate and Phosphate visible on plain X Ray.

5. Mirizzi's Syndrome

- Jaundice d/t Compression of CHD by a large Gall stone

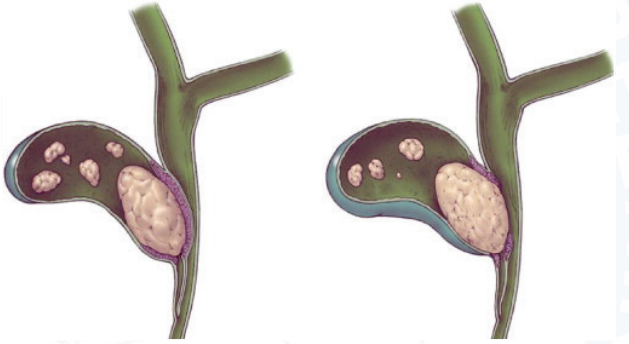
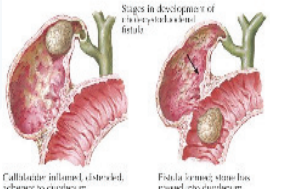


FIGURE- 2.24 Mirizzi's syndrome- Compression of CHD by a large stone in GB/ Cystic Duct

6. Fistula of GB



- MC - D1 - Cholecysto duodenal fistula

↓

Gall stone slip into duod.

↓

Impacted in distal ileum

↓

Int. obsⁿ d/t GS - GS Ileus

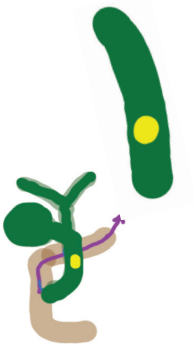
- X - Ray - Rigglar's Triad
  - Multiple Air fluid levels
  - Air in GB
  - Stone in intestine
- Rx - Removal of obsⁿ

Saint's Triad- Gall stones / Hiatus Hernia / Diverticulosis of colon

Active space

RIGHT HYPOCHONDRIUM- CHOLEDOCHOLITHIASIS AND CHOLANGITIS

Bile Duct Stone Disease



- Stone in CBD
  d/t • Slipping of GS into CBD
  • Bile duct infection

Partial obstruction → OPD

↓

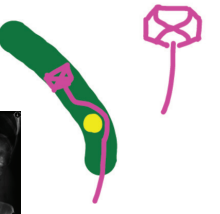
Intermittent Jaundice

↓

LFT, US abd

- IOC - MRCP / ERCP Preferred if any diagnostic or therapeutic Procedure is done
  Rx - Ercp + Sphincterotomy+ CBD Stone extraction c̄ Dormia basket

- Complete Obstruction - Emergency
  Cholangitis Charcot's Triad - Pain, Fever, Jaundice
  Reynold's Pentad PFJ+ hypotension
  Mental Confusion



↓

Basic Resus

IVF, IV Ab

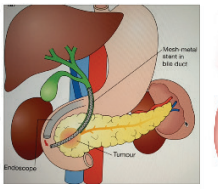
Q July 24

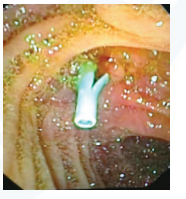
↓

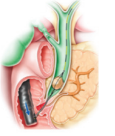
LFT US abd

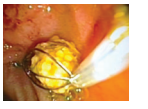
↓

Endobiliary Stenting (ERCP+Stenting)



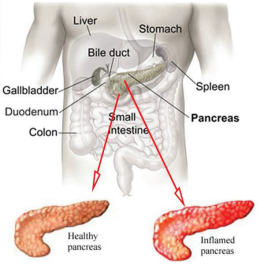






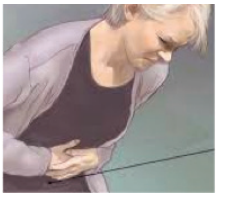
RIGHT HYPOCHONDRIUM- ACUTE PANCREATITIS

- Acute Inflammation of Pancreas - sudden, severe abd pain, shock,
  ↑ Panc. enzymes
- MCC - .....
- 2rd MCC - Alcohol
- Children - Hereditary, Trauma
- Operative - ERCP
- Premature activation of Pancreatic enzymes c̄ in Pancreas → Necrosis of Pancreas



Active space

- Sudden severe upper abd pain, radiating to the back (R/L side), aggravated in Supine position, relieved on sitting c̄ bending forwards



- **Nausea, Vomiting,**

  (Fever is due to Secondary bacterial infection)
- Ascites, Pleural effusion
- Bleeding in Retroperitoneum - Acute hemorrhagic Panc

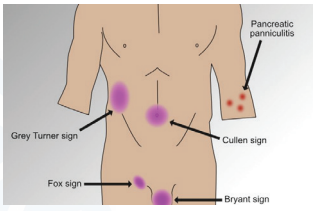
  ↓

  Bluish discolouration

  a) Flanks - GREY TURNER Sign

  b) Umbilicus - CULLEN'S Sign QQ

  c) Inguinal - Fox Sign



- Shock - Tachycardia, Tachypnoea, Hypotension

Q Jan 25
Lipase

- Cl. app : Basic Resus. - IVF ★★

  - No role of antibiotics initially as it is enzymatic necrosis
- Pancreatic enzymes -
- S. Amylase - ↑ immediately, remain ↑ upto 3 days

  Non specific

  > 3 x the normal - favours Acute Panc
- S. Lipase - ....., start ↑ after 3 days
- X-Ray → 1. Colon cut off sign Q

  2. Sentinel loop sign

  3. Renal halo sign

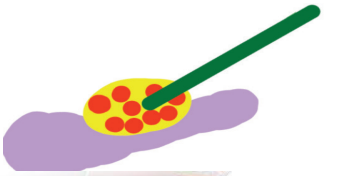
- If in doubt → CT Scan (Radiologic
- Rx - ERCP, sphincterotomy, stone extraction c̄ Dormia basket
- Sx - Not required

Active space

- Indications of Sx :

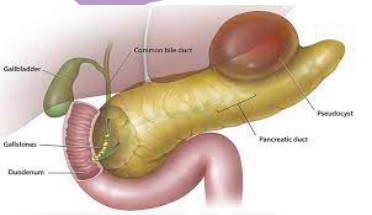
  1. Panc. abscess -

     Percutaneous drainage ↓ US/CT guidance



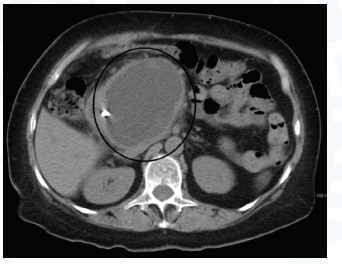
  2. Pseudocyst of Panc -

- collection of Panc. enzymes & fluid around Pancreas



- Pseudo = Not Lined by true cpithelium
- Usually Small, asymp, resolve Spontaneously
- sometimes large - can extend from mediastinum above to Pelvis below
- Epigastric Pain, Malabsorption
- Complicⁿ - Abscess, Rupture.

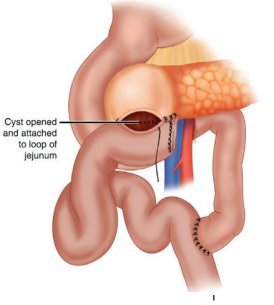
  Malignancy never in Pseudocyst → (No Epithelium)



- IOC - .....
- Rx - If Symp, Complicⁿ, >12 weeks old

  ↓

  ..... → Cysto Jejunostomy / Cysto Gastrostomy



q Sofa Score : QQ

1. RR ≥ 22/min
2. SBP ≤ 100 mm Hg
3. Altered mental status (GCS < 15) ≥ 2 = Sepsis (+)

Sofa Score :

Sequential Organ Failure Assessment

6 Systems -

CNS - 0 - 4

CVS - 0 - 4

Respi - 0 - 4

Liver - 0 - 4

Kidney - 0 - 4

hematology - 0 - 4

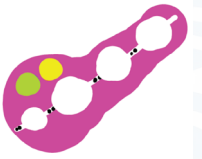
- Other Prognostic Scores in Acute Panc. - APACHE II, Ranson's
- Mortality Rates -

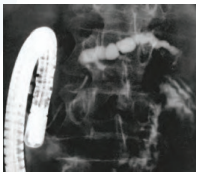
  - Mild Acute Panc. - < 1%

  - Severe/Necrotising Panc. - 20 - 50%

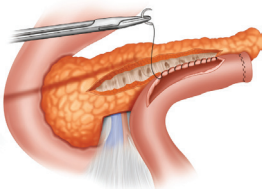
RIGHT HYPOCHONDRIUM- CHRONIC PANCREATITIS

- MCC - Alcohol
- 2nd MCC - Gall Stones
- Irreversible, Panc. Cellular atrophy Calcification, stricture & ectasia of Panc. duct
- 4 x in Males ; > 40 yrs
- Pain - Epigastrium, ↑ after meals
- Malabsn
- DM
- MRCP/ERCP - Chain of Lakes appearance



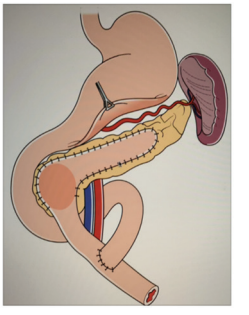


Rx - • Malabsn - Pancreatic enzyme supplements (Pancreatoflat, Creon)

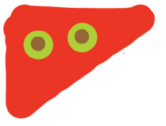


- DM - Insulin
- Pain - xxx Alcohol/Analgesics before meals/ Small frequent meals/Octreotide

Sx → Side to side Pancreatico Jejunostomy (Longitudinal Pancreatico Jejunostomy or Modified Puestow's operation)



RIGHT HYPOCHONDRIUM- LIVER ABSCESS

Pyogenic	Amoebic	Hydatid cyst
Bacteria - Streptococcus milleri E. coli	Entamoeba histolytica	Echinococcus granulosus
Single/Multiple	Usu single	Usu single
• High Grade Fever Chills, Rigors, Jaundice, ↑ Risk of Septicemia	Fever Jaundice Inverted Flask shape ulcers in Sigmoid colon, Caecum	Fever Jaundice
US abd, LFT Best - CECT	US abd CECT	US abd QQ CECT Water Lily Sign
Blood culture Pus culture	Stool Test for isolation of cyst of Entamoeba	ELISA - Ab to Hydatid Ag
Rx : Cephalosporins (Penicillin) + Aminoglycosides + Metronidazole	Metronidazole 400-800 mg TDS x 14 days	Albendozole / Mebendazole
US guided percut. catheter drainage (Pigtail Catheter)	US guided percut Cath. drainage Anchovy Sauce Pus (Chocolate) 	PAIR - Percutaneous Aspiration + Instillation of hypertonic saline + Reaspiration (To avoid risk of dissemination)

Gharbi Classification

I - Fluid

II - Water Lily

III - Honeycomb

IV - Heterogenous

V - Calcified

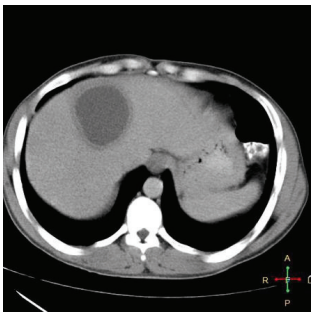


FIGURE- 2.24 Amoebic Liver Abscess

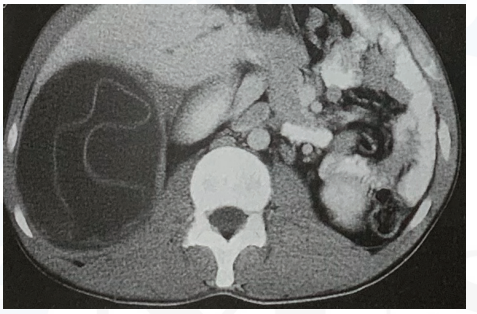


FIGURE- 2.18 Hydatid cyst of Liver

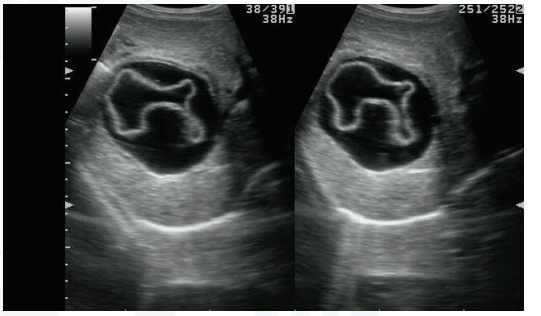
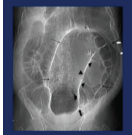


FIGURE- 2.19 Hydatid cyst of Liver- Water Lily sign

RIGHT HYPOCHONDRIUM- SIGMOID VOLVULUS

- Rotation of Sigmoid colon over the Long axis of mesentery
- MC - Anticlockwise
- X- Ray - Coffee bean sign **Q Jan 25**
- Rx - Bird of Prey sign
  - a) - Leave a Flatus tube in Sigmoid colon (24 hrs)
  - b) - If strangul^{n} - Resection & anastomosis

80-year-old lady presented in the emergency with diffuse abdominal pain and distension of abdomen. She is having constipation. On imaging following was found. What is the likely diagnosis?



a. Ulcerative colitis
b. Sigmoid volvulus
c. Tubercular enteritis
d. Intussusception

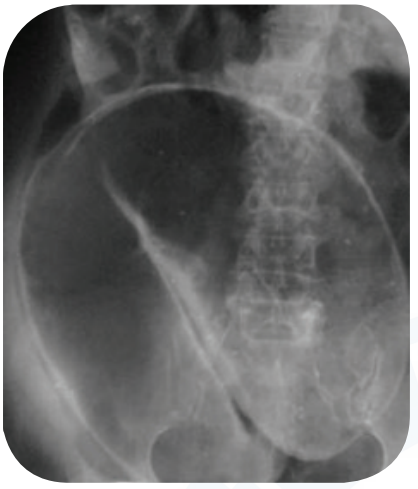


FIGURE- 2.20 Coffee bean sign in Sigmoid volvulus

FIGURE- 2.21 Bird of Prey sign in Sigmoid volvulus

FLANK PAIN- URINARY STONES

- MC Kidney stone - Calcium oxalate (dihydrate)
- MC UB stone - Boys- (Children) - (Adults as well as children)

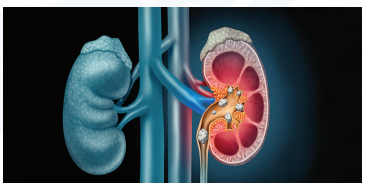
Ca oxalate and Ammonium urate — UB Stones (10 times more common in boys)

Adults - Uric acid

RF :

- MCC - Idiopathic hypercalciuria
- Idiopathic hyperoxaluria
- " hyperuricosuria
- Prolonged Immobilisation
- UTI (Cranberry Juice prevents UTI)
- Def of Vit A
- Hyperparathyroidism

Milk Alkali Syndrome



- Types :

1. Ca oxalate - Monohydrate - 2nd hardest stone
   Dihydrate - MC Envelope Crystals

Active space

- Spiculated - hematuria

Alkaline urine - Ca Po4
Acidic urine - Uric acid

2. Triple PO_4 - Ca, NH4, Mg PO_4 - Struvite

- Alkaline urine - UTI - Proteus

- Silent → can take shape of Pelvicalyceal System →

.....

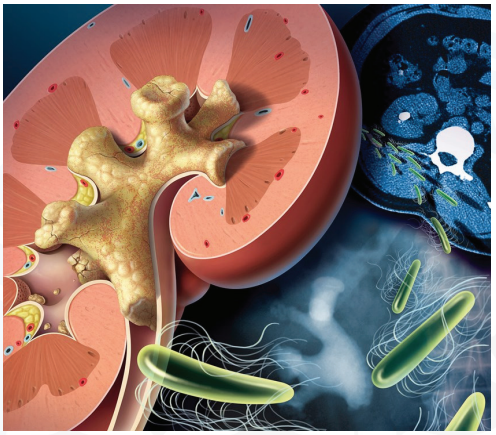


FIGURE- 2.22 Staghorn Stone

FIGURE- 2.23 Staghorn Stone

3. Cystine stone — Benzene / Hexagonal Crystals

- Hardest stones
- Cystinuria
- difficult to be broken by shock waves

4. Uric acid

uricosuria
acidic urine
Radiolucent PG LUX (Not visible on X-Ray)

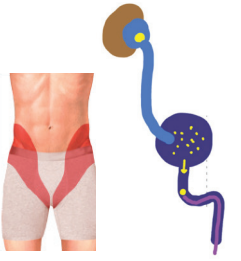
PG LUX Parotid duct Gall bladder Lysine Uric acid xanthine

5. Xanthine, Lysine, Indinavir, Triamterine, Allopurinol

- C/F :
  - MC - .....
  - Earliest - Frequency of micturition
  - Hematuria
  - Interruption of urinary stream →
    (due to large vesical calculus)

Active space

- Pain
- **Kidney St** - Fixed Lumbar pain
  - Upp 1/3 Uret. st - L.P. radiate to < Perineum in ♀ / Testicles in ♂



- Pelvic brim uret st. - L.P. radiate to Inner thigh
- Middle 1/3 uret st. - L.P radiate to Iliac fossa
- Intramural ureter st - Strangury - Painful desire to pass urine
- Bladder st - Suprapubic pain
- Bladder Neck, Urethra - Suprapubic pain referred to the tip of penis

FIGURE- 2.24 Plain X-Ray-KUB- Left kidney stone

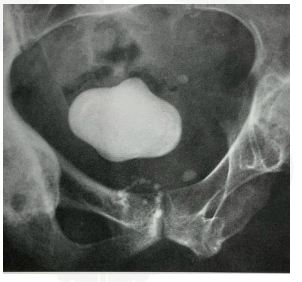


FIGURE- 2.25 X-Ray KUB- Bladder stone (Vesical calculus)

- Cl. app :

  1st - Pain relief - Diclofenac, Tamsulosin

  ↓

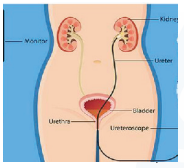
  Plain X - Ray - KUB / US - KUB

  IOC - ..... NEET PG 25

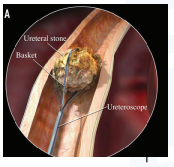
Active space

FIGURE- 2.26 Non contrast CT Scan- Left kidney stone

- Rx : ≤ 5mm stone-Plenty of fluids-90% pass (6-10mm-50% pass)
  ≥ 5 mm stone - Rx OC - ESWL



- Rx OC for Kidney, Upp, Mid Uret Stone - ESWL
- Rx OC - for Low Uret st - URS Ureteroscopic retrieval of stone using Dormia basket



- Rx OC for Bladder st - Litholapaxy - Cystoscope → break the bladder stone → pass spontaneously

- For Urethra st - Push the stone into bladder, Rx like bladder stone

- ESWL - Extra corporeal Shock Wave Lithotripsy
  Shock waves fired onto stone → broken into small fragments c align in ureter → Steinstrasse phenomenon (Street of stones)

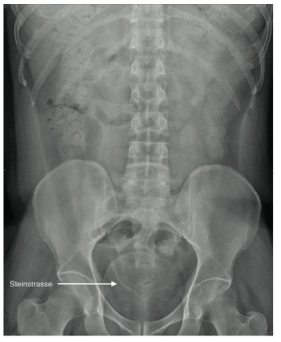
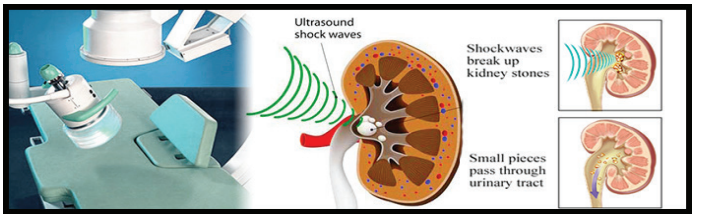


Figure 82.10 Steinstrasse formation after extracorporeal shock wave lithotripsy at the right distal ureter (courtesy of Department of Urology, Christian Medical College, Vellore, India).

→ Pass in urine

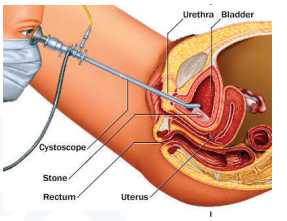


MC Complicⁿ - Bleeding

Avoided in > 2cm stone, Pregnancy, Bleeding diathesis, Distal obstruction, Cystine Stones

Retrograde Intravenal Surgery - RIRS like URS but for Kidney stone Laser

For :
- Renal stones ≤ 2cm
- Lower pole of kidney stone
- Obesity
- Musculoskeletal disorders
- Bleeding disorders



12th Rib approach Postrly

PCNL → Failed ESWL or RIRS

- For > 2 cm Kidney, Upp Uret st/ → PCNL

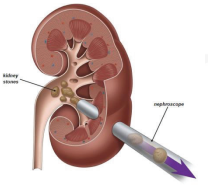
Percutaneous Nephrolithotomy

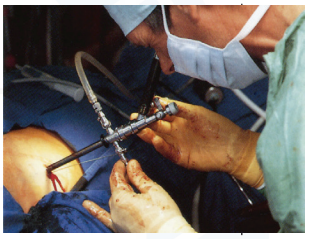
MC Complication - Bleeding

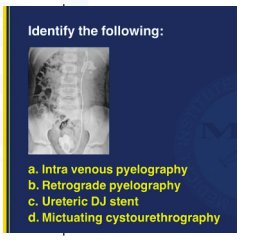
Risk of Injury to Pleura, Colon, Spleen; but not to Lungs

↓

(Hydrothorax/Hemothorax)







- Emergency Urinary Obsⁿ Rx:

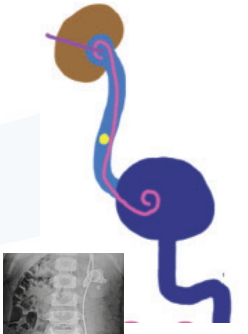
(1) Upper Urinary Tract

a) Endourinary Stenting (DJ Stent)

or

b) PCN - Percutaneous Nephrostomy

If DJ Stent is left inside for > 6 weeks then Cystoscopic removal of this foreign body (DJ Stent) should be done.



QQQQ

(2) Lower Uri Tract

a) Foleys Urethral Catheterization

or

b) SPC = Suprapubic Cystostomy (Supracath)

Q Jan 25

FLANKS- URINARY TUBERCULOSIS

- Hematogenous infection
- Renal TB - Ulcer at Renal Papilla

↓

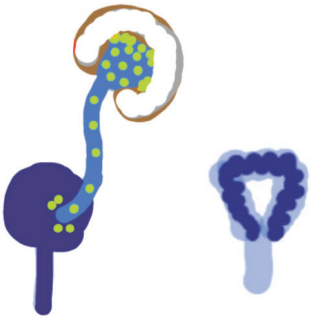
Putty Kidney

↓

Cement Kidney

↓

Auto Nephrectomy



- Bladder TB - Around ureteric orifice

↓

(Narrow, contracted UB) → Thimble bladder

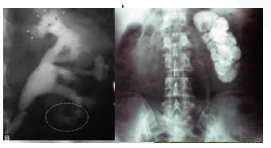
C/f - Frequency, hematuria, Anorexia, Wt loss, low grade evening rise of temp.

- 1st Invⁿ Urine Examⁿ - Mid stream early morning urine sample

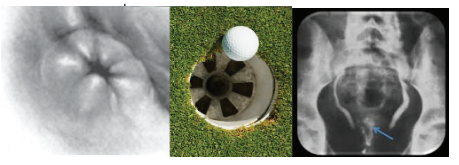
Sterile, acidic, pyuria

- IOC - CECT

Renal TB
- Early-Mothe eaten Calyces (Indistinct Calyces)
- Late - Pseudo Calculi appearance

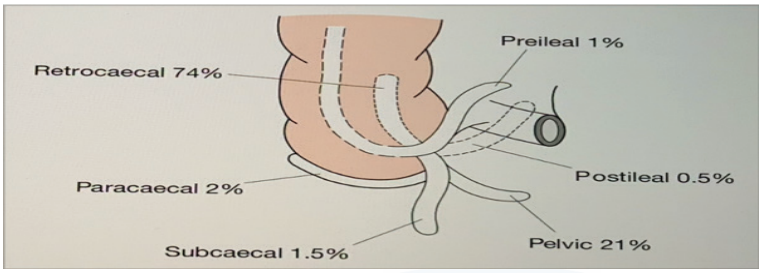


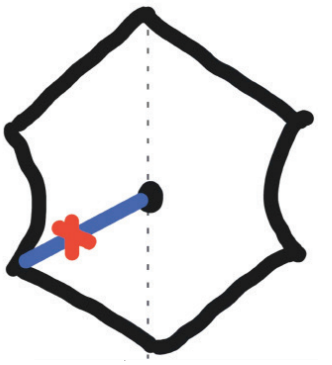
- Bladder TB
  - Early - Golf hole ureteric orifice
  - Late - Thimble bladder



- Rx - ATT

ILIAC FOSSA- ACUTE APPENDICITIS





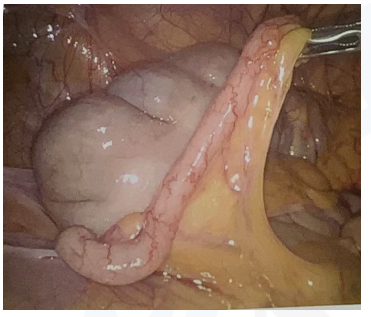
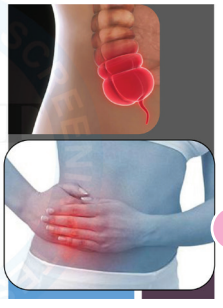


FIGURE- 2.27 Laparoscopic view- Normal Appendix

FIGURE- 2.28 Laparoscopic view- Inflammed Appendix

- MCC - Faecolith obstruction
  - Bacterial - Polymicrobial - Bacteroides, E.coli
  - Parasitic - Pinworms
  - Obsⁿ - Polyp, Carcinoid Tumours
- C/F : ♂ > ♀ ; Late Teens, Early 20's



QQ MANTRELS SCORE (Alvarado Score) QQ

1	M	-	Migratory RIF pain (Shifting RIF pain)
1	A	-	Anorexia
1	N	-	Nausea, Vomiting
2	T	-	Tenderness
1	R	-	Rebound Tenderness
1	E	-	Elevated Temperature
2	L	-	Leucocytosis
1	S	-	Shift for Neutrophils

≤ 5 - Acute appendicitis is ruled out

≥ 7 - high probability of Acute appendicitis

Active space

- Cl. app :

  Pointing sign - Pain path demonstrated by patient

  Pulse - 80 - 90/ min

  Temp - < 100.4° F

  RIF
  - Tenderness RIF - McBurney's Sign
  - Rebound Tenderness (d/t elastic recoil of inflammed Parietal peritoneum) - Shyotkin Bloomberg sign
  - Localised Guarding, Rigidity

- MC Burney's sign - Pain in RIF ; on pressing RIF
- Rovsing's - Pain in RIF; " " LIF
- Obturator's - Pain in RIF; On Internal rotation of flexed (R) hip
- Psoas's - Pain in RIF; On hyper extension of (R) hip
- Dunphy's - Pain in RIF; On Coughing
- No Invⁿ required
- Basic Invⁿ - TLC - 11000 - 14000/cu.mm (4 - 11)- DLC - Neutrophils (Polymorphs)
- Best Invⁿ - CECT abd
- US abd to rule out other Differential diagnosis - If blind ending, tubular, aperistaltic, structure in RIF Diameter > 6mm → Favours Acute App.

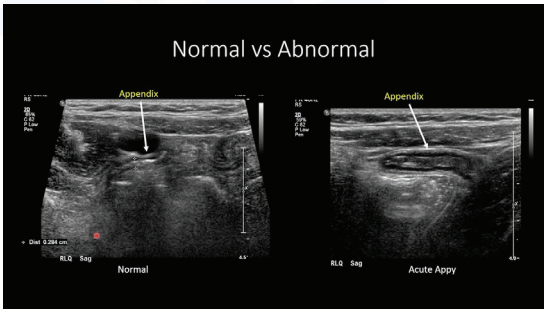


FIGURE- 2.29 Ultrasound showing blind ending tubular appendix with a transverse diameter more than 6mm

Active space

- **Rx : Mild - Conservative c̄ Ab** MIST

  Severe - Operative - Emergency Appendicectomy

  Pregnancy- Appendicectomy (Irrespective of the Trimester)

- Incisions for Appendicectomy :

  1. Ms Splitting Incision :

     a) MC - McBurney's Incision

Q July 24 QQ - (.....Incision)

     b) Lanz .....

     Transverse Incision

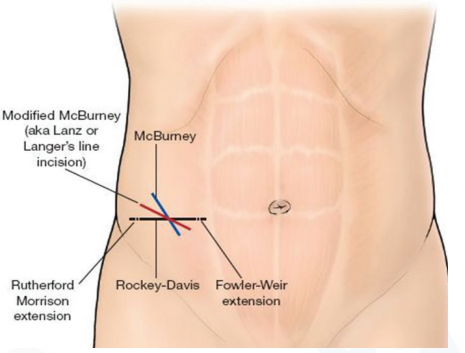


FIGURE- 2.30 Incisions for Appendicectomy- Grid Iron or McBurney's is the most commonly used

  2. Ms Cutting Incⁿ :

     a) Rocky Davis

     b) Fowler Weir (extended medially)

     c) Rutherford Morrison (extended Laterally & upwards)

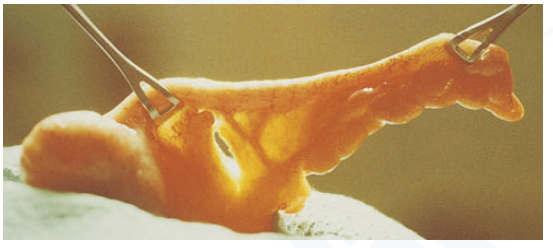
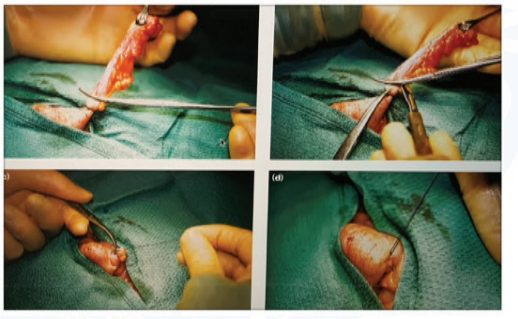


FIGURE- 2.31 Appendix- held with non traumatic Babcock's forceps during Appendicectomy



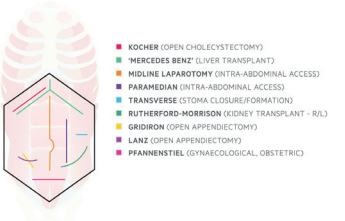
  3. Laparoscopic Appendicectomy - 3 Ports

- MC Complicn of Appendicectomy

  Early - Wound infection on 4th/5th

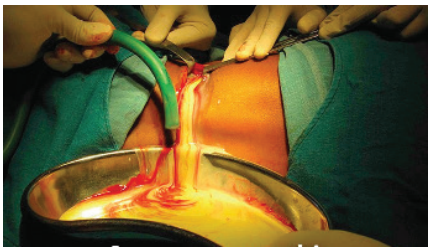
  Postoperative day

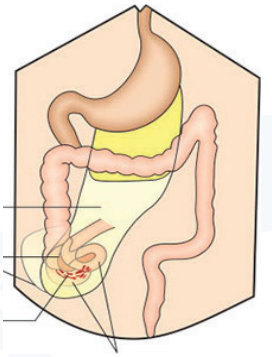
  Late - Adhesive Intestinal obstruction



Active space

- Appendicular abscess -
  Extraperitoneal drainage QQ
- Appendicular Mass





Conservative Rx (Nonoperative) Ochsner Sherren Regime

IVF, IV Ab For 5 - 7 days

↓

Discharge Advice

Interval Appendicectomy after 3 - 6 weeks

Active space

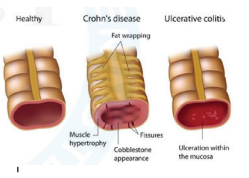
ILIAC FOSSA- INFLAMMATORY BOWEL DISEASES

Crohn's disease	Ulcerative Colitis
Regional Enteritis	
Exact cause ×	×
Autoimmune T. helper I	Autoimmune T-helper II
Western white Rich	Western white Rich
	smoking is protective
20 - 40 y	20 - 40 y
2nd Peak > 60 y	2nd Peak > 60 y
● Any Part of GIT	MC - Start in Rectum
MC - terminal ileum	Spread proximally in continuity
Caecum	Proctitis → Proctosigmoiditis
2nd MC - Anal region	Proctosigmoidocolitis
Skip areas	Pancolitis (Universal,
(Not continuous)	Colitis), Backwash Ileitis
● Transmural (All layers)	
Non caseating granulomas	Involves mucosa, Submucosa
Serpentine ulcers	Non specific inflammatn
Cobble Stone mucosa	Collar button ulcers
Stricture, Sinus, Fissure	Pseudopolyps
Fistula - More common	
● C/F : Pain, Mass - in RIF	Diarrhoea - Mucus/blood
Diarrhoea	Mild - ..... motions/day
Constipation	Mod - ..... " "
	Severe - ..... " "
	Fulminant - ..... " "

- Extraintestinal Manifestations in both Crohn's & UC
  - Joints - Polyarthropathy, Ankylosing spondylitis, Sacroilitis
  - **Eyes - Uveitis, Scleritis**

Q July 25

Active space

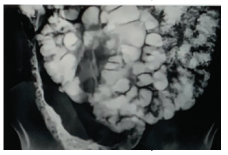


- Skin - Pyoderma Gangrenosum, Erythema nodosum
- Bile duct - Sclerosing cholangitis

  PSC (Primary sclerosing cholangitis) - More common in UC

  - Cholangiocarcinoma

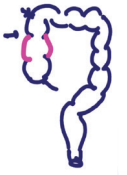
- CECT abd - IOC — Colonoscopy - IOC



Ba meal follow through- — Ba enema-

String sign of kantor — Lead Pipe appearance

(String Sign - TB of Ileum) — (Loss of haustral folds)



- **Rx :**
  - **Main stay - Steroids**
  - 5 - Aminosalicylates in Inflammation
  - Levamisole
  - Infliximab, Rituximab
- Sx - Rarely done in Crohn's (Sx is not curative in Crohn's)

  - Can be " " UC
- Indications of Sx :
  - Failure of Medical Rx
  - Steroid dependent disease
  - Extraintestinal manifestations
  - Risk of Cancer (More in UC)
  - Complications - Toxic Megacolon

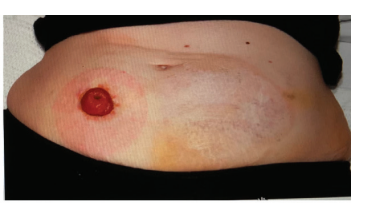
    Haemorrhage

    Perforation

Active space

- **Sx in UC**
- If entire Rectum & Colon is involved → Total Proctocolectomy c̄

  a) Permanent Ileostomy



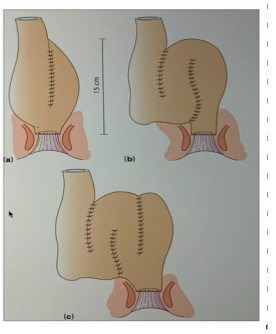
or

b) Ileal Pouch anal anastomosis

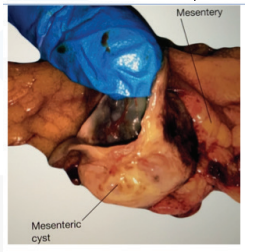
MC Pouch - 'J' Pouch

'S', 'W', 'M'

- Complicⁿ of Pouch Sx - Pouchitis



MESENTERIC CYSTS



- Around 45 years of age
- 2 time more common in females
- 4 Types - Chylolymphatic (Mc)
  - Enterogenous
  - Traumatic
  - Hydatid
- Tillaux Sign → Cyst freely move in horizontal direction, and not in vertical direction
- MC C/F - Painless, Fluctuant abdominal swelling near the umbilicus
- Rx - Chylolymphatic - Enucleation **Q July 25**
- Enterogenous - Removal c̄ resection of segment of intestine

COELIAC DISEASE (GLUTEN ENTEROPATHY)

Immune response to gluten protein in diet (Wheat / Barley / Rye)-

Diarrhoea, Constipation, Wt loss

Most sensitive and specific antibodies → Tissue Transglutaminase A (↑ IgA)

Other Ab → Endomysial IgA, Reticulin IgA

Rx - Avoid Gluten food

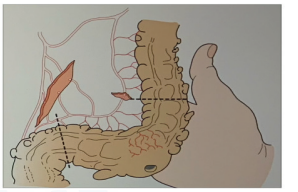
Active space

WHIPPLE'S DISEASE ("INTESTINAL LIPODYSTROPHY")

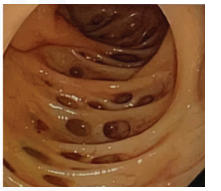
- by bacteria Tropheryma whipplei
- Diarrhoea, wt loss, fever
- Rx - Ab (Ceftriaxone)

ILIAC FOSSA- DIVERTICULOSIS OF COLON

- MC - Sigmoid colon, Mesenteric border more common
- Elderly Acquired - False (do not contain all layers)
- **Constipation**, Pain in LIF, Bleeding PR
- Complicn - Diverticulitis
  Div Perforation



QQ
- Ba enema - .....appearance



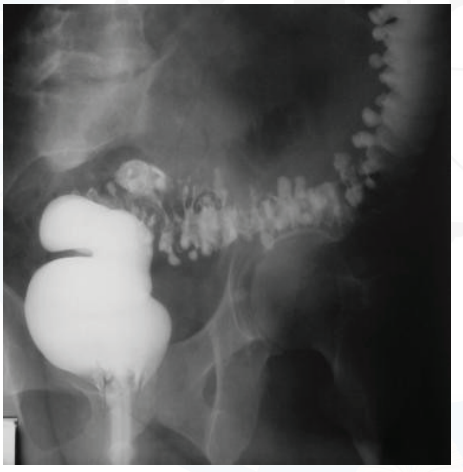


FIGURE- 2.32 Saw tooth appearance on Barium enema in Diverticulosis of colon

- ***IOC*** - .....
- Rx : Initially high fiber diet
  Later - Resection & anastomosis

Active space

PSEUDO OBSTRUCTION (OGILVIE'S SYNDROME)

Acute massive dilatation of colon without mechanical obstruction

Risk Factors :

Trauma / Surgery / Infection / MI / Neurological diseases

Metabolic - Hypokalemia, Electrolyte disturbances

Drugs - Opioids, CCB'S

HYPOGASTRIUM- BLADDER AND URETHRAL TRAUMA

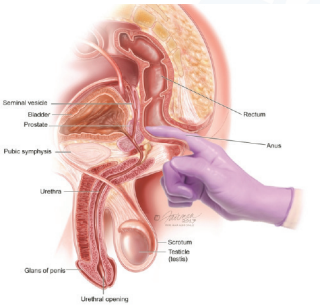
Bladder

Intraperitoneal Bl. Rupture - 20%

- Trauma to Overdistended bladder Beer drinking
- Sudden hypogastric pain, syncope
- Bed side Test - Peritoneal Tap
- IOC - Cystogram
- **Rx** - Laparotomy, Peritoneal lavage, repair bladder, Leave a foley's Catheter for 7-10 days

Extraperitoneal Bl. Rupture - 80%

- Road Traffic Accidents - Pelvic fracture
- Tear drop bladder
- **Rx** - Bladder repair



Urethra

Rupture of Postr Urethra - Membranous urethra

- RTA - Pelvic Fracture

  MC Site - At the apex of Prostate

  DRE - High Lying Prostate

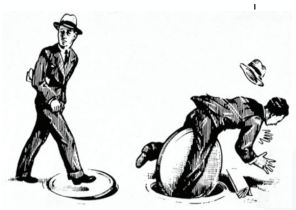
Active space

- Retention of urine
- Pelvic hematoma Urine Collected in true Pelvis
- Blood at Ext. urethral meatus
- RGU (Retrograde Urethrogram)
- **Rx** : Suprapubic Cystostomy (SPC) (Supracath)

Note : Trial of Plain Red Rubber urethral catheterization can be made before SPC

A gentleman give history of falling astride injury, following which he was having difficulty in urination. Blood was present at the external urethral meatus with bruising in the perineum. Which of the following is the most likely injury in this patient?

a. Membranous urethral injury
b. Bulbar urethral injury
c. Penile urethral injury
d. Bladder neck injury

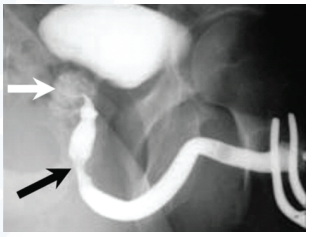


Rupture of Antr Urethra - Perineal (Bulbar)

- **MC** Urethral Rupture
- Injury to Perineum - Loose manhole cover/Bicycle accident/Fall from height **Q Jan 25**
- Retention of urine
- Perineal hematoma (urine gets collected in Perineum/scrotum/ Penis/Lower abd wall)
- Blood at Ext urethral meatus
- RGU
- **Rx** : SPC

IBQ

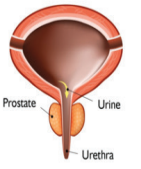
QQ

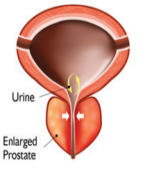


Note : Trial of Plain Red Rubber urethral catheterization can be made before SPC

HYPOGASTRIUM- BENIGN PROSTATIC HYPERPLASIA

- Testosterone $\xrightarrow{5\ \alpha\ \text{Reductase}}$ 1, 5 Dihydrotestosterone
- Narrowing & Lengthening of Prostatic urethra wt of Prostate = 20 gm
- BPH - Transitional zone of Prostate Median, Lateral lobes
- Ca Prostate - Peripheral zone of Prostate Postr lobe
- C/F - 'LUTS' - Low Uri Tract Symp
  - Storage - Frequency (MC), Nocturia, Urgency, Urge incontinence





Active space

- Voiding - Hesitancy, ... (Most Troublesome)
  Intermittency, Dribbling

- Cl. app : - DRE
  - IOC → US abd
  - Vol of Prostate
  - PVR - Post void residue > 100 ml - significant
  - S. PSA -Prostate specific Ag/serine protease/causes Liquefaction of semen / secreted by Prostatic epithelium only
    0-4 ng/ml - Normal
    >10 ng/ml - Suspicious of Ca
    ..... ng/ml - Advanced Prostate Ca

Rx ● Mild Symp (No Complicⁿ)→ Medical Rx

→ Medical Rx oc = Selective Alpha I A blockers - Tamsulosin, Alfuzosin Improve urine flow (Relax Prostatic urethra)

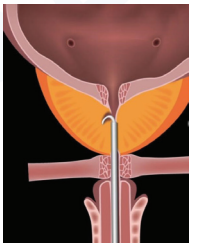
→ 5 α Reductase Inhibitors - Finaesteride, Dutasteride
Reduce the size of Prostate

- Severe Symp, Complicⁿ (Haemorrhage, Retention, B/L hydronephrosis, Recurrent UTI, UB stones, UB diverticula) → Sx
  Sx Rx OC - TURP - Transurethral Resection of Prostate
  Fast Improvement

Other Techniques -

→ Green Light Laser Enucleation of Prostate

→ Holmium Laser Evaporation of Prostate



Complications of Sx

1. CCF - d/t volume overload
2. TUR Syndrome - Water intoxication syndrome
   if water is used as irrigation fluid
   - Dilutional hyponatremia can cause altered sensorium
   Irrigation fluid in TURP-Glycine 1.5%

- Green Light Laser - Bipolar diathermy can be used
  - Irrigation fluid of choice - Normal saline

Active space

3. Haemorrhage
4. Retrograde Ejaculation >75%, -d/t damage to Internal sptincter
5. Urinary incontinence - d/t damage to both Internal + External Sphincter
6. Stricture urethra
7. Sepsis

GROIN/SCROTUM- HERNIA

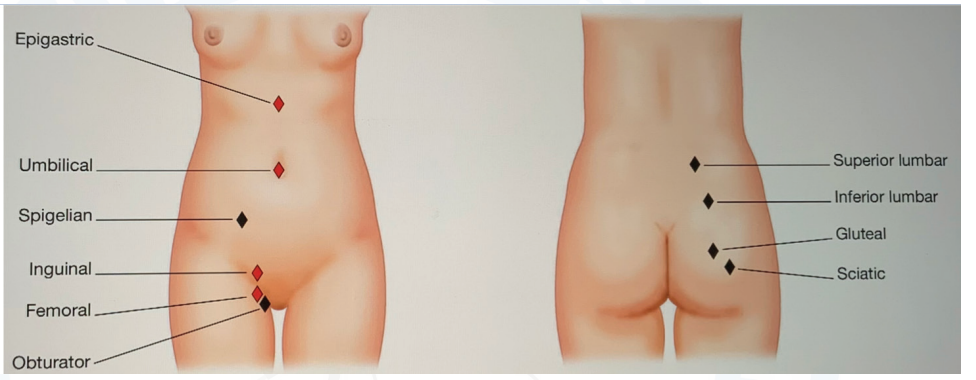


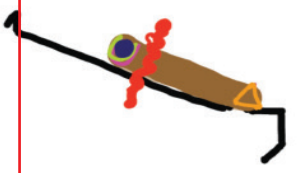
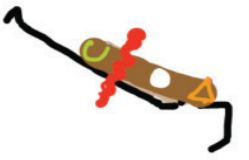
FIGURE- 2.33 Sites of Common Hernias

Risk Factors :

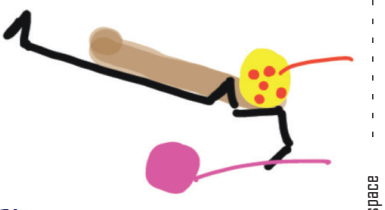
1. Anatomical weakness - Ageing / Nerve, muscle weakness
2. ↑ Intra abdominal pressure
3. Genetic weakness of collagen
4. Trauma
5. Hernia is more common in people with short and wide pelvis

- MC - Indirect Inguinal hernia
- MC h in Young Males - Indirect Inguinal hernia
  " " Elderly " - Direct Inguinal hernia
- Femoral h - 4 times more common in Females
- MC h to undergo Strangulation - Femoral hernia

Active space

	Indirect Ing	Direct Ing
		
Incidence	Common	Less Common
Age	Young	Elderly
Protrude through	Deep Ing Ring	Transversalis Fascia (Hessel bach's Δ)
Relation c̄ Infr Epigastric Vessel	Lateral	Medial
Direction	Downwards Forwards Medially	Forwards
Extension into scrotum	Common	Rare

- Landmark to DD Indirect from Direct Ing h -Infr Epigastric vessels
- " " DD Inguinal from Femoral h - Pubic Tubercle
  Ing h - Above & medial to PT
- Fem h - Below & lateral to PT



European hernia society classification - EHS class .

QQ
1. P or R → Primary, Recurrent
2. L M or F → Lateral (Indirect Inguinal), Medial (Direct Inguinal), Femoral
3. 1/2/3 → 1 Finger Defect / 2 Finger Defect / 3 Finger Defect
   1 Finger diameter = 1.5 cm

Active space

- Nyhus classi :
  - I - Indirect (Small)
  - II - Indirect (Large)
  - III - a - Direct
    - b - Both (Indirect + Direct)
    - c - Femoral
  - IV - Recurrent
- Clinical Classi -
  1. Occult - Not visible
  2. Reducible - visible, ⊕ cough impulse (MC)
  3. Irreducible - visible, cannot be reduced back
  4. Incarcerated - Irreducible ± obstruction
  5. Obstructed - d/t Narrow neck of sac
  6. Strangulated - Obstructed + Ischemia
  7. Gangrenous - No blood supply

  6, 7 - Emergency
- Content classi :
  - Epiploecele - Omentocele - Omentum (Doughy feel)
  - Enterocele - Intestine
  - Littre's - Meckel's div
  - Amyand's - Appendix
  - Richter's - Circumference of intestine

MIST

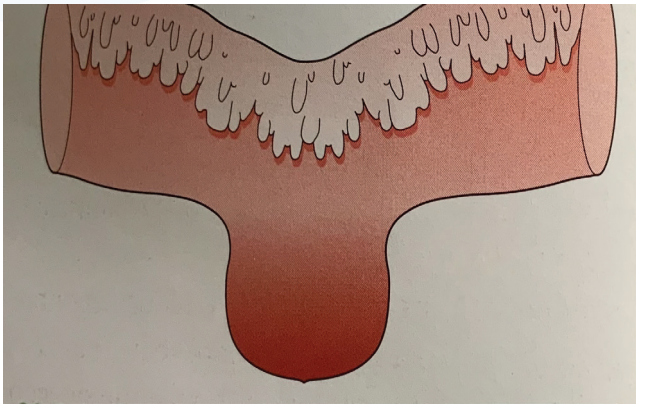
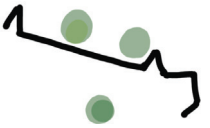


FIGURE- 2.34 Richter's hernia

Figure 64.5 A gangrenous Richter's hernia from a case of strangulated femoral hernia.

Active space

- Maydl's - W loop of bowel
- Sliding - Organ form one wall of h.sac
  MC - Sigmoid colon
  2nd MC - caecum
- Cl. app :
  MC - Inguinoscrotal Swelling
  Cl. examn - Supine
  - Standing
  - 3 Finger Test - Zieman's Test



Key Points in Rx :

- In Elderly asymptomatic - watchful waiting
- Small Umbilical and Epigastric hernia - Conservative Rx
- Narrow neck hernia, femoral hernia, Irreducible[QQ], Obstructed / Strangulated Incarcerated hernias - Always operative repair
- Rx : Aim- Reduce contents, apply a barrier
  Open Sx- Hernioplasty - Tension free Repair (Litchenstein Opn)
  Mesh - Polypropylene
  - Polygalactin
  - Thin thread, Large pore mesh preferred
  (Earlier - Bassini's / Shouldice / Stoppa's operation)

FIGURE- 2.34 Polypropylene Mesh for Hernia Surgery in Adults

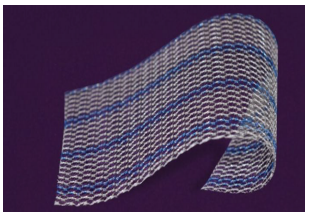
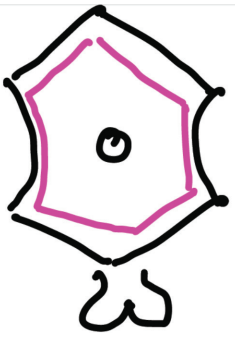


FIGURE- 2.35 Polypropylene Soft Mesh for Hernia Surgery in Adults

QQ

Active space

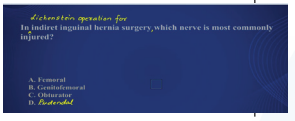


- Lap. Sx - TEP - Totally Extraperitoneal Repair
  TAPP - Transabd Preperitoneal Repair
- Nerves c̄ can be injured in open h Sx - Ilioinguinal nerve. (MC)
  Iliohypogastric / Genital br. of Genitofemoral nerve QQ
- Rx OC for Cong hernia - ..... Q July24
  " " hydrocele - ..... / Age of opⁿ >3yrs of age

Q July 25

Strangulated Inguinal hernia

Femoral Hernia



- - ♀ Elderly, Low wt, Sx - Lothiessen, Lockwood, McEvedy

Umbilical Hernia

- - Blacks (8x) / ♂ = ♀
  No Rx upto 2 yrs
  Sx - > yrs, if large defect

FIGURE- 2.36 Umbilical hernia in a baby

Paraumb. H

Q July 25

- Adults - Obese, Cirrhosis Protusion around umbilicus
- Umb - 'C' Crescent type

FIGURE- 2.37 Umbilical hernia in a baby

- Rx - Sx QQ

Q July 25 + Excision of Redundant skin

Spigelian Hernia

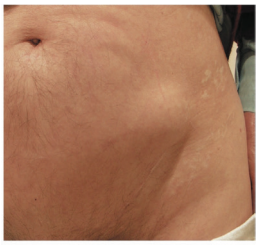


Figure 64.22 Spigelian hernia in the left iliac fossa. Note the scar from a previous left inguinal hernia repair.

- Protrusion through Lateral border of Rectus abdominis muscle, above the arcuate line
- Interparietal hernia/Pain ✓ Swelling ✓
- ↑ Risk of Strangulation
- Rx - Sx

FIGURE- 2.38 Spigelian hernia- at or above the arcuate line

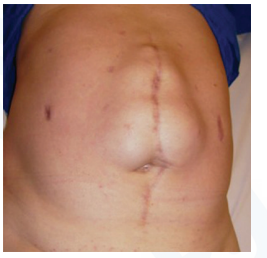
Incisional Hernia Repair

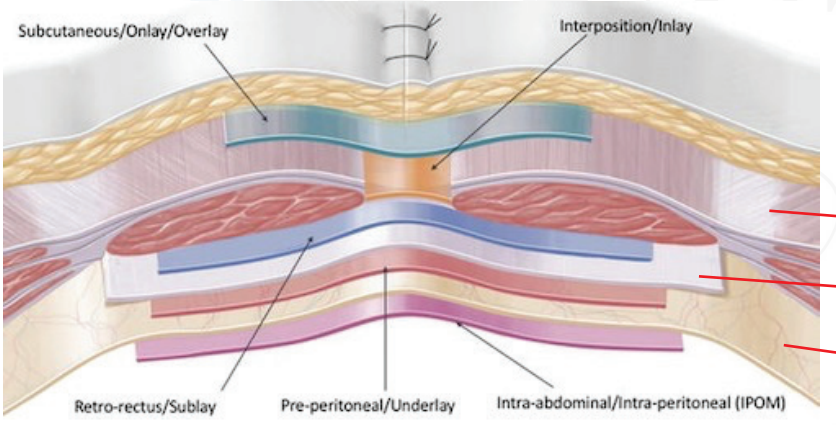
Wound Dehiscence : **Q Jan 25**

- Disruption of any or all layers in the wound.
- MC - 5th to 8th Post operative day

  (Weakest strength of the wound)
- Popping sensation on straining **Jan 25**

Q Jan 24
- Serosanguinous discharge from the wound
- Rx- Ab + Dressings ± Re surgery





Ant Rectus sheath

Posts Rectussheat

P

Underlay - Beneath the fascia & Muscles < Above / Below > the peritoneum

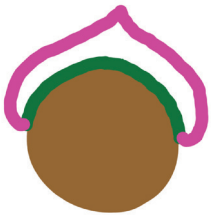
Incisional Hernia Repair

1. Onlay - Mesh put above antr. rectus sheath
2. Sublay - Mesh put above postr rectus sheath
3. Underlay < Extraperitoneal mesh above the peritoneum / Mest put below the peritoneum Intraperitoneal **Q July 24**
4. Inlay - Mesh put to close the defect

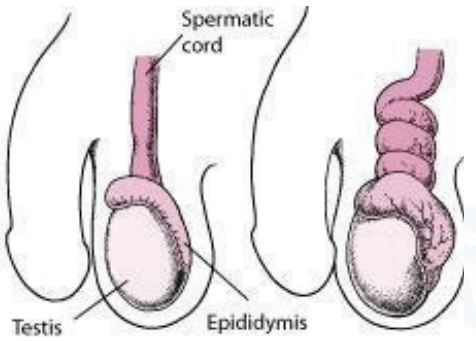
space

SCROTUM- TESTICULAR TORSION, VARICOCELE AND HYDROCELE

Torsion of Testis



Testis hanging like a clapper in a bell



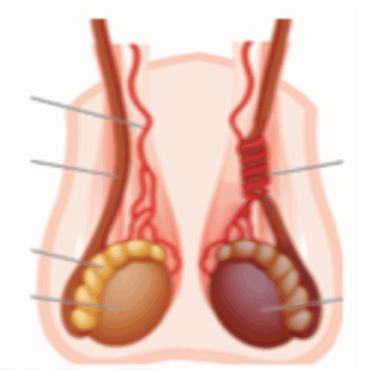


FIGURE- 2.39 Testicular Torsion

Causes : QQ

- MCC-Bell clapper deformity (High Investment of Tunica Vaginalis)
- Simple Inversion of Testis - Sudden ↑ in Intra abd pressure
- Separation of Spermatic cord

C/F :

- MC - 10 - 20 y (10-25y)
- Sudden, Severe Pain, Swelling, Redness in scrotum
- High lying testis, Thick spermatic cord, absent cremasteric reflex

Clinical Test

- Clinical Test to DD from Epididymo orchitis - Prehn's Test (Elevation of scrotum relieve pain in Epididymoorchitis) - Prehn's Test - ⊕ in Epididymo orchitis / ⊖ in Torsion

Q July 25

- Best - Immediate Scrotal Exploration QQ

Rx →

a) **If Testis - Viable (< 6 hrs) -** Derotation, Orchidopexy, Contralateral Orchidopexy

b) **If Testis - Non viable (> 24 hrs)-20% Salvageable** Orchidectomy, Contralateral Orchidopexy

Active space

Management

The management of the case should be determined primarily on clinical grounds. While Doppler ultrasound scanning (*Figure 86.6*) can confirm the absence of the blood supply to the affected testis, false-positive results can be seen so it is not routinely recommended. If there is any doubt as to the diagnosis, then urgent scrotal exploration is indicated. The typical window of opportunity for surgical intervention and testicular salvage is 6 hours from onset of pain. Therefore, early urological surgery consultation upon presentation may be critical even in the absence of confirmatory testing.

Summary box 86.3

Testicular torsion

- If the diagnosis of testicular torsion is possible, then surgical exploration is indicated
- Prompt exploration, untwisting and fixation is the only way to save the torted testis
- The patient should be counselled and consented for orchidectomy before exploration
- The anatomical abnormality is bilateral and the contralateral testis should also be fixed
- Non-absorbable sutures should be used for the fixation of each testis

Varicocele

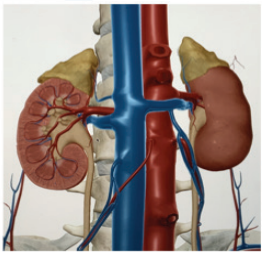
FIGURE- 2.40 Varicocele

- Abnormal dilⁿ of Pampiniform plexus of veins around Testis
- MC - (L) - d/t valvular incompetence in LTV d/t ↑ Pressure

  Tall, thin males c̄ pendulous scrotum
- Vague dragging sensation in scrotum
- Sterility, Bag of worms feel
- **Rx : 1st Line → Percutaneous embolisation of veins**

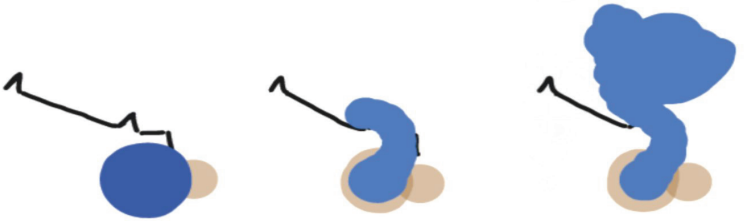
  If fail → Varicocoelectomy

  Alternative Route for venous drainage - Cremasteric vein



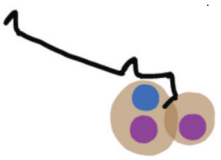
Hydrocele

- Abnormal collection of fluid c̄ in Processus vaginalis QQ Q July24
- MCC - Filariasis Tunica Vaginalis



Active space

- Vaginal — MC (scrotum)
- Infantile (Ing canal) ↓
- Congenital — Communicating (Peritoneal Cavity) QQ



- Encysted hydrocele of Spermatic cord - 3 Testicle appearance
  (Also seen in Spermatocele → Resemble Barley Water
- Hydrocele in ♀ - Hydrocele of Canal of Nuck (Fluid in Ing canal)
- C/F : MC - Scrotal Swelling Transillumination - (+)
  If in doubt - US scrotum (Trans Scrotal FNAC is .....)

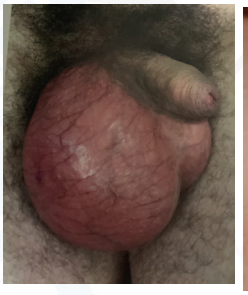


FIGURE- 2.41 Hydrocele

FIGURE- 2.42 Positive transillumination in Hydrocele

FIGURE- 2.43 Hydrocele sac operative picture for Eversion of sac (Jaboulay's operation)

Rx :

- Rx OC - .....(Jaboulay's Opⁿ)
  - Lord's opⁿ - Plication of thin walled sac in small to medium sized hydroceles
  - Cong. Hydrocele - Herniotomy (>3yrs)

Active space

FOURNIER'S GANGRENE

- Progressive infection
- Mixed Bacterial flora - E. coli, Bacteroides, S. pyogenes, Staph aureus
- Rapid progression of cellulitis from genitalia to perineum to abdominal wall with high - Risk of septicemia

Necrotising fascitis of

a) Scrotum is Fournier's gangrene

b) Abdominalwall is Meleney's gangrene

Mortality risk is 30-50%

- Surgical Emergency
- Aggressive Antibiotics and surgical debridement

(Testis Spared)

PENIS- PHIMOSIS, PEYRONIE'S DISEASE, PRIAPISM

Phimosis and Paraphimosis

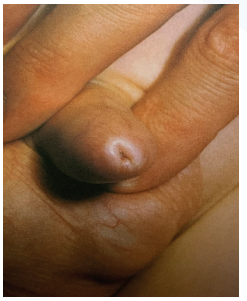


FIGURE- 2.44 Phimosis

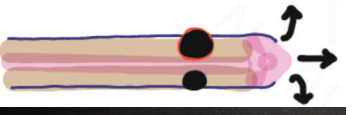
FIGURE- 2.45 Paraphimosis

Active space

Phimosis	Paraphimosis
Inability to retract Prepuce onto glans	Inability to bring back the retracted prepuce onto glans
Physiological - upto 6yrs No Rx	
Risk - Poor hygiene Penile Cancer	Rx -
Rx - Circumicision	• Manual Reduction of Prepuce (Icebags Compressn to ↓ Swelling
(MC - Religious reasons Jews / Muslims)	• Dorsal Slit Opⁿ • Circumcision

Peyronie's Disease

- hard fibrous plaque in one or both Corpora Cavernosa
- asso c̄ Trauma / Dupytreyn's Contracture
- Erect Penis bend towards plaque side
- Spontaneous resolution can occur
- Reassurance
- ESWL
- If persist for 18-24 months → Nesbit's opⁿ (Placing a non absorbable suture in opposite cavernosa)



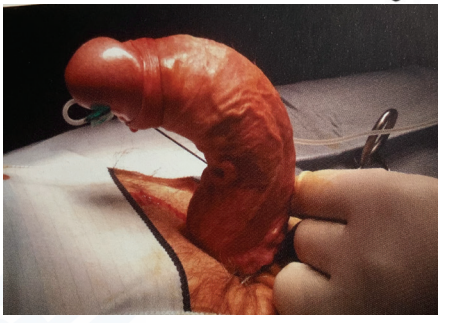


FIGURE- 2.46 Peyronie's disease

Priapism

- Persistent Pathological Erection of Penis > .....hrs
- Low Flow - Ischemic - MC
- MCC - Idiopathic

Sickle Cell anemia
Leukemias — Hypercoagulable states

Overuse of Sildenafil

- Penile blood → ↓ O_2
  ↑ CO_2

FIGURE- 2.47 Priapism

Active space

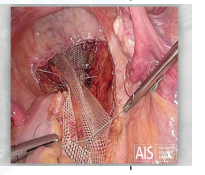
- Rx - Aspiration of blood from cavernosa / Adrenaline injⁿ
  - If required - Divert the blood from cavernosa by Shunt
- High Flow - Trauma - Penile blood - Arterial (↑ O_2 / ↓ CO_2)
- Rx - Angioembolisation

HAEMORRHOIDS (PILES)

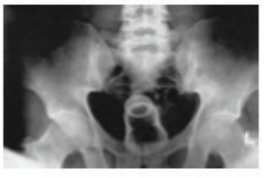
Rectal Prolapse

- ↑ Intraabd pressure / Weakness of Pelvic floor

- Partial - Only Mucosa protrude out
- Complete - All layers of Rectum protrude out
  - > 4cm ; Hernia en glissade
- Rx : Children - Digital Repositioning
  - Adults ◉ Perineal opn - Thiersch wiring
    - Delorme's
    - Altemier's



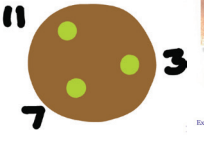
- ◉ Abdominal opⁿ - Resection Rectopexy

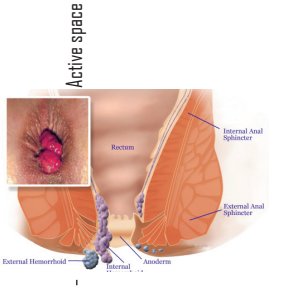


- Well's Opⁿ
- Ripstein's Opⁿ
- Lahaut's Opⁿ

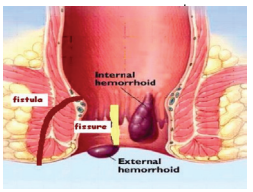
Haemorrhoids (Piles)

- Prolapse of Anal cushion
- Internal - MC - Above dentate line, ...
  - 3, 7, 11o'clock position





- External - Below dentate line, ...
- Thrombosed Ext. hemorrhoid - 5 day, painful, Self Limiting disease
- Grading of Int. hem. -



I° - Mass does not protrude out
II° - Mass protrude out on straining, goes back spont
III° - " " " " " does not go back spont, has to be reposited manually
IV° - Mass permanently outside

- MC C/F - Bleeding PR (splash in the pan)
- DD - Anal Fissure - Longitudinal Streak of blood on the postr surface of stool

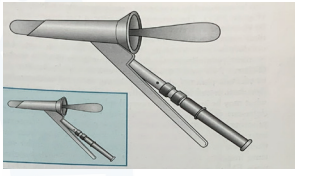
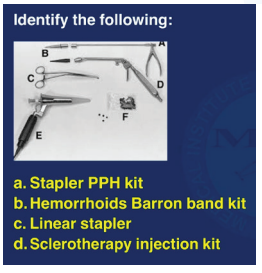


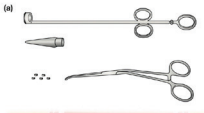
FIGURE- 2.48 Proctoscope

- MCC of Bleeding PR - Haemorrhoids (Piles)
- MCC of BPR - Infant - Intussusception
  Child - Juvenile Polyp
  Adult - Haemorrhoids (Piles)
  Elderly - Colorectal Cancer
  - Diverticulosis of colon

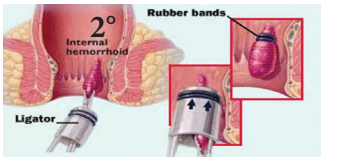
Jan 26



- MCC of BPR - (R) colon - Angiodysplasia of colon (Elderly)

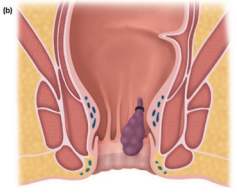


- IOC - Proctoscopy
- Rx : - 1° - Sclerotherapy

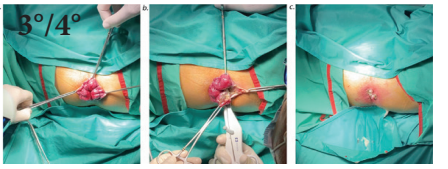


5% Phenol in Almond oil

- 2° - Banding -Mass falls off in 5-10 days **Q Jan 25**



- 3°/4° - Sx - Hemorrhoidectomy



- Complicⁿ of Sx -

  Early - Reactionary hemorrhage, Urinary retention

  Late - Secondary hemorrhage, Anal Stenosis

Other Techniques

- THD - Transanal hemorrhoidal dearterilization
- Stapler Anopexy — Circular Stapler →

  - PPH - Procedure for Prolapsed hemorrhoids

RECTUM AND ANAL CANAL- ANORECTAL ABSCESS AND FISTULA IN ANO

Anorectal Abscess

- MC - Perianal absccess
- MCC - Infection of anal gland
- Rx - Incision & Drainage, Ab

  Cruciate Incision (Diamond Shaped)

FIGURE- 2.49 Perianal abscess

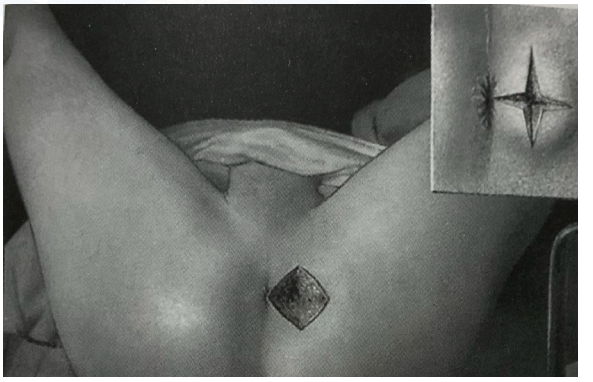
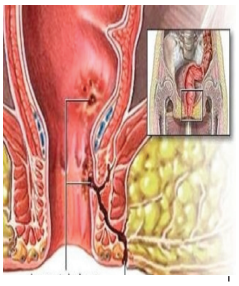


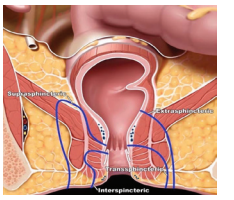
FIGURE- 2.50 Cruciate (Diamond shaped) Incision for Perianal abscess

Active space



Fistula in Ano

- Abnormal communication between — Skin / Anal canal / Rectum



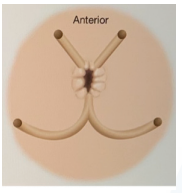
Park's classification for Fistula in Ano

MC - Intersphincteric **Neet PG 25 IBQ**

MCC - Infecⁿ of anal gland

Internal opening may be at crypt at dentate line

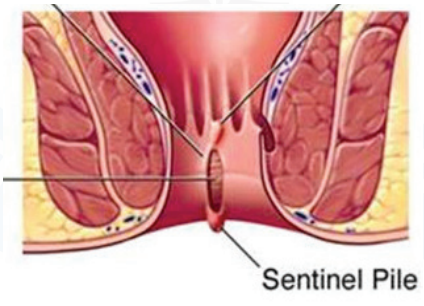
- Good sall's Rule -

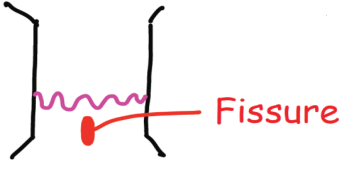


- Antr fistula = Direct type, straight track
- Postr fistula = Indirect, curved track, open in midline postly
- > 2.5 cm from anal verge-6o' clock
- IOC- Gold Standard Invⁿ for fistula in ano = ...
- Rx
  - Fistulotomy
  - Seton placement

MIST

RECTUM AND ANAL CANAL- FISSURE IN ANO





- Laceration in anal canal, just proximal to anal verge, below dentate line, midline, postrly, at 6o'Clock position
- Constipation, Anal stenosis ; Extremely painful
- Longitudinal streak of blood on the posterior surface of stool

- Sentinel skin tag - Chronic anal fissure

Active space

- Rx

  - Rx OC - Conservative - Aim - Make stool soft, relax anal sphincter

- Sitz bath twice a day (Lukewarm water)
- - Avoid Constipation - High fiber diet
- - Stool Softeners
- - Muscle relaxant cream for local application

  or < Diltiazem (twice a day)
  Nitroglycerine (thrice a day)

QQ

If fails

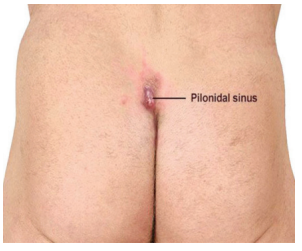
- Sx - Sx Rx OC - Lateral Sphincterotomy / (Internal sphincterotomy)
- Obsolete - Lord's 4 - 8 finger anal dilation

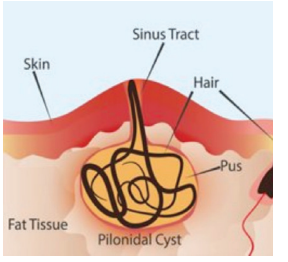
RECTUM AND ANAL CANAL - PILONIDAL SINUS

- Jeep's disease
  - Prolonged sitting, 20 - 29 y, 4x ♂, hairy gluteal skin
  - Dead hairs create abrasion, hole in skin of intergluteal cleft
  - Tuft of hair settle in subcutaneous tissue
  - Serous discharge from skin opening
- Rx - Plastic Sx reconstruction - Rx OC - Limberg Rotational Flap (Rhomboid Shape incision) / Karydakis Flap (Rotational Flap

  Excision with Secondary healing by granulation tissue

Q July 24





Z Plasty

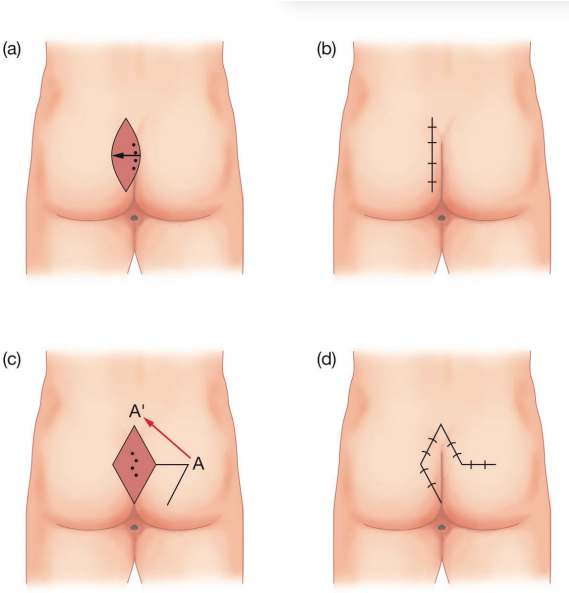
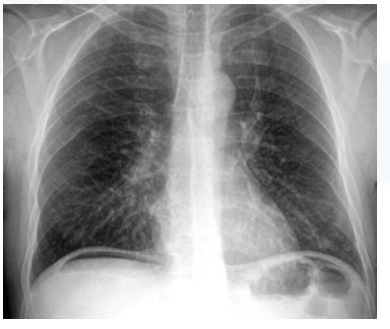


Figure 80.9 Off-midline closure techniques for pilonidal sinus. Karydakis's operation **(a)**: an off-midline incision is made around the sinus complex, which is excised, and a contralateral flap is mobilised to allow tension-free off-midline closure **(b)**. The Limberg flap **(c)**: the sinus complex is excised using a rhomboid incision and a measured flap is rotated (A) to (A') to achieve tension-free closure **(d)**.

Workbook Questions

1. Which of the following pathology is shown in the image?



A. Sub acute intestinal obstruction

B. Paralytic ileus

C. Pleural effusion

D. Hollow viscus perforation

Active space

2. Patient has undergone cholecystectomy few days back and now the patient has presented with upper abdominal pain. MRCP was done which showed dripping of bile from the cystic duct stump. Patient is hemodynamically stable. Which of the following should be done in the management of this patient?

   A. CECT

   B. ERCP and stenting

   C. Exploratory Laparotomy

   D. Conservative treatment with antibiotics

3. A chronic alcoholic patient presented in the emergency with sudden onset of upper abdominal pain radiating to the back of with anorexia, nausea and occasional vomiting. Serum amylase was raised. Which of the following is the most likely diagnosis in this patient?

   A. Acute cholecystitis

   B. Acute pancreatitis

   C. Acute appendicitis

   D. Chronic hepatitis

4. A patient presented with red, tender swelling in the inguinal region. Which of the following is more appropriately done in this patient?

   A. First Angiography to confirm the diagnosis

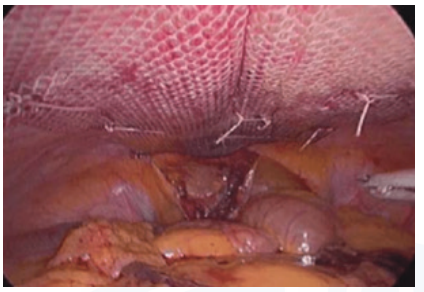
   B. First antibiotics and observation

   C. Incision and drainage

   D. None of the above

Active space

5. 5. Which of the following method is shown in the surgical procedure to close the abdominal wound with the help of a mesh ?



A. Onlay

B. Sublay

C. Inlay

D. Intraperitoneal (Underlay)

6. A patient presented in the OPD with serous and pus discharge from the opening in perianal region diagnosed as Fistula in ano. Gold standard investigation for fistula in ano is-

A. USG

B. CT SCAN

C. MRI

D. Sinogram

Answers: 1. (D), 2. (B), 3. (B), 4. (C), 5. (D), 6. (C)

Active space

ENDOCRINE SURGERY

THYROID GLAND

Goiter

- Enlargement of Thyroid Gland
- MCC - I_2 deficiency

  $I_2 \downarrow \rightarrow \downarrow T_3\ T_4 \rightarrow \uparrow$ TRH, $\uparrow$ TSH

  TSH Promote release of Preformed T_3 T_4 and cause enlargement of follicle

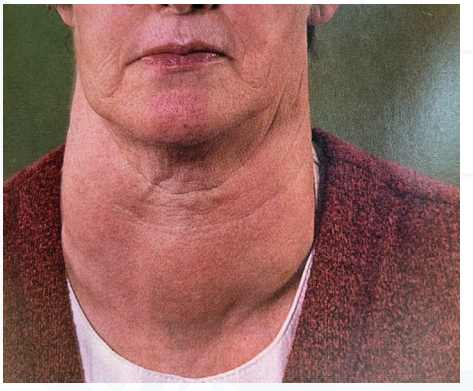


FIGURE- 6.2 Multinodular Goitre

1. Diffuse Goiter - Reversible c̄ I2 / Thyroxine
2. Nodular Goiter - Irreversible
   - Single nodule
   - Multi nodular Goiter

- Nodular Goiter - Usu - Asymp

  s/t - Pressure Symp - Dyspnoea, Dysphagia

  - Complications -
    - Secondary thyrotoxicasis
    - Follicular thyroid Cancer

Active space

- Cl. approach -

  Cl. examⁿ

  US guided FNAC

- Rx : If Symp, complications, cosmetic,

  patient's wish → Sx removal

  ↓

- Hemithyroidectomy -

  Rem of one lobe + Isthumus

- Total thyroidectomy

  Rem of complete thyroid

- Thyroid Sx → Make patient Euthyroid before Sx to avoid Thyroid Storm or Thyrotoxic crisis.

  Drugs which can be used :-

  a) Antithyroid drugs → Carbimazole, Methimazole, Propylthiouracil Given upto evening before Sx

  b) Iodides → Reduce vascularity of gland: upto morning before Sx

  c) B blockers → Inhibit Peripheral Conversion of T4 into T3,. Given for upto 7 days after Sx.

- Complications of Thyroid Sx:

- 1. Respiratory distress →

  (a) Inside O.T. → Damage to both Recurrent Laryngeal nerves during operation

  → Anaesthetist will not be able to extubate the patient after opⁿ.

  (b) few c̄ in hrs after the opⁿ. Reactionary hage due to slipping off a ligature from Supr Thyroid artery → Immediate Rx → opening of skin sutures

  (c) 2-5 days after opⁿ. → Late → Hypocalcemia (Carpopedal Spasm) Neet PG 25

Active space

- 2. Injury to nerves →
  - MC- External branch of supr Laryngeal nerve
  - 2nd MC → RLN
- 3. Hypothyroidism → Require Life Long Thyroxine replacement
- 4. Hypoparathyroidism → Rare →
  - MCC → Ischemic injury
  - 2nd MC → Accidental removal of Parathyroid. Reimplant the Parathyroid back into muscle → Sternomastoid, forearm, arm muscle.

1. Primary h' age → bleeding during opⁿ
2. Reactionary h' age → bleeding c̄ in 4-24 hrs after opⁿ
   - Slipping off a ligature • Dislodgement of clot • Rise in BP
3. Secondary h' age → bleeding > 5-7 days of opⁿ
   - Infection - necrosis or erosion of the wall of blood vessel.

Retrosternal Goiter

- Enlargement of Thyroid gland (Lower pole) behind sternum - Dyspnoea, Dysphagia
  Engorged neck & facial veins
  Stridor

- Pemberton's Test - Flushing of face on raising both hands above the head multiple times
- Rx -Removal via neck incision (cervical incision)

FIGURE- 6.3 Pemberton's sign in Retrosternal Goitre

Active space

MASTALGIA AND BREAST ABSCESS

Mastalgia Breast Pain

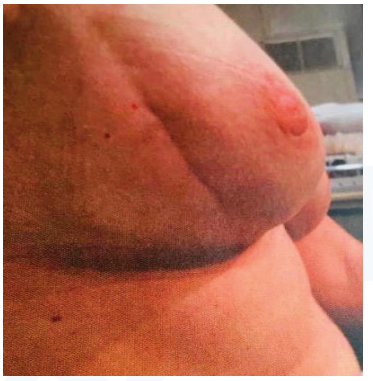


FIGURE- 6.4 Monder's disease of Breast- Thrombophlebitis

FIGURE- 6.5 Mastitis

Cyclical

↓

d/t hormonal influence

↓

Varies c̄ phases of menstrual cycle

Rx - Reassurance

↓ Young girls

Tab. oil of evening primerose

↓

Danazol

Bromocriptine

↓

Tamoxifen

Non Cyclical

↓

Not d/t hormones

↓

Inflammation of

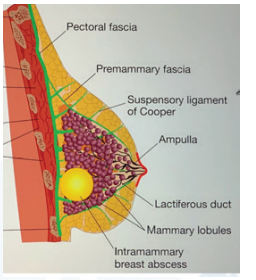
- Breast - Mastitis
- Veins - Monder's disease

Active space

Mastitis

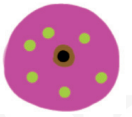
QQ

- Pregnancy, Lactation
- MCC - Staph aureus - from baby's mouth through cracks in the nipple (mothers)
- Initially - Redness, Swelling, ↑ Temp. Pain
- Rx - Antibiotics, Breast support QQ

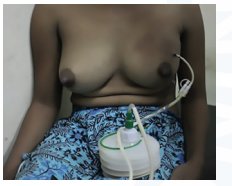


↓

- If fails can result in Breast Abscess QQ
- Early Stage - Small Pus pockets inside breast
  Fluctuation - Negative
- Rx - Ab + Us guided multiple Needle aspirations



- Late stage - Good amount of pus collection is present
  Fluctuation - Positive
- Rx - Ab + US Guided catheter drainage

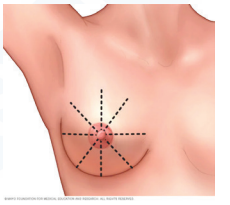


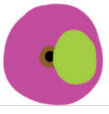
Size > 3cm or Volume > 30ml → Vacuum assisted Suction drainage (Romovac)

Breast feeding to be continued

Earlier Incision & drainage

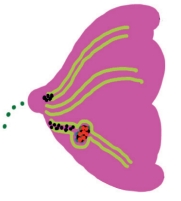
- Best Incision - Radial Incision
- Cosmetic " - Periareolar or Circumareolar Incision





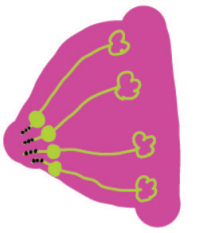
DISCHARGE FROM NIPPLE

- MC discharge from nipple - Milk
- MC Patho " " " - Serous
- MCC Green " " " - Duct Ectasia
- MCC Blood " " " - Duct Papilloma/Duct Carcinoma



Active space

Duct Ectasia



- Periductal Mastitis, Plasma cell Mastitis
  - Abnormal dilatation of L. ducts
  - Usu multiple ducts involved
  - d/t dehydration - Smokers/Alcoholics
- MC C/F - Green discharge from nipple (Brown, Black or bloody)
- Slit Like retraction of nip (Circumferential - Cancer)
- Chronic subareolar mass or abscess (Zuska's disease)
- Cl. approach

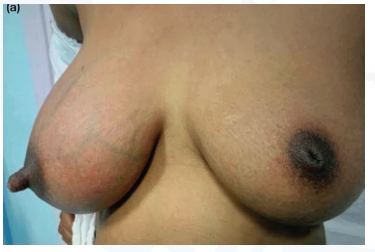
Green discharge

↓

No Invⁿ

↓

Rx - of Duct Ectasia



Blood discharge

↓

1st → Cytological evaluation of discharge to rule out cancer

↓

Radiological Invⁿ

↓ US < 40y

mammography > 40y

Ductography

(Ductoscopy)

↓

Rx - of Duct Ectasia

Lump Triple Assessment

↓

Clinical — C R P

Radio US < 40y

Mammo > 40y

Patho

Biopsy - Best

FNAC

↓

Rx - of Duct Ectasia

Active space

- Rx : Plenty of fluids, Breast massage

  Rx oc - Conservative

  ↓ fails

  Microdochotomy

  ↓ fails

- Hadfield's operation (Excision of mul. L. ducts) [Conical Incision]
- Excision of Single L. duct - Microdochectomy [Tennis Racquet Incision]

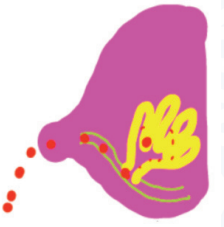
Duct Papilloma

- Abnormal Papillary growth from L. duct
- Usually Single L. duct inv.

Q Jan 25

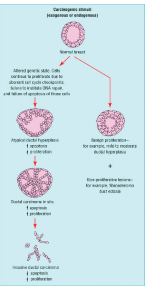
- MC - Blood discharge from nip Premalignant
  - Single duct, No Atypia - Risk of Cancer ↓
  - Mul ducts, Atypia ⊕ - " " ↑
- Rx : Microdochectomy

  (If multiple L. ducts involved → Hadfield's opⁿ)

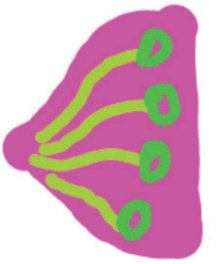


Active space

BREAST LUMP- FIBROADENOMA AND PHYLLOIDES TUMOUR

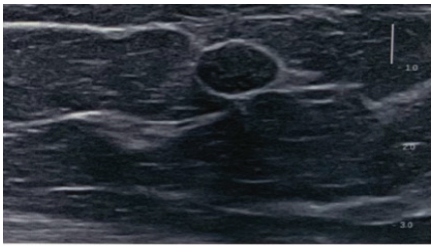


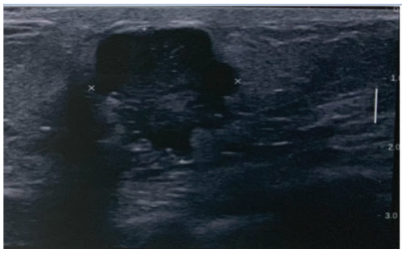
- If controlled hyperplasia → Benign Tumour
- " Uncontrolled " " → Atypical hyperplasia
  Ca. in situ
  Invasive Ca
- MC Benign Breast Tumour - Fibroadenoma
- MC Breast Tumour < 30 y - Fibroadenoma
- MC " " > 30 y - Invasive ductal Ca
- MC Benign Breast disease - Fibrocystic disease of Breast



Fibroadenoma

- Breast Mouse QQ
- hyperplasia of a single lobule
- 15 - 25 yrs
- Small, Asymp, can resolve spont by 30 y
- ..... → Giant fibroadenoma
- Complex fibroadenoma → ↑ risk of Lobular Ca
- MC C/F- Breast Lump >25yrs → Triple Assessment
- Approach - C R P
- Mammography - Popcorn Calcification in fibroadenoma
  (on CXR - Popcorn cal - Pulmonary hamartoma)

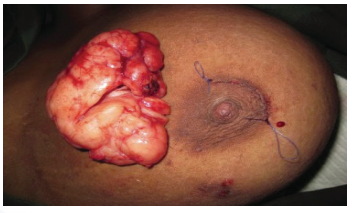




Active space

Rx - Tamoxifen/Ormeloxifene

- Rx : Wait upto 30 y - Conservative ✓
- Indications for Sx -
  - Suspicious on imaging / histolgy
  - Age > 30 yrs
  - Size > 5 cm
  - Patient's wish
  - Family h/o of Ca



- Rx - Excision
  (Periareolar / Circumareolar Incision) QQ

Cystosarcoma Phylloides

- Phylloides Tumour -
- Serocystic disease of Brodie
- Hyperplasia of Stroma
- > 30 yrs
  Diffuse enlargement of breast
  c̄ variegated, bassellated appearance
  Leaf like appearance on histopathology **(Q Jan 25)**

FIGURE- 6.7 Cystosarcoma Phylloides

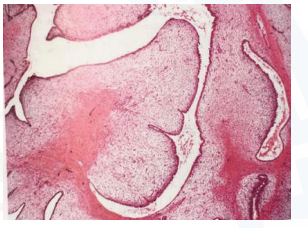


FIGURE- 6.8 Classical microscopic leaf like appearance of stromal hyperplasia in Cystosarcoma Phylloides

- If malignant - Sarcoma
  ↓
- does not spread via .....
- Rx - Wide Local Excision c̄ 2cm margin + P. major ms
  eg :
  - Liposarcoma
  - Lymphangiosarcoma
  - Osteosarcoma
  - Malignant GIST
  - Desmoid Tumour -

♀ Below Umb.
MS/ Aponeurosis of Antr abdwall

Active space

Workbook Questions

1. A lady presented to OPD with complaint of pain in the breast. On clinical examination cracks in the nipple were present and fluctuation was positive. Most likely diagnosis in this lady will be?

   A. Breast engorgement
   B. Breast abscess
   C. Acute Masitis
   D. Atypical hyperplasia

2. 20-year-old lady, 2cm single, mobile, asymptomatic breast lump with well-defined margin, noticed by the patient herself on self-breast examination, presented to the OPD. She was diagnosed as having a benign lesion. You are the doctor and what would you recommend to this patient

   A. Reassure, continue to follow-up the patient
   B. Surgical Excision
   C. Biopsy
   D. Give Danazol

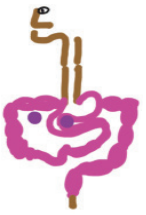
Answers: 1. (B), 2. (A)

Active space

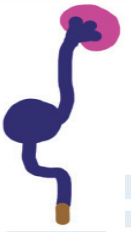
ONCOSURGERY

A patient presented with the following as shown in the image. Which of the following will have less priority in calculating the prognosis in this patient?

a. Duration
b. Site
c. Depth
d. Histological Grade



Squamous
Adeno Ca
Transitional



GIT CANCERS- ORAL CANCER

- MC - Squamous Cell Ca
- MC Site in India => Gingivobuccal Sulcus (Buccal mucosa)
- MC Site in Western => Lateral border of tongue (Middle 1/3)
- RF - Tobacco Chewing

Areas near natural openings of the body - HPV **Q July 25**

Premalignant lesions → Leukoplakia → White patch

Erythroplakia → Biopsy QQ (Red Patch - 17x more malignant than Leukoplakia) **Q July 24**

Prognosis depends on -

1. Stage (TNM) 2. Site of Ca
3. Depth of Ca (Duration can be excluded)

ESOPHAGEAL CANCER

- MC - Squamous cell ca → Upper2/3 Eso-Middle 1/3
- MC - Western - Adenoca → Lower 1/3 Eso
- RF :
  - Smoking, Alcohol, Increased consumption of smoked food
  - Betel nut Liquid
  - GERD, Barret's
  - Achalasia cardia
  - Zenker's div
  - H P V
  - Radiation
  - Tylosis - Palm, Soles
  - Syn - Plummer Vinson Syndrome or Paterson Brown Kelly syn (Iron def, Dysphagia, Eso webs - Middle aged ♀ SCC) → in Postcricoid-Upp esophagus)

Active space

- Patho - Polyp
- C/F - Dysphagia - Initially for .....
  - Gradually for both Soilds & Liquids
  - Regurgitation - Aspiration - Pulm complicⁿ
  - Hoarseness of voice - d/t involvement of RLN (Recurrent Laryngeal nerve)
  - Anorexia, wt loss
- Cl. app :
- 1st Ba Swallow - Filling defect
  Rat Tail appearance
  Apple core appearance

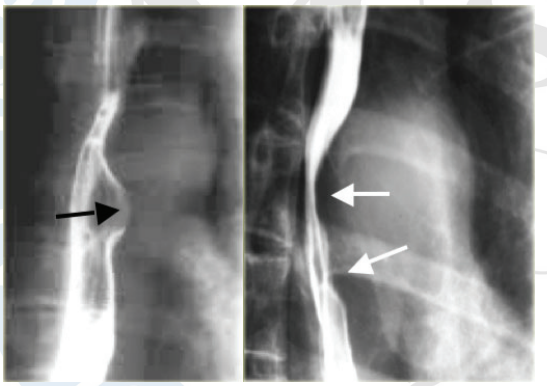


FIGURE- 3.1 Filling defect and Rat tail appearance in Cancer esophagus on Barium swallow

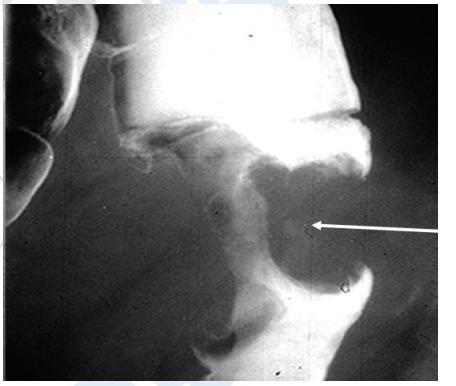


FIGURE- 3.2 Ba Enema- Apple core appearance in Colon Cancer

- IOC - UGIE + Biopsy to diagnose
- Staging IOC - CECT Thorax and abdomen

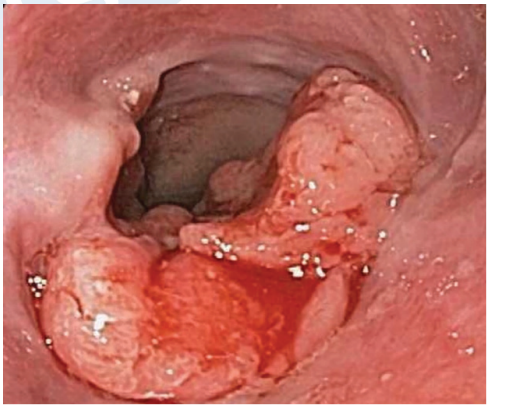


FIGURE- 3 .3 Esophageal cancer on Upper GI Endoscopy

- IOC for depth of
- Tumour - Endo US

Q Jan 25

Active space

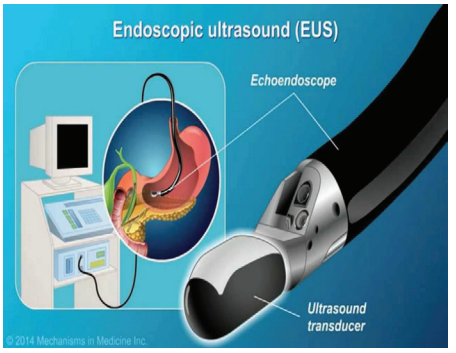


FIGURE- 3.4 Endoscopic Ultrasound equipment

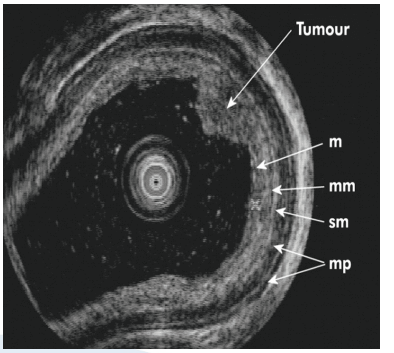
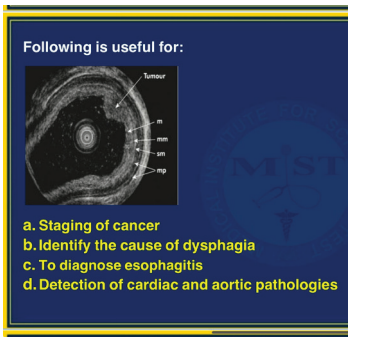


FIGURE- 3.5 Each layer visible on Endoscopic Ultrasound



- IOC for distant metastasis - PET Scan (Positron Emission Tomography)
- 18F - FDG - Fluorine - Fluoro deoxy glucose (T1/2 = 120 min)

FIGURE- 3.6 PET CT equipment

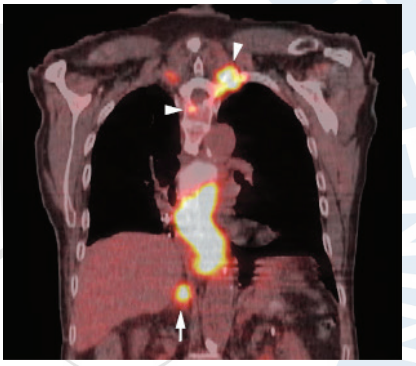


FIGURE- 3.7 Golden FDG uptake on PET Scan

- Staging : T=Tumor ; N=Lymph node ; M=Metastasis
- TNM In GIT N = No: of L.N.
  - T_1 - Submucosa
  - T_2 - Muscle
  - T_3 - Adventitia
  - T_4 - Adjacent organs - T4a- Pleura, Pericardium and Diaphragm
    - T4b- Trachea, Bone and Aorta
  - N_1 - 1 - 2 LN
  - N_2 - 3-6 LN
  - N_3 - 7 or more than 7 LN
  - M_1 - Mets

Active space

Prefix

C = Clinical Stage

Y = Staginig after neoadjuvant therapy

P = Pathological Staging

Suffix

Tx = Tumor cannot be assessed

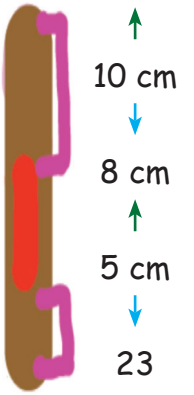
To = No evidence of tumor

Ro = No Residual tumor

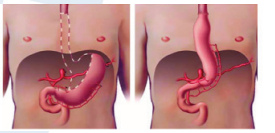
- Rx : Mainstay - Sx -
- Removal c̄ Margin - Proximal - 10 cm, Distal - 5 cm

Transthoracic Total Esophagectomy (Ivor Lewis opⁿ)

c̄ Eso reconstruction



- Stomach is used for eso reconsⁿ
- Early Complicⁿ = Anastomotic Leak (Pulmonary Complicⁿ)
- Late Complicⁿ = Stricture Eso
- Chemotherapy - 5 Fluorouracil, Cisplatin



Neoadjuvant therapy	- Rx given before main therapy
Adjuvant therapy	- Rx given after main therapy

GIT CANCERS- STOMACH CANCER

- MC - Adenocarcinoma
- MC - Site → Prepyloric (Antrum)

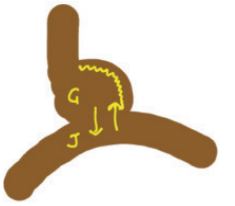
  MC - Western - Cardia
- RF : Smoking, Alcohol

  ↑ Salt, ↑ Pickle, ↑ Smoked food,

  Canned food, ↓ Vitamins, antioxidants
- H. Pylori
- EBV - Ebstein Barr Virus

Active space

- Autoimmune/Atrophic Gastritis
- Menetrier's disease
- Reflux Gastritis - Gastric Sx - Alkaline G, Intestinal metaplasia
- Ga ulcer
- Adenomatous Polyps,
- Genetic - P53, SMARCA 1, RHOA
- Patho :



Diffuse Ga Ca
↓
Leather Bottle Stomach
Linitis Plastica

FIGURE- 3.8 Leather bottle stomach- Linitis Plastica (Diffuse Gastric cancer)

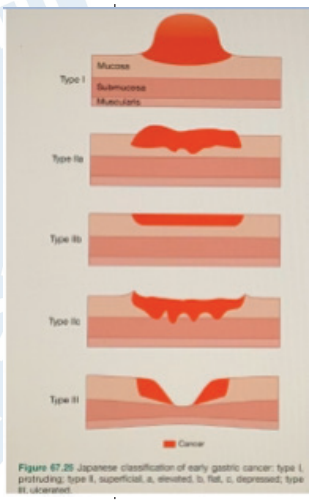


Figure 67.25 Japanese classification of early gastric cancer: type I, protruding; type II, superficial; a, elevated; b, flat; c, depressed; type III, ulcerated.

- Early (Japan) - Muc., Submuc ± LN inv
- Bormann ← Advanced - Muc, Submuc, Muscle ± LN inv
- C/F :
  - Malena, Hemetemesis
  - Epigastric mass
  - Anorexia, wt loss
  - Trousseau's sign - Migratory Thrombophlebitis

  MC in Pancreatic Cancer

  Can be seen in Gastric Cancer

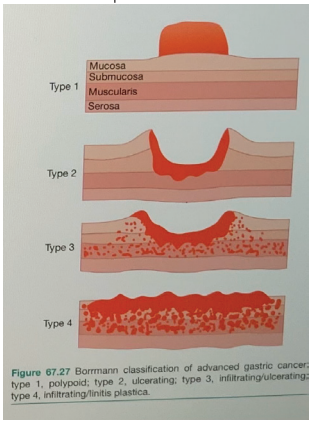
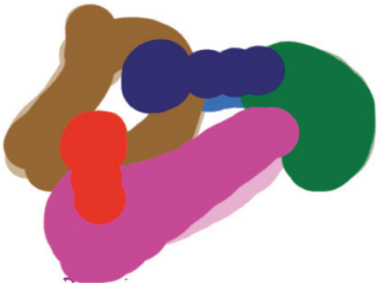


Figure 67.27 Borrmann classification of advanced gastric cancer: type 1, polypoid; type 2, ulcerating; type 3, infiltrating/ulcerating; type 4, infiltrating/linitis plastica.

- Spread :

1. Local



Proximal Ca - Spleen

Distal Ca - Pancreas

2. Lymphatics -Principal LN - Coeliac LN
   - Enlarged (L) Supraclavicvlar LN - Troisier's sign and (Virchow's node)
   - Enlarged Axillary LN - Irish node
3. Blood - Liver, Lungs
4. Transcoelomic Spread - Direct Peritoneal spread

MIST

Peritoneal seedlings or deposits

QQ

Periumb - Sister Mary Joseph Nodule

Ovarian - krukenburg Tumour

Rectal wall - Blummer's shelf

- Clinical approach
- Malena → 1st Stool for occult → blood → (+) → UGIE + Biopsy
- Mass → 1st → US abd → UGIE + Bx
- IOC - UGIE + Bx
- Staging IOC - CECT abd
- IOC - depth of Tumour - Endo US
- IOC - distant mets - PET Scan

FIGURE- 3.9 Virchow's node-enlarged left supraclavicular lymph node in a patient of Gastric cancer

Active space

- **TNM Staging - N = No : of LN**
  - T1a - Lamina
  - T1b - Submucosa
  - T2 - Muscle
  - T3 - Subserosa
  - T4a - Perforate the serosa
  - T4b - Adjacent organs
  - N1 - 1-2 LN
  - N2 - 3-6 LN
  - N3a - 7-15 LN
  - N3b - More than 15 LN
  - M1- Distant metastasis

- Rx : Best - Sx - Margin - 5 cm
- Proximal Ga Ca -

  Total Gastrectomy

  c̄ Splenectomy (for adequate lymphatic clearance)
- Distal Ga Ca -

  Partial or Subtotal

  Gastrectomy

Active space

GIT CANCERS COLORECTAL CANCER- AETIOPATHOLOGY AND CLINICAL PRESENTATION

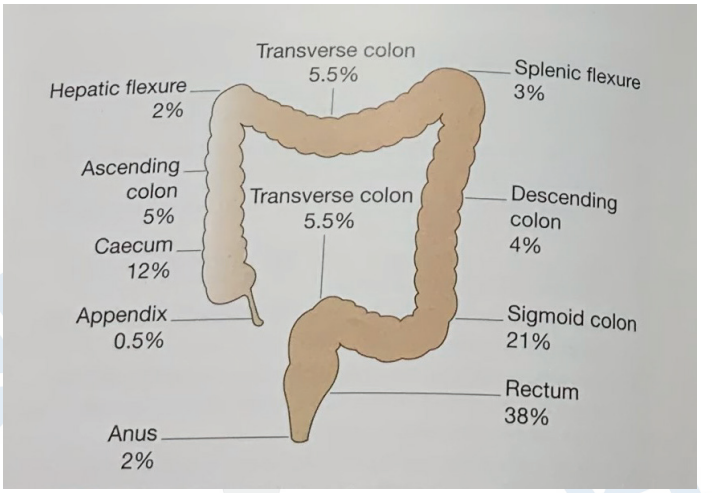


FIGURE- 3.10 Sites of colorectal cancer- percentage distribution

- MC - Adenocarcinoma
- MC Site - Rectosigmoid
- RF :
- Diet : ↑ Red meat/↑ Animal fat/↓ Fiber/↓ Vitamins, antioxidants/ Smoking/Spirits
- Inflammatory bowel ds (UC > Crohn's)
- Cholecystectomy
- Radiation
- Genetic - APC gene
- Polyps - Adenomatous, FAP
- Gardner's, Turcot's syn
- Lynch syn - Hereditary Non Polyposis Colorectal Ca - MMR Gene mutation

Q July 25

Familial Adenomatous Polyposis -

- Autosomal dominant
- APC gene mutation
- 50% - Cong. hypertrophy of Retinal Pigment Epithelium

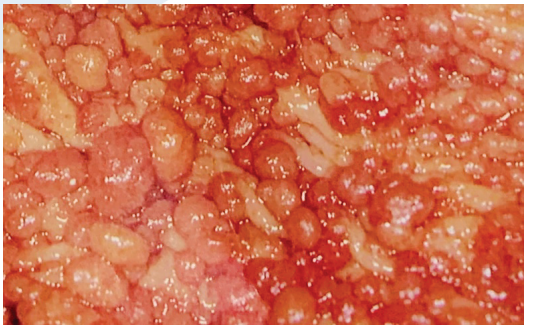


FIGURE- 3.11 More than 100 adenomatous polyps in Familial adenomatous polyposis

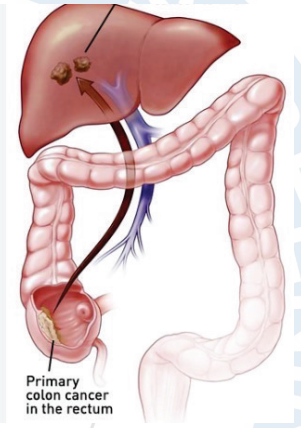
Active space

- >100 adenomatous polyps is diagnostic
- Gardner's syn - FAP + Epidermoid cysts, Desmoid tumours
- Turcot's syn - FAP + Brain Tumours
- ↑ Risk of Cancer

Familial hamartomatous Polyposis Q July 25

- - Peutz Jeghers syn - Multiple Juvenile polyps in intestine + Melanin Pignentation of lips, buccalmucosa, hands, forearms

- C/F : (R) Colon Ca - Ulcerate→Bleeding
  (L) Colon Ca - Annular→Obstuction
- Rectal Ca - MC - Bleeding
  PR
- 1/3 (33%) of colorectal Ca present c̄ Liver mets



GIT CANCERS- COLORECTAL CANCER- MANAGEMENT

- IOC - Colonoscopy (Whole Colonoscopy) + Biopsy
  Screening Invⁿ - Colonoscopy once every 10 years

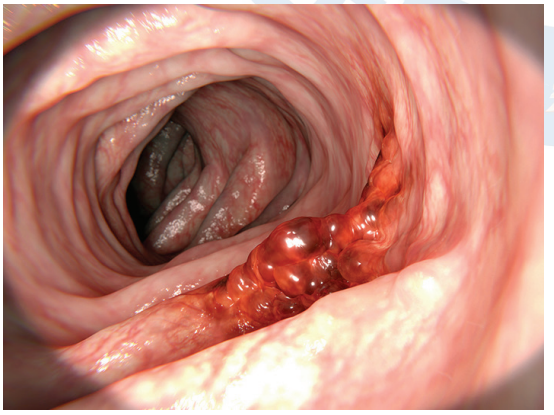


FIGURE- 3.12 Colon cancer on Colonoscopy

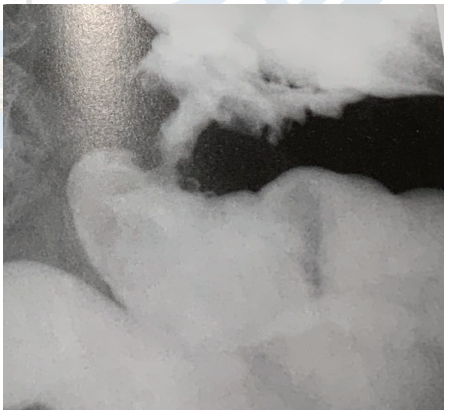
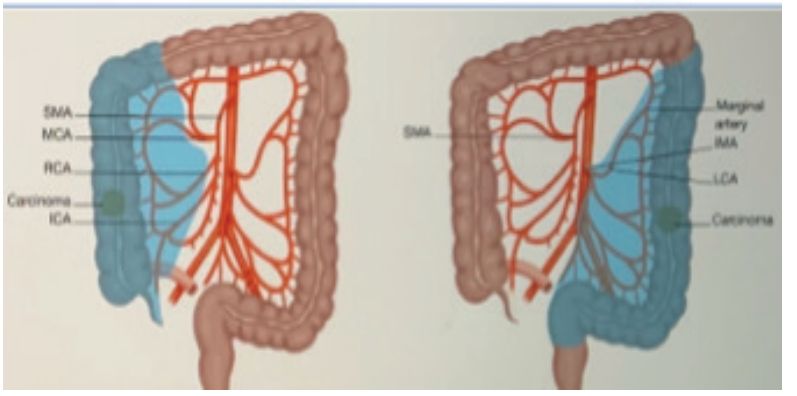


FIGURE- 3.13 Apple core appearance on Barium enema in Colon cancer

Active space

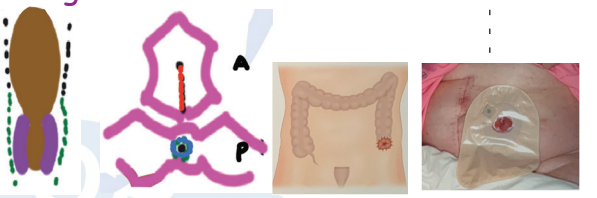
- Staging IOC - CECT (For Rectal Ca - MRI)
- IOC depth of Tumour - Endo US
- IOC distant mets - PET Scan
- TNM Staging : N = No : of LN
  - T1- Submucosa
  - T2- Muscle
  - T3- Subserosa
  - T4- Adjacent Organs
  - N1- 1-3 LN
  - N2- 4 or more than 4 LN
  - M1- Distant metastasis
- Rx : Best Sx - Margin - 5 cm for colon Ca
  - 2 cm for rectal Ca
  - (distal margin can be 1 cm also)
- → Ⓡ hemicolectomy
- → Ⓛ hemicolectomy
- Extended Ⓡ hemicolectomy



Active space

Rectal Cancer Surgery

- Antr Resection
- Low Antr resection
- Endo circular stapler sphincter preserving Sx
- If anal Sph cannot be preserved → APR - Abdomino Perineal Resection and Permanent End Sigmoid colostomy

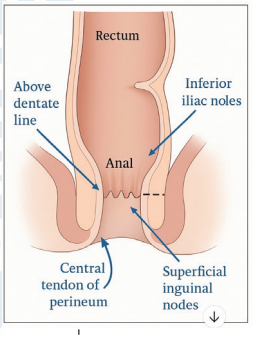


Prognosis of colorectal malignancies depends best on -

1. Stage (TNM)
2. LN Status

GIT CANCERS- CARCINOMA ANAL CANAL

- HPV
- Squamous cell Ca.
- Superficial Inguinal LN involved **Neet PG 25**
- Rx OC - CTh + Rth Chemoradiation - (.....Regime) QQ
- CTh in Colorectal Ca - 5 Fluorouracil, Oxaliplatin
- Targeted therapy in Rectal Ca - Cetuximab



GIT CANCERS- CARCINOMA OF GALL BLADDER

- MC site - Fundus, Adenoca
- RF - Gall stones, Porcelain GB, Adenomatous Polyps, Typhoid carriers
- C/F - Cholecystitis, mass, Jaundice
- IOC - CECT abd
- Rx - Radical cholecystectomy (Rem of GB, LN)

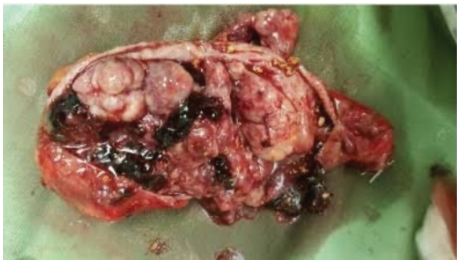


FIGURE- 3.14 Carcinoma Gall bladder- growth at the fundus of GB

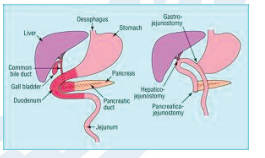
Active space

GIT CANCERS- CARCINOMA OF PANCREAS

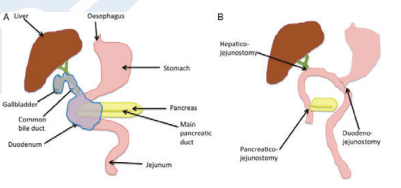
- MC - Adenoca
- MC - Ca head of Pancreas
- 2nd MC - Periampullary Ca
- RF : Smoking, Chronic Pancreatitis
  DM, Hereditary

- C/F ⊖ Jaundice - Ca HOP - Late, Progressive Jaundice
  Periamp - Early, waxing, waning Jaundice
- ⊙ - Marked wt loss
- ⊙ - Corvoisier's Law - Neet PG 2025
  GB always palpable if CBD obsn is d/t Cancer
  GB not " " " " " Stone

- IOC - Ca HOP - CECT (Endo us guided biopsy)
- Periamp - ERCP + Biopsy
- T. marker for Pancreatic Ca - CA 19-9
  2nd - CEA (Carcinoembryonic Antigen)
  (T marker for colon Cancer - CEA)



- Rx : Sx - Pylorus preserving
  Pancreatico duodenectomy
- Whipple's opn - Triple Bypass opn
  PJ, HJ, GJ



GIST- GASTRO INTESTINAL STROMAL TUMOUR

- Any part of GIT- MC- Stomach; 2nd MC- Ileum
- From interstitial cell of Kajal
- C-Kit gene mutation; overexpression of CD 117
- If become malignant then it is Sarcoma (Sarcomas does not spread by lymphatics)

Active space

- Rx - Wide local excision like partial Gastrectomy
  - LN removal is not required
- If GIST is massive/ malignant or recurrent- then Imatinib (Glevec) can be given lifelong.

QQ Q July 24

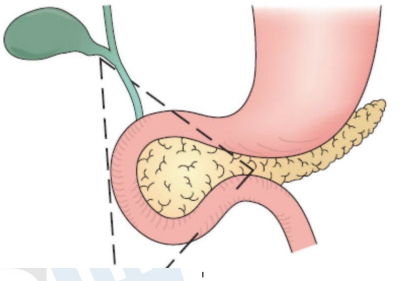
- Gastrinoma- Zollinger Ellison Syndrome

  Gastrinoma Triangle- Triangle of Passaro-

  Superiorly- Junction of Cystic duct and CBD;

  Inferiorly- Junction of 2nd and 3rd part of duodenum;

  Medially- Junction of Neck and body of Pancreas



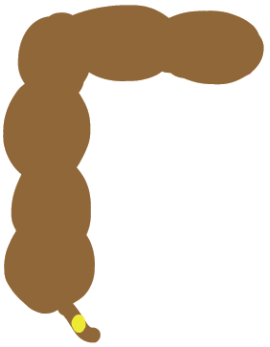
- Autosomal dominant
- Asso with MEN I Syndrome
- Asso with Non Beta Islet cell tumor of Pancreas
- Raised HCl secretion due to increased Gastrin produced by the tumour, lead to severe GI ulcerations and abdominal pain
- Lab Tests- Best -
  1. If Fasting S. Gastrin > 10x and Gastric P^H < 2 - it is diagnostic
  2. Secretin provocative test- Rise in Serum Gastrin- >110 pg/ml
- Radiological tests 68 Ga labelled SSA with PET CT is best
  - CECT Abdomen / Endoscopic Ultrasound
- Rx- Aggressive PPI's / Removal of tumour (If size >2cm)

Gallium labelled Somatostatin analogue

CARCINOID TUMOR (AGRENTAFFINOMA)

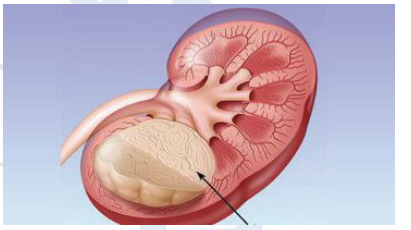
- Arise in argentaffin tissue (Kulchitsky cells of crypts of Leiberkuhn).
- MC site - Appendix
- Clinical Features- Cyanosis / Flushing / Diarrhea / Borborgymi / Asthmatic attacks

Active space



- 5 -hydroxytryptamine (serotonin)
- 5 -hydroxyindoleacetic acid in urine QQ **Q July 24**
- Plasma marker - Chromogranin A/
- Rx- for appendiceal carcinoid tumor -
  - Appendicectomy
  - If tumor 2cm or more in size / If caecal wall is involved / lymph nodes involved, Then-Right hemicolectomy.
  - Rx For Ileal Carcinoid → 1st SSA 2nd → PRRT Peptide Receptor Radionuclide therapy

UROLOGY CANCERS- KIDNEY TUMOURS



1. Wilm's Tumour - Nephroblastoma

- < 5 yrs. MC Tumor ofchildhood
- Inactivation of WT suppressor genes
- WT_1 - WAGR syn, Denys Drash syn
- WT_2 - Beckwith Weidman syn
- MC - Lower Pole of Kidney
- C/F - Mass (Lumbar region)
- Pyrexia
- Hematuria
- IOC - CECT abd - Claw sign
- Rx - Sx + Cth
- Sx - Radical Nephrectomy

2. Adult Kidney Ca - Hypernephroma

- MC - Adenocarcinoma
- MC - PCT Epithelium
- MC - Clear cell type
  - RF : Smoking
  - Dyes, chemical - Aniline dyes Naphthyamine/Benzedine dyes/ Azo dyes
  - Long Term Analgesic use

Industrial Cancers - Kidney / bladder

Active space

- Long standing stone disease
- VHL syn - von Hippel Lindau syn - Mutationin VHL gene located on chromosome 3p Q July 25

- C/F - MC - Painless hematuria
- Pain
- Mass - Lumbar
- (L) RCC → Varicocele (L)
- Paraneoplastic syn -
  - Hypercalcemia
  - Erythrocytosis
  - ↑ ESR
- 1st Invn for Painless hematuria -

  Urine Cytology for malignant cells if ⊕

  ↓

  Cystoscopy ⊖

  ↓
- IOC - CECT abd
- Rx - Sx alone
- T < 4 cm - Partial Nephrectomy
- ≥ 4 cm - Radical Nephrectomy
- Immunotherapy - BCG

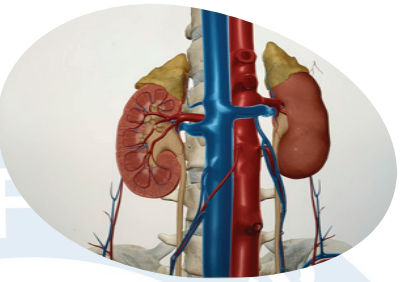


FIGURE- 3.15 Left Testicular vein open in Left Renal vein- Left RCC can cause Left Varicocele

FIGURE- 3.16 Radical Nephrectomy specimen

Indications for Partial Nephrectomy / Nephron sparing

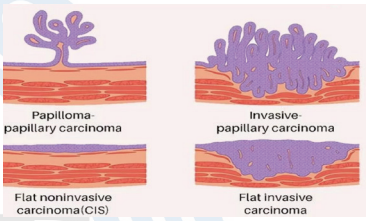
Sx : - QQ.

1. T < 4cm
2. T 4 - < 7cm - in selected cases
3. Bilateral Kidney tumor
4. T in solitary Kidney
5. Patients c̄ Renal dysfunction

Active space

UROLOGY CANCERS- BLADDER TUMOUR

- MC - Transitional Cell Ca (90 – 95%)
- Sq. cell Ca - Schistosomiasis (Bilharziasis)
- Adenoca- Dome or Fundus of UB (Congenital Uninary bladder diverticula) Midline, Anterosuprly
- MC Site - Lateral bladder Wall, Trigone
- RF - Smoking, Dyes, Chemicals
- Patho :



1. MC - Non muscle invasive
2. Ms invasive
3. Flat, Non invasive - Multicentric - Poor Prognosis

- C/F - MC - Painless hematuria

1st - Urine Cytology for malignant Cells ⊕

↓

IOC - Cystoscopy + Biopsy

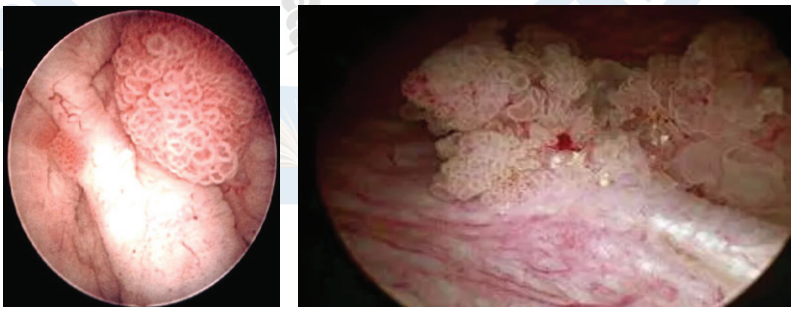
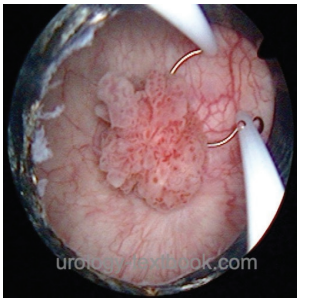


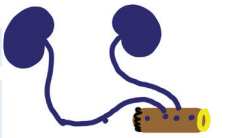
FIGURE- 3.17 Bladder cancer on Cystoscopy

Active space

- Staging IOC - MRI
- Rx : Non ms invasive - TURBT + Intravesical
  Cth c̄ Mitomycin C /
  (Transurethral Resection of bladder tumor)
  Note → We can also use Intravesical Immunotherapy c̄ BCG



- Ms Invasive - (1) Neoadiuvant Cth + Radical Cystectomy
  (Cystoprostatectomy)
  a) Creation of a Neobladder (Ileum)
  or b) Urinary diversion = Ileal Conduit / Colonic Conduit
  (2nd) Option → Cth + Rth



UROLOGY CANCERS- PROSTATE CANCER

- MC Ca in Eldery males
- 70% of Males, > 80 yrs - Prostate Ca
  Adenoca ; Peripheral zone (Postr. lobe)
- RF : ?? Androgen
- Asymp
- Spread -
  1. Local - upwards, Laterally Seminal vesicles involve early
     Urethra involve Late
  2. Lymphatics - Obturator LN → Int. Iliac LN (Pelvic LN)
  3. Blood - MC - Bones - Lumbar vertebra
     through Bateson's Vertebral plexus of veins
     Bony mets in Prostate Ca → Osteoblastic
     " " Breast Ca → Osteoclastic + Osteoblastic

Q Jan 25

Elderly gentleman diagnosed with prostate cancer presented with low backache. What is the likely route of bony metastasis to lumbar vertebra in this patient?

a. Arteries
b. Veins
c. Lymphatics
d. Direct

- Bony mets are common in PBKBT (Prostate, Breast, Kidney, Bronchus, Thyroid)

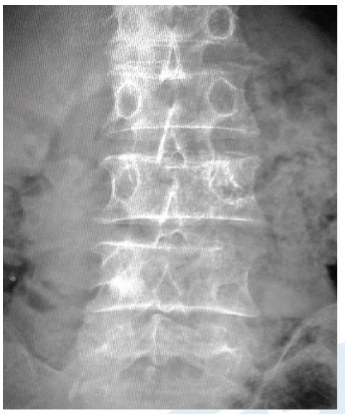


FIGURE- 3.18 Prostate cancer bony osteoblastic metastasis- Lumbar vertebra

FIGURE- 3.19 Prostate cancer bony osteoblastic metastasis- Femur

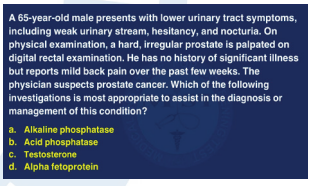
- IOC - Transrectal us guided prostatic biopsy (12 biopsies)
- Adenoca Gleason's grading (Gleason's score = 2 to 10)

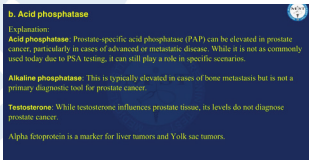
  (Breast Ca - Bloom Richardson Grading)
- Staging IOC - mp MRI - Multi parametric MRI
- IOC distant mets - PET Scan Ga68 PSMA PET Scan
- T. marker - S. PSA Q Jan 25

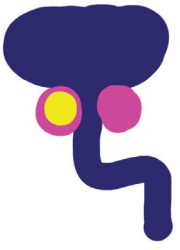
Prastate specific membrane Ag

Uses of PSA -

1. Risk of disease progression
2. Diagnosing Pathologies of Prostate
3. Monitoring Response of Pr Ca Rx QQ
4. Asscss Recurrence of Pr. after Ca Rx







- Rx :

  (1) If Prostate Ca confined c in Prostate

  a) If Life expectancy

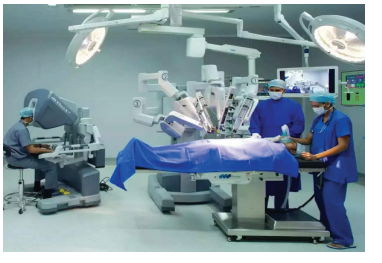
  > 10 yrs / Age <70 yrs -

  Radical Prostatectomy - Robotic

  or External beam Radiotherapy

  Rth - I ^{125}/Palladium ^{103}

  b) If life expectancy < 10yrs/Age> 70 yrs → Active Surveillance



Active space

2 Advanced Prostatic Ca - Palliative Px

Hormonal Therapy

↓

Androgen Ablation

↓

Medical Castration ✓	Surgical Castration
↓	↓
• LHRH agonists → Goserelin ✓/ Buserlin • LHRH antagonists → Degarelix • Androgen Receptor blocker → Enzalutamide • Testosterone production blocker → Abiraterone	Bilateral Subcapsular Orchidectomy

An elderly gentleman undergoing evaluation for prostate cancer is found to have serum PSA levels of 15 ng/mL. What is the most specific clinical use of PSA in the management of prostate cancer?

a. Assessing the survival and prognosis of prostate cancer
b. Confirming the diagnosis of prostate cancer
c. Monitoring treatment response in prostate cancer
d. Distinguishing between prostate cancer and benign prostatic hyperplasia (BPH)

UROLOGY CANCERS- PENILE CANCER

- MC - Squamous cell Ca
- MC Site - Glans/Prepuce
- RF : Poor hygiene, Smoking,
- HPV, Long Standing genital warts
- Multiple Sexual Partners
- PIN - Penile intraepithelial neoplasia

↓

Glans	Shaft
↓	↓
Erythroplasia of Querat	Bowen's disease

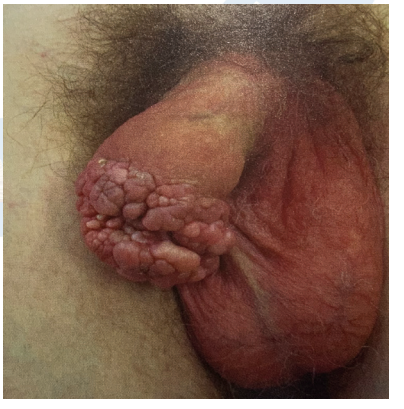


FIGURE- 3.20 Premalignant genital warts over the Penis

Active space

- C/F : Ulceroproliferative, Foul smelling growth on the Penis
  Enlarged Inguinal LN (50% - Infection 50% - Tumour)

Figure 85.24 A squamous cell cancer of the penis.

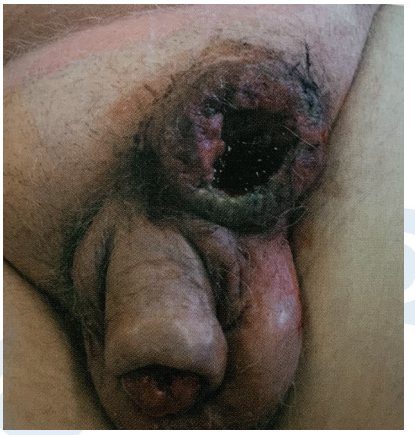
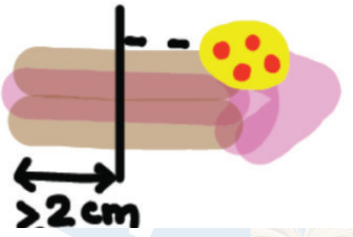
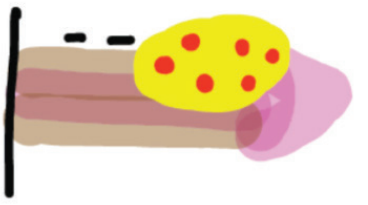


FIGURE- 3.21-A Ulceration of skin by enlarged Inguinal lymph nodes in Penile cancer

FIGURE- 3.21-B Penile cancer

- Death - d/t Torrential hemorrhage
  d/t erosion of femoral / Iliac vessels by enlarged inguinal LN
- IOC - Biopsy
- Staging IOC - MRI
- IOC distant mets - PET Scan
- Rx : Best - Sx - Margin - 2 cm





Partial Penectomy
(If preserved Penile Stump ≥ 2 cm)

Total Penectomy + perineal Urethrostomy

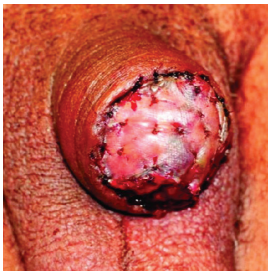


FIGURE- 3.22 Partial Penectomy for Penile cancer

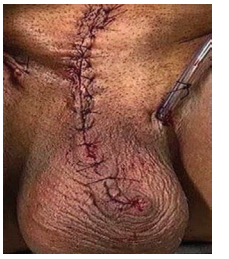


FIGURE- 3.23 Total Penectomy for Penile cancer

- Rx for LN - Antibiotics - 3 weeks - If persist- Removal

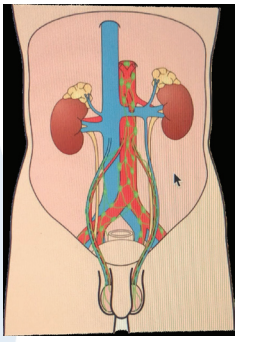
Active space

UROLOGY CANCERS- TESTICULAR TUMOURS

- MC Ca in Young males
- MC - Seminoma

Germ Cell Tumours

Seminoma	Non Seminoma
MC	• Embryonal Ca • Yolk sac Tumour • Teratoma



- MCC - Undescended Testis
- C/F : MC - Scrotal Enlargement
- heaviness in scrotum
- Enlarged Pre/Paraaortic LN → Epigastric Lump
- Hemoptysis in Testicular Ca - Pulmonary mets (Canon Ball deposits)
- (Seminoma spread mainly by Lymphatics
  Teratoma " " " Blood)

Ⓡ Testicular Ca → Inter aorto caval LN

Ⓛ Testicular Ca → Ⓛ Para aortic LN

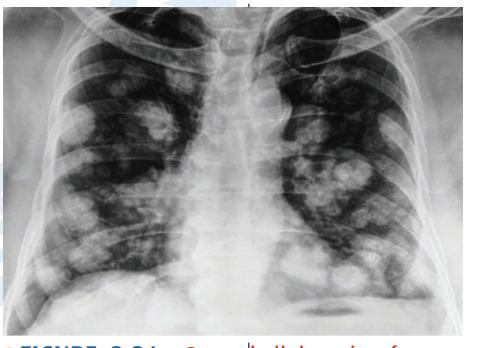


FIGURE- 3.24 Canon ball deposits of metastasis in Lungs in a patient of Testicular cancer

- Cl. app : Transillumination - Ⓝ
- Trans scrotal FNAC / Biopsy - Contraindicated

↓

US scrotum → Yolk sac tumor

↓

T. Markers - α FP ✓ (Alpha Fetoprotein)

Chorioca ← HCG ✓ (Human chorionic gonadotrophin)

Seminoma ← LDH ✓ (Lactate dehydrogenase)

Seminoma ← Placental Alkaline PO4

↓

CECT abd - 10C

Active space

Staging :

I - Testis

II - L.N. below the diaphragm

a - 2 cm

b - > 2-5 cm

c - > 5 cm

III - L.N. above the diaphragm

IV - Metastsis

A 28-year-old male presents with a painless swelling in the right testis, which he noticed two months ago. On examination, the swelling is firm and non-tender. There is no transillumination, and the contralateral testis appears normal. An ultrasound confirms a hypoechoic mass in the right testis. Additionally, imaging reveals enlargement of the para-aortic and pre-aortic lymph nodes. Tumor markers, including AFP, β-hCG , and LDH, are elevated. The physician suspects testicular cancer. What is the next appropriate step in management?

a. Biopsy of the testicular mass.
b. Biopsy of the pre-aortic lymph nodes.
c. Orchidectomy of the affected testis.
d. High inguinal orchidectomy of the affected testis.

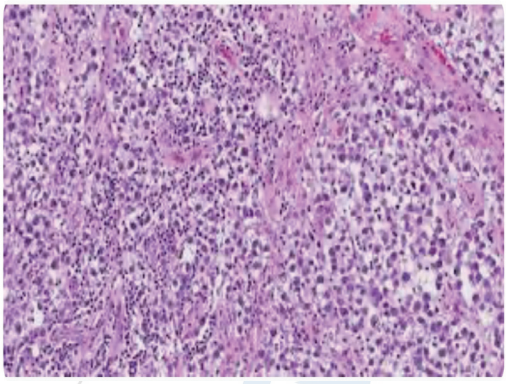


FIGURE- 3.25 High Inguinal Orchidectomy specimen for Testicular Tumour

↓

High Inguinal Orchidectomy **(Q Jan 25)**

- Rx - Cth - BEP - Bleomycin, Etoposide, Cisplatin
- Seminoma - Radiosensitive

(Q July 24) - Prognosis of Seminoma is better than Teratoma

ENDOCRINE CANCERS- THYORID CANCER

- MC - Differentiated Thyroid Cancer
  - Papillary - MC
  - Follicular = 2nd MC
- Medullary Thy Ca (Para follicular c-cells)
- Mets to Thyroid from Kidney, Breast Ca

Active space

	Papillary	Follicular
Incidence	Common	Less Common than Papillary
Age	< 40 - 50 y	> 50 y
MCC Q	● Radiation to neck	Long standing endemic goitre
● Patho Q	● Thyroglossal cyst	Encapsulated
Q	● Orphan-Annie eyed nuclei	FNAC Cannot distinguish Follicular odenoma from Follicular Ca as FNAC
	● Psammoma bodies	Cannot comment on Capsular & Vascular invasion
● C/F	Thyroid Swelling	
● Spread	Mainly Lymphatics	Mainly blood
● T. marker	Serum Thyroglobulin	
● Rx	total thyroidectomy c̄ removal of involved lymph nodes	
● Prognosis	Better	Poor

Psammoma Bodies (PSM)

Papillary Renal cell Ca / Papillary Thyroid Ca / Pituitany adenomas

Serous cystadennoma of ovary / Salivary adenomas / Meningiomas

Q Jan 25

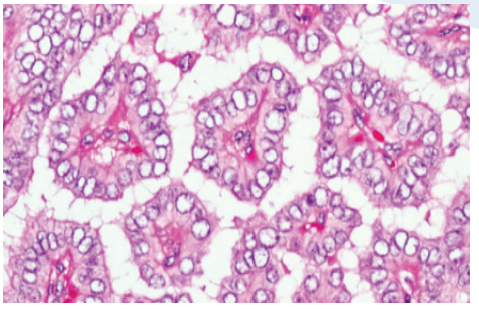


FIGURE- 3.26 Orphan empty eyed nuclei in Papillary Thyroid cancer

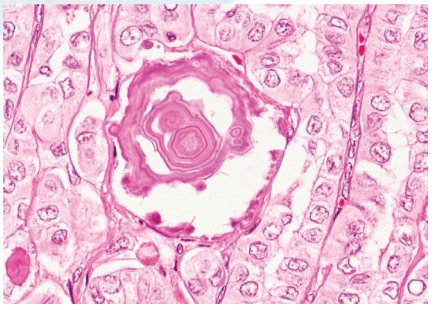
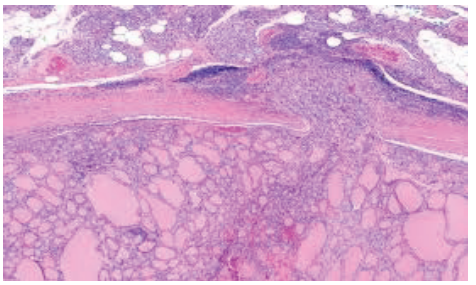


FIGURE- 3.27 Psammoma bodies in Papillary Thyroid cancer

Active space

A 60-year-old female presents with a solitary hard thyroid nodule. Fine-needle aspiration cytology (FNAC) reveals follicular cells with abundant intracellular eosinophilic cytoplasm. The nodule is non-tender and mobile, with no significant lymphadenopathy. Thyroid function tests are within normal limits. What is the most likely diagnosis?

a. Papillary thyroid carcinoma
b. Follicular thyroid carcinoma
c. Medullary thyroid carcinoma
d. Anaplastic thyroid carcinoma



Q Jan 25

FIGURE- 3.28 Invasion of capsule in Follicular Thyroid cancer

FIGURE- 3.29 After Total Thyroidectomy for Thyroid cancer

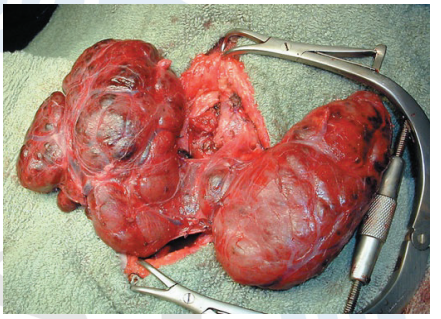


FIGURE- 3.30 Specimen of Total Thyroidectomy

Medullary Thyroid Cancer

- Arise from Parafollicular C-cells
- Not dependent on TSH ; S. thyroglobulin is not the Tumor marker
- Patho - Amyloid Stroma - **NEET PG 25**
- Aut. dominant
- Asso c̄ MEN II syn (Multiple Endocrine Neoplasia)
  RET Protooncogene **Q July 25**
- Neck Swelling, enlarged LN
- 25 - 30% - Diarrhoea (Serotonin)
- T. marker - S. Calcitonin ; 2nd - S. CEA
- Rx ; Total thyroidectomy c̄ removal of involved LN

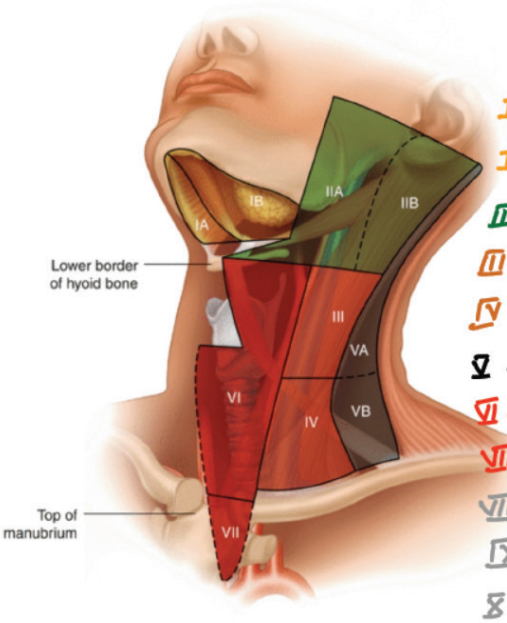
RND → I-V LN + Sternomastoid ms, IJV, Spinal accessory nv.

MND → I-V LN c̄ Preservation of any or all out of Sternomastoid ms / IJV / spinal accessory nv.

SND → MND but c̄ preservation of 1 or more LN groups out of I to V

Active space

Cervical Lymph Nodes



IA - Submental
IB - Submandibular
II - Upper Jugular
III - Middle "
IV - Lower "
V - Posterior triangle
VI - Pretracheal (Delphic)
VII - Mediastinal
VIII - Parotid
IX - Buccofacial
X - Retroauricular and occipital

Q Jan 25

NECK DISSECTIONS (ND) :

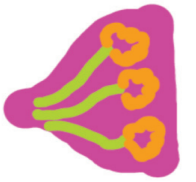
1. RND - Radical - Level I to I LN + IJV, Spinal accessory nerve, Sternomastoid muscle
2. MIND - Modified - Radical but preservation of any or all out of IJV, spinal accessory nerve Sternomastoid ms.
3 SND - Selective - Modified but preservation of one or more LN groups out of I to V

Variants - a. Anterior ND - Removal of VI
- B. Supra Omohyoid ND Removal of I, II, III
- C. Lateral ND Removal of II, III, IV
- D. Posterolateral ND Removal of II, III, IV, V

4 ERND - Extended Radical-Radical ND + Removal of more LN groups (I to V+more) and /or more structures like corotid artery / hypoglossal nerve / vagus nerve etc.

Active space

ENDOCRINE CANCERS- BREAST CANCER- AETIOPATHOLOGY AND CLINICAL PRESENTATION



- MC - Invasive Ductal Ca
- MC Ca in females QQ
- MC - Adenoca

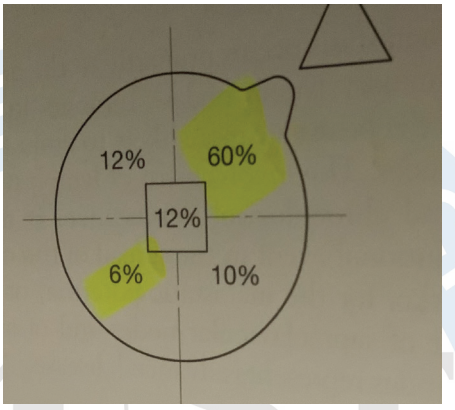


FIGURE- 3.31 Sites of Breast cancer- MC- Upper outer quadrant while least common site is Lower inner quadrant

- MC site - Upper Outer Quadrant (LC-Lower, Inner Quadrant)
- **RF :**
  1. Familial-Maternal, Paternal
  2. Genetic-BRCA 1 } Both BRCA 1 & BRCA 2 for female Breast Ca
     BRCA 2 }
     Only BRCA 2 for Male Breast Ca
  3. Hormonal-Estrogen exposure
- Early Menarche, Late Menopause
- Nulliparous
- Post menopausal obese ♀
- Hormone Replacement Therapy
- OCP's - high estrogen content OCP for a long period
- Less Phytoestrogens in diet (Beans/Pulses)
- Protective - Late Menarche, Early Menopause / Multipara / Prolonged breast feeding / Having 1st child at an early age

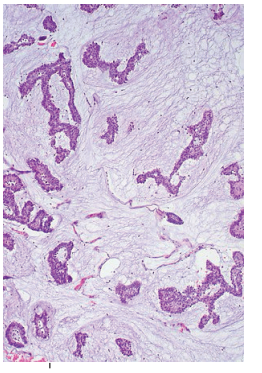
Active space

- Patho :

Duct (90%) → In situ (DCIS) / Invasive ductal Ca MC

Lobular (10%) →

- Bilateral, Multifocal
- e - cadherin Ab Negative



Mucinous breast cancer

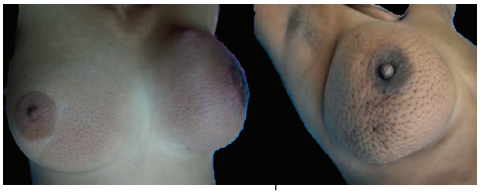
Q Jan 25

Note : LCIS - Lesion c̄ ↑ risk of cancer

→ Signet ring cell breast Cancer -

- Signet rings can be seen in both Lobular and ductal breast cancer.
- Poor Prognosis
- Bloom Richardson Grading :
  based on PMT - Pleomorphism
  Mitotic Rate
  Tubule formation
- Well diff / Mod. diff. / Poorly diff.
- In ductal breast Ca - NST - No Special Type
  Colloid/Papillary/Tubular - Good Prognosis
  Medullary
  Inflammatory - Worst Prognosis
- C/F : MC - Lump - UOQ - hard, irregular, ± Fixed ± LN inv.
- Dimpling or Indrawing of skin does not mean skin inv by breast Ca
  d/t → Involvement of Suspensory Ligaments
  (Astley/cooper's Ligaments) It is not T4b Q July 25

Indrawing (Dimpling) → One Cooper's Ligament is involved
Puckering or Tethering of skin → Multiple cooper's Ligament involved



- Circumferential nipple retraction
- Blood discharge from nipple
- Skin involvement by Breast Ca -

- Peau de orange (Orange peel app) ← NEET PG 25 IBQ

Q.Q.QQ d/t blockade of Superficial, Subdermal Lymphatics of skin

T4b - ≤ 1/3 surface area of br.
T4d - Redness or Inflammation c̄ > 1/3 surface area of breast

- Satellite nodules,
- Skin ulceration also suggest Skin inv
- Lymphoedema of arm - d/t blockade of Axillary Lymphnodes.
  - Can complicate to Lymphangiosarcoma
  → Bluish, Purplish subcutaneous nodules over the forearm. Steve Treves Syndrome Q

FIGURE- 3.32 Lymphoedema after MRM for Breast cancer

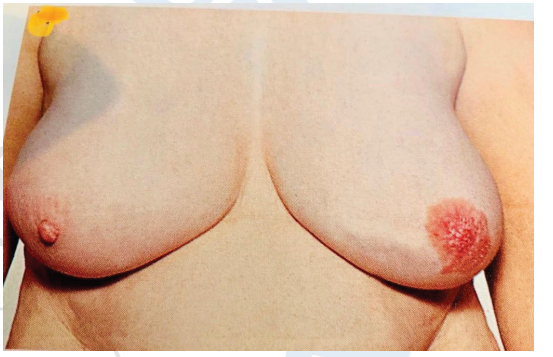


FIGURE- 3.33 Paget's disease in Breast cancer- Nipple is eroded by cancer cells due to intraductal spread.

- Spread :
  1. Local - Skin / Pectoralis ms, Ribs, Intercostal ms
     Inv of Ribs, intercostal ms → Inv of Chest wall
  2. Lymphatics - Axillary LN
     Internal mammary LN
  3. Blood - MC - Bones - Lumbar vertebra
     through Bateson's vertebral plexus of veins QQ
     - Liver, Lungs, Brain

P B K B T

Active space

ENDOCRINE CANCERS- BREAST CANCER- INVESTIGATIONS AND STAGING

- Cl. app : Triple Assessment
  - C Clinical - history, examⁿ
  - R Radiological - US (<30-40Y); Mammography (>30-40y)
  - P Patho - Biopsy - Needle (IOC)
    FNAC
- Mammo - Irregular, spiculated microcalification

Q July 25

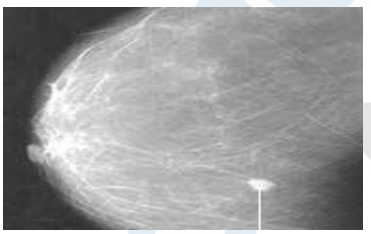


FIGURE- 3.34 Mammographic lesion in Breast cancer

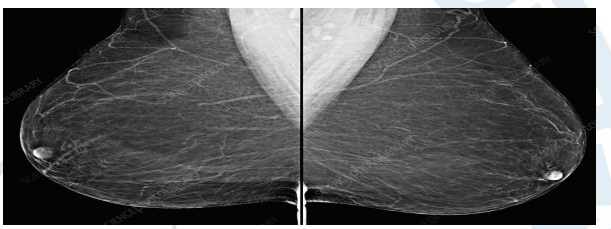


FIGURE- 3.35 Normal Mammography in less dense Breast

FIGURE- 3.36 Mammographic lesion in Breast cancer

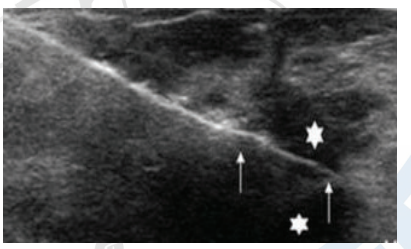


FIGURE- 3.37 Mammographic lesion in Breast cancer

- Staging IOC - MRI
- IOC distant mets - PET Scan

- TNM Staging :
- T_1 → ≤ 2 cm
- T_2 > 2 - 5 cm

T_1micro = ≤ 1 mm
T_{1a} = > 1 - 5 mm
T_{1b} = > 5 - 10 mm
T_{1c} = > 1 - 2 cm

Active space

- T_3 -> 5 cm
- T_4 - Any size - if inv of chest wall - T_{4a}
  Skin - T_{4b}
  both - T_{4c}
  Inflammatory - T_{4d}
- N_1 - Ipsilateral, mobile, Axillary LN
- N_2a - " " , fixed, " "
- N_2b - " " , internal mammary LN
- N_3a - " " , Apical + Axillary LN
- N_3b - " " , Axillary + Internal Mammary LN
- N_3c - " " , Supraclavicular LN
- Bilateral or Contralateral LN inv → Metastasis
- M1 - Distant mets

Stage	TNM
Stage 0	Tis, N0, M0
Stage IA	T1, N0, M0
Stage IB	T0, N1mi, M0
	T1, N1mi, M0
Stage IIA	T0, N1, M0
	T1, N1, M0
	T2, N0, M0
Stage IIB	T2, N1, M0
	T3, N0, M0
Stage IIIA	T0, N2, M0
	T1, N2, M0
	T2, N2, M0
	T3, N1, M0
	T3, N2, M0
Stage IIIB	T4, N0, M0
	T4, N1, M0
	T4, N2, M0
Stage IIIC	Any T, N3, M0
Stage IV	Any T, Any N, M1

ENDOCRINE CANCERS- BREAST CANCER- TREATMENT

- Rx : Sx - Breast Conservative Sx (WLE - Wide local excision)
  Simple Mastectomy
  Modified Radical Mastectomy - MC
- CTh - ACT - Adriamycin, Cyclophosphamide, Taxanes
  → CMF - Cyclophasphamide, Methotrexate, 5 FU
- RTh
- HTh - If hormone receptors- ER, PR are positive
- Premenopausal ♀ - Tamoxifen 20 mg OD for 5-10 yrs
  Postmenopausal ♀- Aromatase Inhibitors (Anastrozdes)

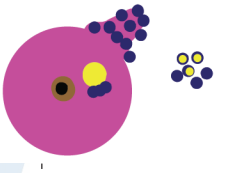
Active space

- Targeted therapy - If Her 2 neu gene positive →
  Trastuzumab (herceptin)/Pertuzumab

NAST = Neoadjuvant Systemic therapy Cth/Rth/Hth/Tth

NACT = Neoadjuvant Cth

- BCS - Breast Conservation Sx - WLE
  Removal of Tumour c̄ 1 cm margin, + Adjuvant Rth
- Restoration of architecture of breast by volume displacement or → volume replacement
- Simple Mastectomy - Rem of breast
- MRM - Rem of breast + LN
  (Modified Radical Mastectomy)
- Sentinel LN Biopsy - For LN not inv in Cancer - preoperatively
- Dye injected around Tumor-Peritumoral/Around areola Subareolar



Dye used in SLN Bx

1. Methylene bluet Radioisotope technetium - 99 labelled Albumin / Sulphur colloid or
2. Fluoroscein or Indocyanine Green (No Radio isotope)

- If SLN is positive for Ca
  → Removal of LN by separate incision
- After mastectomy - breast reconsⁿ =

- MC - Silicongel
- Ms flaps - TRAM - Transverse Rectus Abdominis ms
  LD - Latissimus dorsi ms

DIEP - Deep
Infr Epigastric
Perforator flap

FIGURE- 3.38 Some females are happy with reconstruction after Mastectomy- MC- Silicon gel implant for reconstruction

FIGURE- 3.39 Some females are happy without reconstruction after Mastectomy

- Early breast Cancer: T_1 / T_2 ; No/N_1 ; M_o

  BCS/MRM ± Rth ± Hth ± Tth ± Cth

- Locally Advanced breast Cancer : T_3 / T_4 ; N_2/N_3 ; M_o

  Neoadjuvant Cth → MRM →
  Completion Cth ± Rth ± Hth ± Tth

- Metastatic breast Cancer ; Any T, Any N, but M1

  Curative Rx not possible / only Palliative Rx
  (Simple mastectomy)

Active space

- Prognosis - Nottingham Prognostic Index

QQ depends on • Stage (TNM) • LN status

Molecular classi :	ER	PR	Her 2 neu	Ki 67
QQ Luminal A -	⊕	⊕	⊖	Low
Luminal B	⊕	⊕	⊖	High
Her 2 neu enriched-	⊖	⊖	⊕	High
Basal (Triple Negative)	⊖	⊖	⊖	High
Claudin Low	⊖	⊖	⊖	Claudin Low

If ER/PR ⊕ → Good

If Her 2neugene ⊕ → Bad

If Ki 67 ↑ → Bad

→ Best Prognosis

If Claudin Low → Bad

→ Poor Prognosis

QQ • Luminal A - has the best prognosis Q Q Q Q

Indications for Genetic Risk Evaluation -

1. Breast Cancer ≤ 50 yrs
2. Triple Negative Breast Cancer ≤ 60 yrs
3. Two breast cancer primaries
4. Male Breast cancer
5. Breast cancer at any age but if -
   a) 2 or more relatives affected c̄ Breast cancer
   a) 1 or more relatives affected c̄ Breast cancer at ≤ 50 yrs of age / invasive ovarian / pancreatic / prostate / male breast cancer

Not for Postmenopausal breast cancer

Active space

Workbook Questions

1. 75-year-old male presented to the OPD with complaint of difficulty in swallowing initially for solids and later for liquids as well. There is history of significant weight loss. What is the best diagnostic confirmatory investigation in this patient?

   A. Esophageal Endoscopy
   B. Manometry
   C. CECT
   D. Barium study

2. Which of the following isotope is used in the treatment of Prostate cancer?

   A. I131
   B. I125
   C. I123
   D. I121

3. 60 yr old male is diagnosed with testicular cancer. Which of the following is true about spread of testicular tumor?

   A. Involvement of the obturator lymph nodes
   B. Involvement of para aortic lymph nodes
   C. Involvement of inguinal lymph nodes
   D. Involvement of precaval lymph nodes

4. Peau de orange appearance in breast cancer is due to blockade of which of the following?

   A. Superficial lymphatic channel
   B. Deep lymphatic channel
   C. Superficial veins
   D. Deep veins

Active space

Answers: 1. (A), 2. (B), 3. (B), 4. (A)

SURGICAL TRAUMA

PRINCIPLES OF TRAUMA SURGERY

- Disaster Management: NDMA - National Disaster Managment Authority - New Delhi
- 1st S.T.A.R.T. — Simple Triage and Rapid Rx

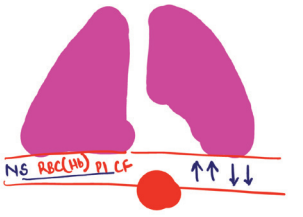
  Paramedical Staff

  Disaster Site → First Aid → Transfer

- 2nd E.S.I. — Emergency Severity Index

  Doctors

  Hospital → Save Life ; Triage (segregation of patients into different categories)



PRIMARY AND SECONDARY ASSESSMENT

- Primary Assessment —

  A B C D E Airway, Breathing, Circulation, Disability and Exposure

  ↓

- Secondary Survey → History, Head to Toe evaluation, Special Inv, Definitive Rx

Primary Assessment —

- Airway — stabilisation of cervical spine

  clear debris, mud, dirt Foreign bodies, Any fractures

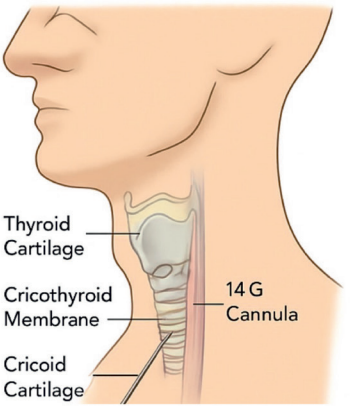
  - LMA, Laryngeal mask airway - Endotracheal Tubes
  - Nasal Intubation is C.I. in Antr cranial fossa fractures

Active space

- Tracheostomy, Cricothyroidotomy, Needle Cricothyroidotomy (Wide bore needle)

Needle Cricothyroidotomy

A procedure using a large bore needle to create a temporary emergency airway when endotracheaal intubation fails or is not possible.



Indications
- Children < 12 years
- „Can't intubate – can't ventilate" scenario

Contraindications
- Age >12 (prefer surgical cricothyroidotomy)
- Laryngeal or cricoid injury

- Breathing: 14 Gauze
  - Pleural Collection, (Pneumo/Hemothx) Drain collection to achieve Lung expansion
  - Ventilatory Support for Resp ms Paralysis
- Circulation:
  - Tachycardia - Earliest sign of hypovolemia
  - Cutaneous Vasoconstriction - Early sign of hypovolemia
  - Hypotension (Fall in SBP) - Late sign of hypovol. (> 30% blood loss)
- IV Fluids - Initial fluid of choice
  - Isotonic Crystalloids - Normal saline / Ringer lactate
- Fluid Challenge - 1L NS c̄ in 1 hr (through 2 wide bore IV canula)
- 14 Gauze - Orange - Wide - Shock
- 16 " - Grey - " - "

Active space

- 18 " - Green - Normal - Adults - ♂
- 20 " - Pink - " - " - ♀
- 22 " - Blue - " - Children
- 24 " - Yellow - " - Infants (0-1 yr)
- 26 " - Violet - " - Neonate (≤28days of birth)

FIGURE- 4.2 IV Canula Colour codes

- Monitor Fluid Resus - Urine Output

- UOP = > 0.5 ml / kg / hr = Adult
  (Min) > 1 ml " " " = Children
  Minimum Uop > 2 ml " " " = Infant

- If > 30% blood loss= Blood Transfusion RBC's - O Negative
  Platelets, Fresh Frozen Plasma
  (Clotting factors)
  [Leukodepleted Blood]

- Injn of Trenexaemic acid → 1st dose c̄ in 10mins (3hrs), 2nd dose with in 8 hrs
- If needed - Immediate Surgical Exploration to control bleeding
  Best method to control bleeding = Direct Pressure
- Shock Index = ... $\frac{HR}{SBP}$ ..... (≥ 1) = Hemodynamically unstable
- If Hemodynamically stable → Investigations & Proceed (Non operative)
- If hemodynamically unstable → Save Life (Basic resuscitation ABCDE)
  ↓
  Immediate Surgical Exploration & Direct Pressure to stop the bleed
- Disability - Loss of consciousness - Glasgow Coma Scale
- Exposure - Undress taking care of Hypothermia, Cover c̄ blankets, Warm IVF

Glasgow Coma Scale

	Max	Min	
E Eye	4.	1	MILD = 13-15
V Verbal	5.	1	MOD = 9-12
M Motor	6.	1	SEV = ≤ 8 (Coma)
	15	3	Dead = 3

- Eye = Spont 4 Voice 3 Press 2 None 1
- Verbal = Conscious 5 Confused 4 Words 3 Sound 2 None 1
- Motor = Obeys 6 Localise 5 Normal Flexion 4 (Withdrawl)
  Abnormal 3 flexion Extension 2 None 1

If non testable = nt

Most useful for Prognosis in GCS = Motor response

If variable response in 2 sides => Better response is considered

A patient was brought to the hospital after road traffic accident. He was conscious but confused and was able to open his eyes on voice command. On clinical examination, he was able to localize the stimulus. What can be the most likely Glasgow coma scale in this patient?

a. 6
b. 8
c. 12
d. 15

- **Triage**

  RED - Immediate intervention reqd
  YELLOW - Early intervention reqd
  GREEN - Ambulatory patient, can wait
  BLUE - Moribund
  BLACK - Dead

- Secondary Survey -

  Detailed H/O -

  A - Allergies

  M - Medication history .....

  P - Past Illness/Pregnancy

  L - Last Meal

  E - Events of Injury

  Stop

  Aspirin → Platelets

  Warfarin → Vitamin k, FFP **Q July 24**

  QQ

  Heparin → Protamine sulphate

- Head to Toe Examⁿ - Evaluation
- Special Invⁿ - X-Rays, CT Scan, contrast studies Endoscopies
- Definitive Rx.

Shock :

- Inadequate organ perfusion & Tissue oxygenation ↓ O_2 Supply to tissues hypoperfusion of tissues
- Cardiogenic - ↓ contractility - Direct Trauma to heart
- Hypovolemic Shock -
- Distributive - Warm shock - d/t Infections / Neurogenic
  Trauma to Brainstem, Cervical/Thoracic spine
  Loss of Sympathetic tone
- Obstructive - Cardiac tamponade
- Endocrine -

Active space

HEMORRHAGE AND BLOOD TRANSFUSION

Haemorrhage

- Measurement of blood loss :
- Blood clot in a clenched first = 500 ml
- Closed # - Humerus/Tibia = 750 ml
  Femur = 1500 ml
  Pelvis = 2000 ml
- Categories of Blood Loss :

I = 0-15% - 0-750 ml
II = 15-30% } - 750 - 1500 ml
III = 30-40% } - 1500 - 2000 ml
IV = >40% - > 2000 ml

	I	II	III	IV
Pulse	<100	100-120	120-140	>140
RR	<20	20-30	30-40	>35
SBP	N	N	↓	↓↓
UOP	N	N	↓	↓↓ NIL
GCS	N	N	↓	↓↓

Blood Transfusion

- Storage Temp of blood = 4 ± 2°C (2-6°C)
- SAG - M Blood = Sod. chloride, Adenine, Glucose, Mannitol
  Shelf life 5 weeks
- Complications of BT :

1. Transfusion reactions -

ABO Incompatibility - Mismatched BT

MCC - Clerical error

Ag - Ab reacⁿ → Large clumps → Renal Failure

MC Symp → Pain & heat sensation along the vein

MC Sign → Oliguria > Hemoglobinuria

MC in anaesthetized patient → Generalized oozing of blood

Rx - Diuretics, Dialysis

2. Infections - Hep B, Hep C, HIV, Malaria, Syphilis

Active space

- Massive BT

  > 10 Units of packed RBC'S $\bar{c}$ in 24 hrs

  or > 4 Units of packed RBC'S $\bar{c}$ in few hrs MIST

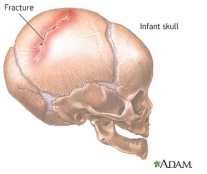
- Complic^{n} of Massive BT:

  1. CCF - volume overload - Anemics, Elderly
  2. Hyperkalemia - d/t lysis of RBC's in stored blood
  3. Hypokalemia Hypocalcemia, Hypomagnesemia **(Q July 25)**
  4. Hypothermia
  5. Thrombocytopenia
  6. DIC - Disseminated Intravascular Coagulopathy dilutional coagulopathy - Rx - FFP, Platelets - (FFP = Stored at (-40^{0} C) rich in clotting factors Shelf Life = 2yrs)
  7. Metabolic Acidosis - Stored blood pH 6.3 - acidic

HEAD TRAUMA PART 1

Scalp Bone Brain

- CSF, SKull - Protect brain
- **Scalp** : Most dangerous zone - Loose areolar tissue (Emissiary veins)



- **Bone** : #
  - Linear - Blunt Trauma
  - Depressed - Penetrating Trauma → Hammer

Active space

FIGURE- 4.2 Depressed Skull Fracture

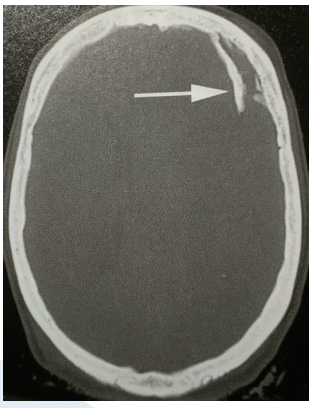


FIGURE- 4.3 Depressed Skull Fracture- NCCT Head

- Base of Skull - Antr, Middle, Postr Cranial Fossa #
- ACF# - MC Part → Cribriform plate (Ethmoid bone) CSF Rhinorrhoea

B2 tranferrin

CSF

Periorbital hematoma- Raccoon eyes or Panda's sign

- MCF# - MC Part → Petrous part of temporal bone. CSF Otorrhoea

Battle's Sign → Bruising/brownish discolouration of Mastoid

- Pond's# - children
- Orbital Blow out # - MC → Floor (Infr wall) of orbit

  2nd MC - Medial wall of orbit

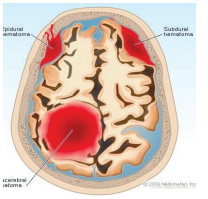
- **Brain :**

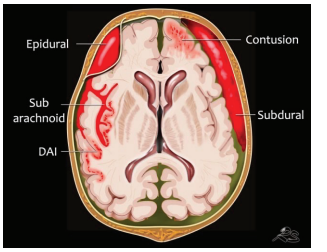
  Coup Injury - At site of impact

  Counter coup Injury - Remote from site of impact, head is in motion

Active space

- Brain hematomas
  1. Intracerebral hematoma
  2. Extra dural hematoma
  3. Sub dural hematoma
  4. Sub arachnoid hemorrhage
  5. Cerebral herniation





- EDH - Epidural h. - Extradural hematoma
  blood outside duramater, beneath periosteum
- MC Site - Temporoparietal region
- MCC
  - Injury to middle Meningeal artery - branch of Maxillary artery (Site → Pterion)
  - Transverse/Sagittal Sinuses
- Lucid Interval - Transient period of consciousness after trauma
  - can carry out normal activities before losing consciousness
- IOC- CT Scan - hyperdense. Biconvex (Lenticular)
  NCCT

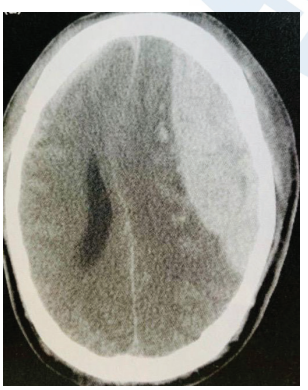


FIGURE- 4.4 Convex Hyperdense- EDH

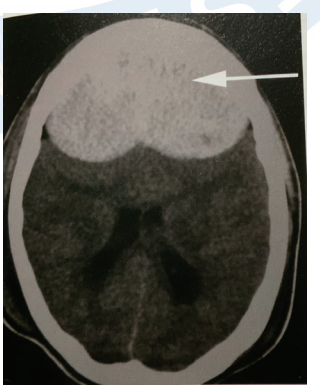


FIGURE- 4.5 Convex Hyperdense- EDH

Active space

- SDH - Boxer's hematoma
  blood beneath dura
  Injury to cortical bridging emissiary Veins
- CT - hyperdense, Concavo-convex QQ
  Acute - 0-3 days (Crescent)
  Subacute - 4-21 days
  Chronic - > 21 days

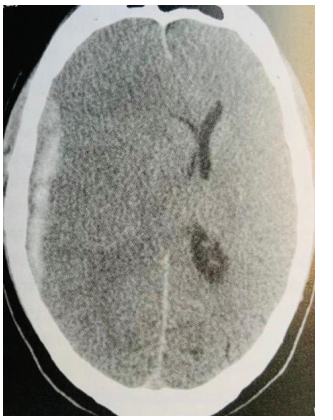


FIGURE- 1.12 Concave Crescent Hyperdense- SDH

Diffuse Axonal Injury

- Etiology : Trauma → Rapid acceleration / deceleration
  : Shearing of axons in white matter from strain on brain tissues
- Areas affected :
  - Corpus callosum
  - Gray-white matter junction
  - Dorsolateral pons
- Exam : Loss of consciousness (Prolonged in severe cases), persistent coma
- CT : Scattered punctate hemorrhages, blurring of the gray-white interface
- Treatment : Supportive

HEAD TRAUMA PART 2

- Management of H.I.
- Aim - ↓ Intracranial Pressure (ICP);
  ↑ Cerebral Perfusion Pressure (CPP)

Active space

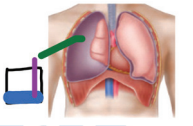
- (N) ICP = 10 mm Hg ; > 22 = dangerous
  Aim = ↓ 20 mm Hg
  CPP = Mean Arterial Pressure — ICP
  Aim = ↑ > 60 mm Hg
- Rx : • Diuretics **Mannitol (Osmotic Diuretics)**
  Should be used c̄ caution in intracranial bleeds

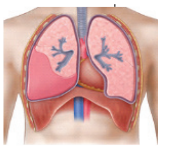
  Sx → Burr holes, Decompressive Craniotomy / Lumbar Puncture is avoided QQ

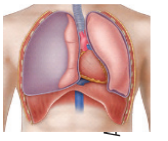
  • IVF = Isotonic Crystalloids = NS, RL
  = Hypertonic Saline
  Dextrose (Glucose) based solutions avoided

CHEST TRAUMA

- MC - Rib # - (L) > (R) - 4th - 9th Ribs
- Pneumothorax
- Hemo "
- Hemopneumo "
- Tension pneumothx
- Sucking wound (Open Pneumothx)
- Flail chest





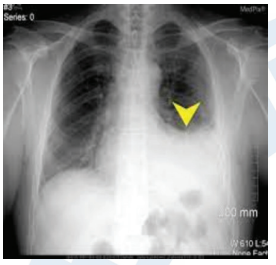


Pneumo/hemothx -

- Dyspnoea - Mild difficulty in breathing
- Percussion - Mid finger on Mid finger, wrist joint
  Resonant (Tympanic) → Air (Pneumothorax)
  Dull → Fluid (Hemothorax)

- Auscultation - Breath sounds diminished
- Chest X-Ray - Hyperlucency (Black) - Pneumothx
  (1st Invⁿ) - White, haziness, blunting of Costophrenic angle in Hemothx

- IOC - CT Scan
- Rx - Aim - (N) breathing and Adequate Lung Expansion

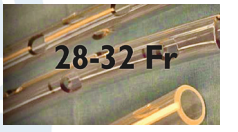


Drain Collection + Breathing exercises

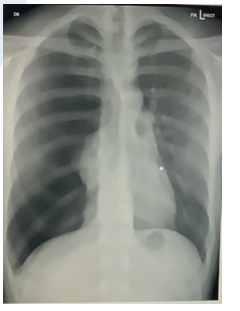
↓ ↓

Intercostal Tube drain
4th/5th Intercostal Space →
Just antr to mid axillary line
in Δ of Safety →
QQ along the upper border of lower rib
→ Connected to underwater Seal bag
Size of catheter = 28-32 French

- Spirometer
- Football bladder (Lung Expansion)



- Indications of Thoracotomy -
  1. > 1500 ml of blood at one instance in first hour - (> 20 ml/kg in 1 hour)
  2. > 200 ml/hour for more than 2-4 hours - (> 3 ml/kg/hour)
  3. Major Tracheobronchial, Eso ruptures



- **Tension Pneumothx - *Q Jan 25***

Real Sx Emergency → Severe Resp. distress /Massive Air

Collecⁿ → Ipsilateral Lung collapse

→ Mediastinal Shift/Opposite lung under pressure

Asso c̄ Engorged, dilated neck veins, Hypotension, breath sounds diminished

Active space

QQ Immediate Needle thoracocentesis - 14 G needle QQ ***QJuly25***

2nd ICS, Mid clavicular line

There after ICTD + Breathing Exercises

- DD c̄ Cardiac Tamponade -

  Pericardial fluid collection / Engorged, dilated neck veins / Hypotension / Muffled heart sounds - BECK'S TRIAD

  Immediate - Open Sx

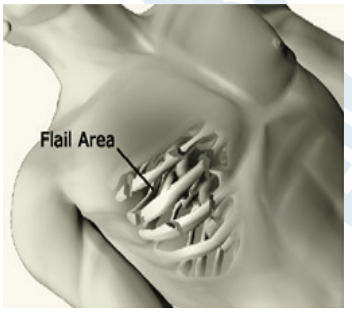
- **Sucking wound - Open Pneumothx -**

  Communicating wound b/w Pleural cavity & outside atmosphere > 3cm size

  Rx -Immediate Rx → Valvular dressing c̄ taping on 3 sides

  Thereafter ICTD c̄ Breathing exercises

- Flail Chest :



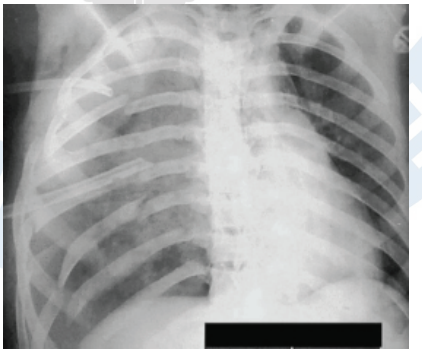


FIGURE- 4.9 FLAIL CHEST

≥ 3 Consecutive Rib # at ≥ 2 sites

Flail segment shows paradoxical respiration

Rx : Sx Emergency, Intermittent positive Pressure Ventilation, Sx

Active space

ABDOMINAL TRAUMA

- Blunt - MC - .....; 2nd.....
- Penetrating - MC - .....; 2nd.....
- Seat Belt Abd Trauma - Whiplash Injury - Mesentery/Intestinal rupture
  (Gut Ischemia)
- Blast Injury -
  MC - Air filled cavities - Ear drum, Trachea, Lungs, Intestine

* Blunt Abd Trauma

↓

Hemo.

Stable	Unstable
↓	↓
Admission for observation	Save the life-Basic Resus A B C D E
↓	↓
1st - e-FAST Best - CECT	e- FAST
↓	↓
Rx-Conservative (Non operative)	Immediate Laparotomy by Midline Incision

- Sequence in Damage control Sx :
  1. Patient selection
  2. Control of bleeding, Prevent infection
  3. ICU Resuscitation → Stable
  4. Definitive Sx

Active space

- Balanced hypotension: Try not to increase too much pressure in the pipeline as it may not allow the bleeding points to seal better
  Target → 90 - 100mm of Hg

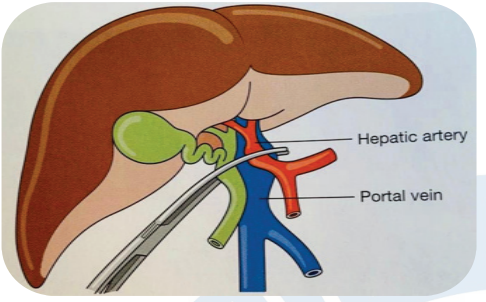


FIGURE- 1.20 Pringle manouver- clamp in hepatoduodenal ligament to temporarily block Portal vein and Hepatic artery to control Liver bleeds

- FAST - Focussed Assessment by Sonography in Trauma
  4 Areas
  - Epigastrium - Pericardium
  - RHC - Liver / (R) Kidney
  - LHC - Spleen / (L) Kidney
  - Pelvis - Free Fluid in Peritoneal cavity
- e-FAST Now → Extended Fast -FAST+Thoracic evaluation for Pleural collecⁿ -

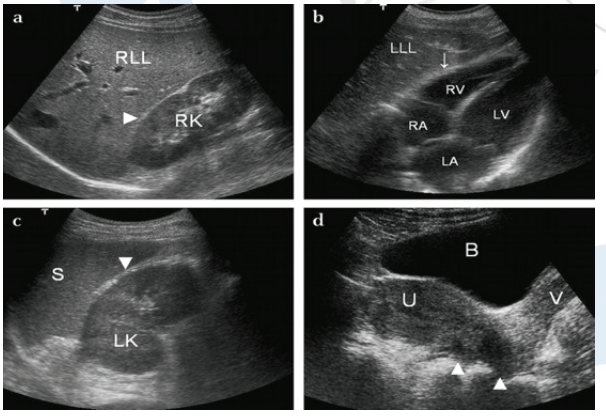
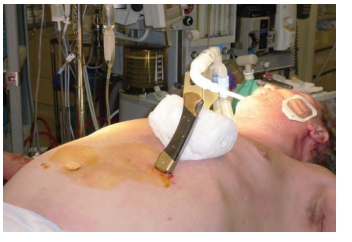


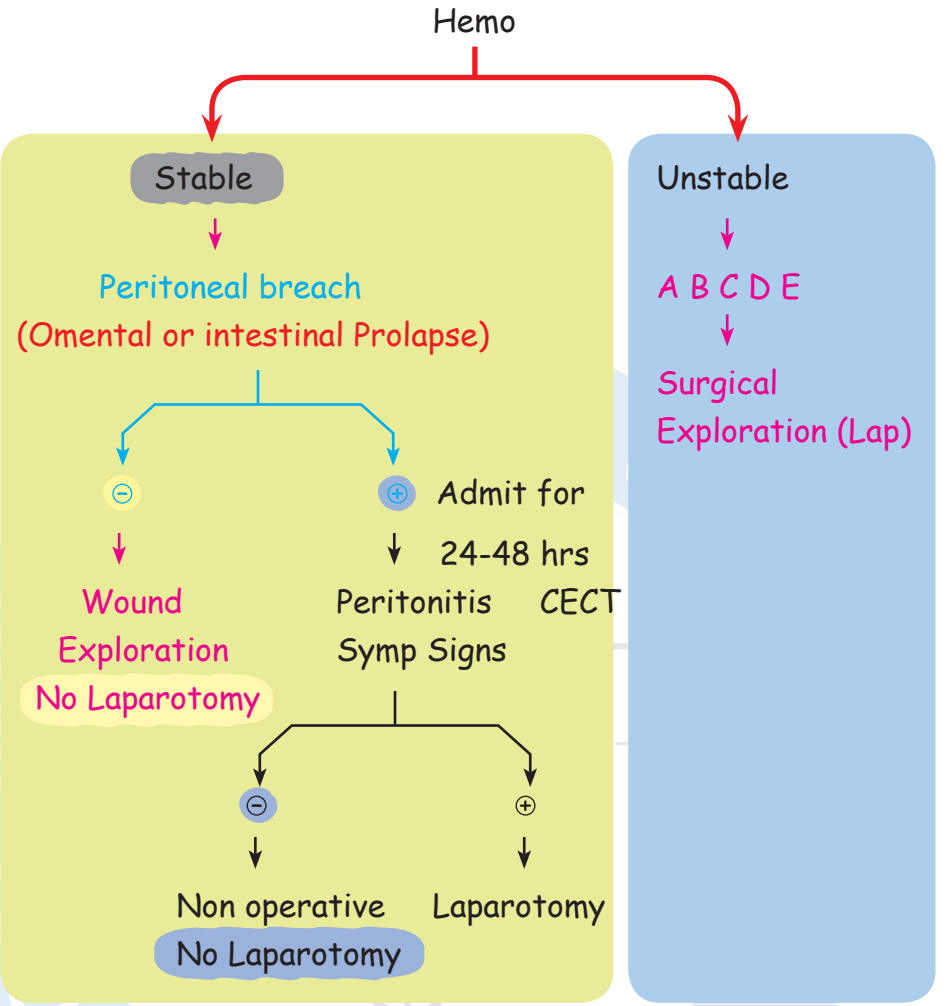
FIGURE- 1.21 FAST- Focussed assessment by Sonography in Trauma

DPL - Diagnostic Peritoneal Lavage Sensitive but invasive - Obsolete

✱ Penetrating Abd Trauma
FB should be removed under complete control in O.T.



Active space



✱ Retroperitoneal Trauma

I - Central zone
(Aorta, IVC)

II - Lateral zone
(Kidney)

III - Pelvic Retroperitoneum
(Pelvic organs)

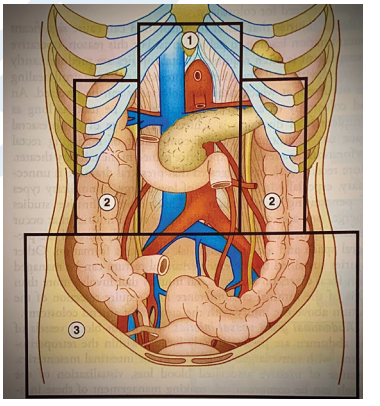


FIGURE- 4.10 Retroperitoneal Trauma Zones- I, II, III

- Zone I - Always Require Surgical Exploration even if hemodynamically stable
- Zone II, III - Require Surgical Exploration only if hemodynamically unstable

Active space

THERMAL TRAUMA- BURNS

Thermal Burns

- MC - Day to day - Scald (Hot Liquid)
- MC - Hospital - Flame Burns

- Pathophysio - Hypoxia, Hypovolemia, Sepsis
  MC Infecⁿ - Pseudomonas
  2nd Infecⁿ - Staph aureus

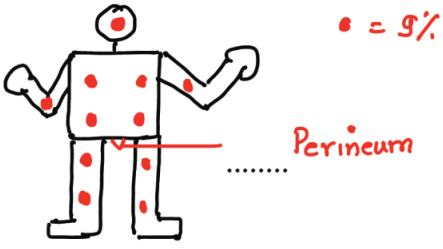
- Assessment of Severity - %, Depth
- Management ⊖ Prehospital
  - Hospital
    - Basic Resus
    - Definitive care

Assessment of Severity :

1. % - TBSA Total Body Surface Area Burn in %

- MC - Wallace Rule of 9

QQ



- Palm Rule = Palmar surface of hand = Palm + Fingers → .....

- In children - head & neck burns
  bear larger surface area as compared to adult

Active space

- 1 yr of age - Head & Neck Burn = 18% ; Reduce 1 % Every year
  So that at 10 years of age, it is 9%
- In Children - Lund & Browder charts

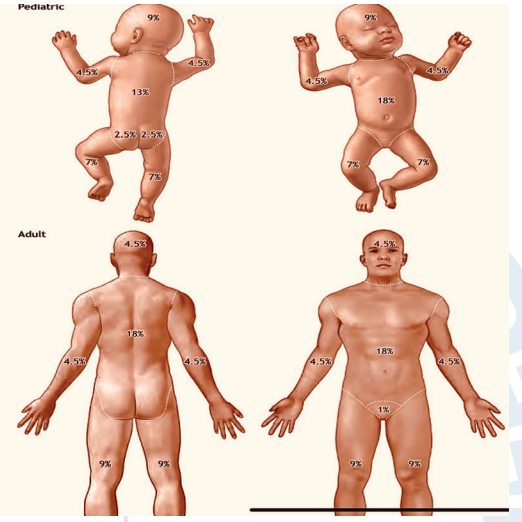


FIGURE- 1.23 TBSA- Total Body surface area Burn in %- Child and Adult

3 Depth of Burns -

depends on Temp of source & duration of contact

① - Epidermis - No Rx, Exposure Sunburns

② - Dermis

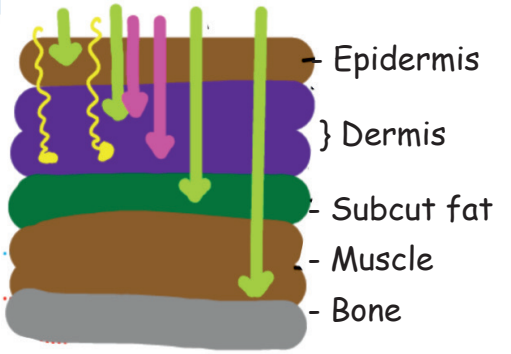
2° Sup - Sup dermis - Painful, Blisters

2° Deep - Deep dermis - Painless

Blister, Red, blotchy, velvety

③ - Subcut fat Muscle / Bone

- Painless, Black / White, Leathery



Active space

Management:

- Pre hospital - Ensure rescuer safety first, Stop, Drop, Cool,
  Drop c̄ burning surface down
  Cool - c̄ Running water at room temp.
  Ice X- hypothermia → Transfer

- Hospital :

→ 1st - Basic Resus - A B C D E

A = Early Endotracheal Intubation preferred in suspected airway burn

B = If Resp ms paralysed = Ventilator

C = Initial fluid of choice for basic resuscitation Isotonic crystalloids → Ringer Lactate, Hartman's solution, Normal Saline (Colloids should not be given in 1st 12 hrs)

Parkland's formula-..... × % × wt(kgs)

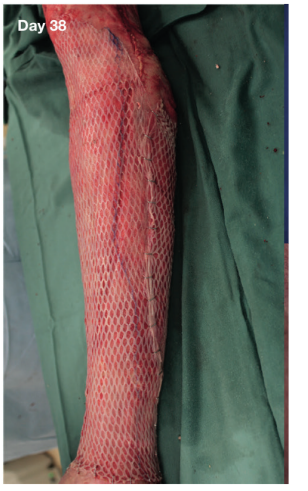
Q July 25

- Monitor fluid resus by Urine output - If less → ↑ Fluid infusion

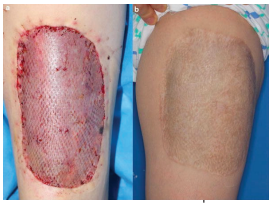
→ Definitive Rx :

1. Burn wound -

- 1° Burns (Sunburns) - No Rx, Exposure
  1° Burns are not included in calculating the %
- 2° - Dressing
  - Silver Sulphadiazine cream (1%) - MC
  - Mafenide acetate - Painful, Met. Acidosis
  - Silver Nitrate solution
- 3° - Debridement (Removal of dead skin) → Skin Grafting :
- Auto G - Same Person
- Iso G - Identical Twins



- Allo G - Same Species
- Xeno G - Different Species
- Instruments - Humby's Knife, Wolfe's blade



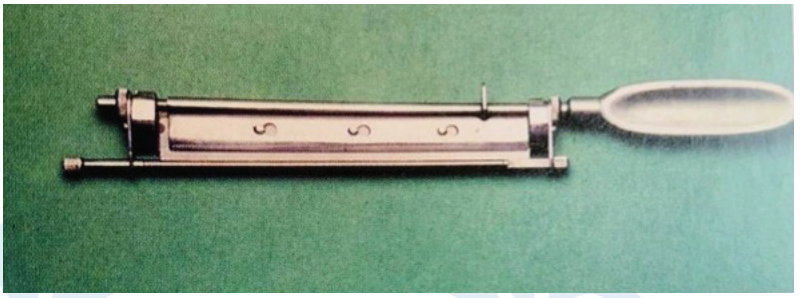


FIGURE- 1.24 Humby's Knife for Skin Grafting

- Dermatome QQ Q July 24

1. Partial Thickness SG (split SG/ Thiersch's G)
   - Composed of Epidermis + Sup. Dermis
   - MC donor site - Thigh (Anteromedial > Antero lateral)
   - Donor site Covered c̄ Dressing
   - Can be expanded to cover a large area
   - Graft contraction more - Cosmetic Outcome Poor

2. Full Thickness SG (Wolfe's G)
   - Composed of Epidermis + Complete Dermis
   - MC donor site - Postauricular / Supraclavicular area
   - Donor site covered by suturing
   - Can Cover small areas
   - Graft contraction less - Cosmetic outcome Good. (Face)

Active space

3. ● Composite Graft → Skin + Fat / Cartilage

eg Ala of nose eyebrow

- SG survive
  - 1st 48 hrs d/t Plasma Imbibition, of nutrients from underlying surface
  - 3rd day-Inosculation
  - By 5th day - SG settle at recipient site
- Contraindications For SG :

  1. Infection c̄ ß Hemolytic Streptococci

  2. Bare Bone / Cartilage / Tendon NEET PG 25

- Escharotomy - in Eschar acting as Tourniquet
- Fasciotomy in Compartment Syndrome (Electrical Burns)

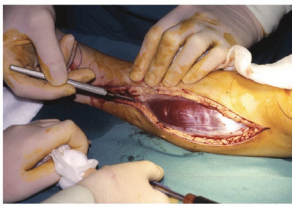


Figure 3.13 Fasciotomy of the leg.

FIGURE- 4.11 Escharotomy- Incising the whole length of limb in mid axial line down to the subcut fat

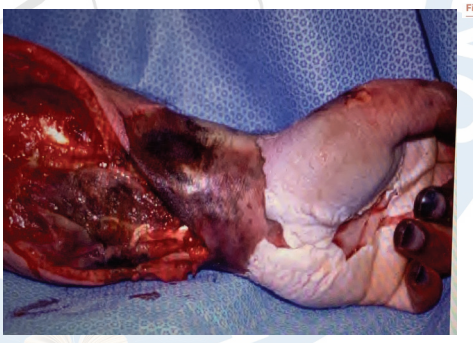


FIGURE- 4.12 Fasciotomy for Compartment syndrome in Electrical Burns- Incising whole length cutting the deep fascia upto the muscle

- Nutrition - Enteral Preferred
- Role of Antibiotics - No Role of Prophylactic Ab
- Physiotherapy - started on day 1

  Q July 25 except in SG ; Started > 5 days

Active space

- MCC of Immediate Death = Suffocation (Asphyxia)
- " " Early " = Hypovolemia
- " " Late " = Sepsis

Electrical Burns

- Muscle - Rhabdomyolysis → Myoglobinuria
  → Acute Renal Failure
- Heart - Interference in rythm of SA node/Direct Myocardial damage
- Rx : • Aggressive Fluid Mt, ECG mandatory
  • Mt of Compartment Syn - Urgent Fasciotomy

Chemical Burns

- Rx -Irrigation c̄ running water for Long duration (20 - 30 mins)
- No Role of Neutralizing agents
- In Powder chemical burns → 1st Remove powder c̄ a brush

Cold Burns :

FROST BITE

NON FREEZING INJURY
TRENCH FOOT

FIGURE- 4.13 Late Frost Bite

FIGURE- 4.14 Early Non Freezing Injury

Active space

- Rx :
- Rewarming - 40°C in warm water bath for 20 - 30 mins
- Analgesia - IV Narcotics- Aspirin - essential
- Surgery - delayed for weeks - months unless a clear line of demarcation develop between viable &non viable parts

Workbook Questions

1. A patient after head trauma presented in the emergency. On examination he was able to open his eyes on painful stimuli, shows withdrawal movement and used inappropriate words. What is the Glasgow coma score in this patient?

   A. E3M5V4
   B. E2M4V3
   C. E1M3V3
   D. E2M3V2

2. qSOFA Score includes all except:
   A. Respiratory rate >22/min
   B. Systolic blood pressure < 100 mm of Hg
   C. Altered mental status (GCS<15)
   D. UOP

Active space

3. A patient suffered burns following which the surgical procedure was done as shown in the image. Which of the following best describe the likely procedure done?

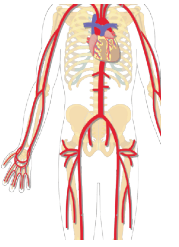
A. Full thickness skin graft
B. Thiersch skin graft
C. Biological mesh
D. Artificial membrane

Answers: 1. (B), 2. (D), 3 (B)

Active space

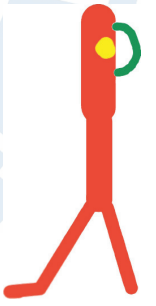
VASCULAR SURGERY

Arteries	Veins	Lymphatics
↓	↓	↓
Gradual obsⁿ	Sup. veins (Varicose Veins)	Lymphoedema
		Primary-
Sudden obsⁿ	Deep veins	Secondary-
(Embolism)	(Deep vein thrombosis)	

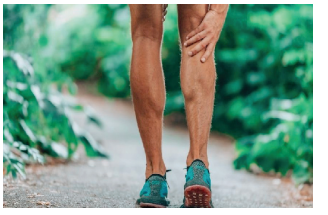


CLINICAL PRESENTATION OF GRADUAL ARTERIAL OBSTRUCTION

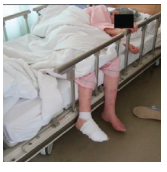
- MCC - Atherosclerosis (> 50yrs)
  Buerger's disease (Thromboangitis obliterans) (< 45yrs)
- C/F :
- Intermittent Claudication - d/t anaerobic muscle metabolism
  - Cramp Like Pain, in Calf, on walking a certain distance.
  - Distance walked is called claudication distance
  - Not present at rest Usually - Lower Limbs
  - S/t - Upper Limbs - Writer's spasm
  - Boyd's Grading :



→ I - Calf Pain on walking, c is relieved on further walking
→ II - " " " ", Continue to walk c̄ pain .
→ III - " " " ", Cannot walk further, has to take rest
→ IV - Rest Pain



- Rest Pain - Pain in foot, at rest



- Coldness, Numbness, Paresthesias - foot

  Mod - Severe arterial insufficiency
- Pregangrene
- Gangrene

DIAGNOSIS OF GRADUAL ARTERIAL OBSTRUCTION

- Cl. Examⁿ :
- Examⁿ of Pulses :
- Distal to obsⁿ - Low volume (feeble)/absent
- Proximal to obsⁿ - Normal / high volume
- Always compare c̄ Pulse of other (N) Limb
- Disappearing Pulse - d/t Collaterals
- On Initial Examⁿ - Distal Pulse - Normal
- On exercise - " " " - Disappear
- On Rest - " " " - Reappear
- Auscultation -
- Bruit → Turbulence (Stenosis/obsⁿ)
- ABPI = Ankle Brachial Pressure index

$$\frac{\text{SBP at Ankle}}{\text{SBP at Arm}}$$

- 1.1 ± 0.2 = Normal
- < 0.9 = Intermittent claudication
- < 0.5 = Rest Pain
- 0.3 = Critical Limb Ischemia
- 0.1 = Gangrene

Active space

FIGURE- 5.3 Ankle Brachial Pressure Index Measurement

FIGURE- 5.4 Hand held Doppler

Reflection of an ultrasound wave from moving objects such as red blood cells causes a change in the frequency of the ultrasound wave. By measuring this frequency change, it is possible to calculate the speed and direction of the movement. This principle forms the basis of Doppler ultrasound, whereby

- FONTAINE'S CLASSIFICATION :

  I = No Symp

  II = Intermittent claudication
  - II_A - Mild
  - II_B - Moderate

  III = Rest Pain

  IV = Gangrene

- IOC = Colour Doppler

  - Angiography = Invasive

Hand held Doppler → For measurement of blood flow.
If combined c sphygmomanometer → Measure SBP

QQ **Q July 24**

TREATMENT OF GRADUAL ARTERIAL OBSTRUCTION

- Rx
- Aim - Relieve Symp - by ↑ distal blood supply

1. Lifestyle modification - X smoking; ↓ wt; Modify diet

   Brisk walk for 40 mins
   - morning
   - evening

   Improvement in 6 month d/t formⁿ of good collaterals

2. Medical Rx
   - Vasodilators - Cilostazole, oxypentiphyline
   - Anti platelets - Aspirin (Long Term)
     Clopidogrel (Short Term)
   - Statins

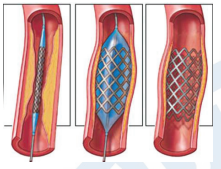
Active space

- 3. PTA- Percutaneous Transluminal Angioplasty

  P T C A

  Percutaneous Tranluminal Coronary Angioplasty

  Fogarty catheter in Vascular Surgery



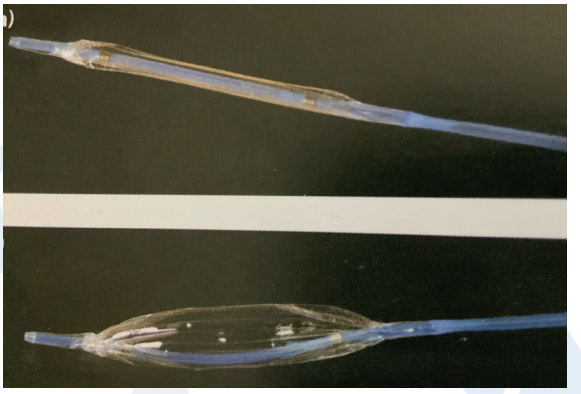
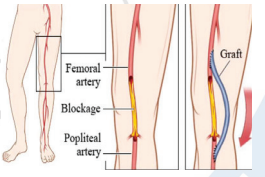


FIGURE- 5.5 Fogarty catheter for Vascular Surgery

- Balloon A
- Stent A - Drug eluted stents
  - Sirolimus / Everolimus



- 4. Surgery -

(a) Bypass Graft -

- Autologous LIMA,- LSV, RIMA, Gastroepiploic artery, Radial artery

Best long term patency results - for CABG is with LIMA(2nd with LSV)

- Synthetic
- Textile - Dacron better - Woven ✓ / Knitted
- Non Textile - PTFE (Polytetrafluoroethylene)

- MCC of Early Graft failure => Faulty Technique
- MCC of Late Graft failure => Graft Occlusion
- MCC of Graft Infection => Staph aureus

Active space

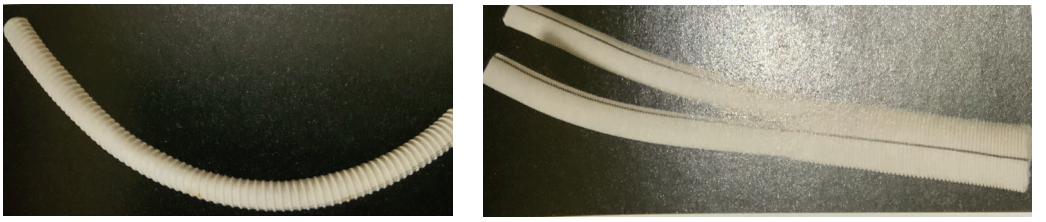
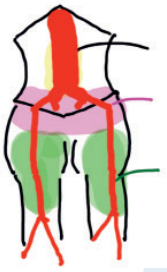


FIGURE- 5.6 Synthetic Dacron graft for Bypass Surgery- Note the V shaped graft to be used for Aorto-iliac obstruction



CABG - LIMA QQ

Aorto Iliac obsⁿ - Supra Inguinal obsⁿ → Dacron Graft

Ileofemoral obsⁿ - Inguinal obsⁿ → Angioplasty

Femoro Popliteal obsⁿ - Infra Inguinal obsⁿ → LSV Graft

b) Lumbar Sympathectomy - Foot arterioles - dilated

Indicated in Rest Pain

Contraindicated in Intermittent Claudication

c) Amputation - in Gangrene best pain relief.

BUERGER'S DISEASE AND RAYNAUD'S DISEASE

Buerger's Disease

- TAO - Thromboangitis obliterans ✓
- Thrombosis, Thrombophlebitis of small - medium Sized vessels
  - Associated c̄ Raynaud's syndrome
- < 45 yrs (<30 yrs) QQ
- 95% Males, smoking (tobacco use)
- Usually - Lower Limbs
- Involve artery, vein. Can involve nerves
- Does not involve Lymphatics
- C/F - Rest Pain (Small - medium sized vessels)
- Rx - Mainstay - Lifestyle modification

Active space

" Cigarettes or Legs "

- Medical Rx
- Lumbar Sympathectomy - to relieve Rest Pain

Raynaud's Disease

- Spasm of ..... on cold exposure
- Young Females
- Upper Limbs
- Pulses - Normal
- Primary R - No other associated disease
- Secondary R - Associated c̄ some other disease

SLE (Systemic Lupus Erythematasus, RA (Rheumatoid arthritis), Scleroderma, Thoracic outlet syndrome-Cervical Rib - Adson's Test

QQ Raynand's phenomenon- sequence of colour changes
White → Blue → Red

FIGURE- 5.7 Raynaud's phenomenon- White-Blue-Red Primary Raynaud's- No Cause Secondary Raynaud's- Asso with SLE, RA, Scleroderma, Thoracic outlet syndrome (Cervical Rib)

- Rx
- Lifestyle modification - ✓
- Avoid cold exposure, wear warm clothings
- Medical Rx - CCB'S, Nitrates
- Rarely - Cervical Sympathectomy

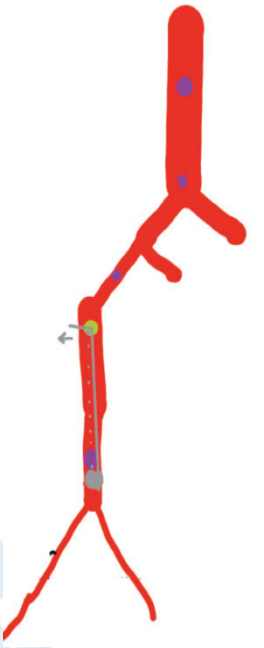
Active space

SUDDEN ARTERIAL OBSTRUCTION- EMBOLISM

- Sudden occlusion of a vessel by a substance which is foreign to bloodstream

- MC Source - Heart - Mural thrombi from heart
  No Collaterals
- Acute Limb Ischemia - Vascular - Emergency
- P's - Pain (severe, out of proportion to clinical signs)
  Pallor, Pulseless, Paresthesia, Paralysis (Poor)
  - Poikilothermia

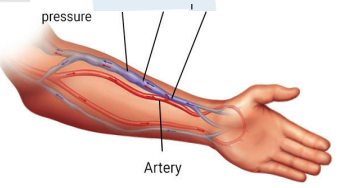
Worst Signs



- Rx : Emergency -
- 1st - Low mol. wt Heparin - Prevent Extension of thrombus
  ↓
- Thrombolysis - Standard - Tissue Plasminogen Activator
  Streptokinase / Urokinase / Alteplase
- Thrombectomy - Embolectomy

AV FISTULA AND ANEURYSM

AV Fistula Arterio - Venous Fistula



- MCC - Penetrating Trauma
- Iatrogenic- for Dialysis
- Infections - Bacterial, Viral, Parasitic
- C/F : Abnormal dilⁿ, engorgement of vein, Risk of bleeding, skin ulceration QQ
  Pulsatile Varicose Veins Q July 24
  ↑ Venous return → ↑ Cardiac output QQ 17 Jan 26
- Pressure over artery proximal to fistula reverses
  Changes of Av fistula - Branham's Sign
  or Nicoladoni's Sign
- Rx - ✓ Angioembolisation (Therapeutic Embolisation)

Aneurysm

- Abnormal dilatation of an artery
- Elderly
- True → All layers are in Aneurysm
- False → Not all layers in Aneurysm
- MC Site of Peripheral A = Poplited artery
- MC Site of Pseudo A = Femoral artery

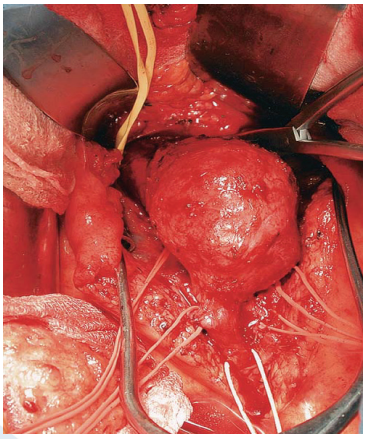
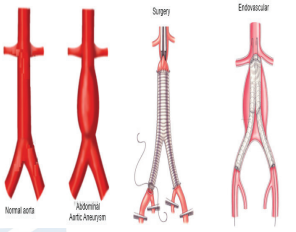


FIGURE- 5.8 Abdominal Aortic aneurysm



- Usu - Asymp
- IOC - CT Scan
- If Symp or > 5.5cm (Abd Aorta) → Risk Rupture ↑

↓

EVAAR - Endo vascular Aortic Aneurysm Repair

(For Popliteal artery > 2.5cm) Risk of rupture ↑

VENOUS DISORDERS- VARICOSE VEINS

- Sup. V → Varicose Veins
- Abnormal dilated tortuous veins
- MC - Lower Limb
- d/t Valvular incompetence - MC - Saphenofemoral

  MC Vein - LSV (GSV)

Active space

LSV
(GSV)
QJuly25

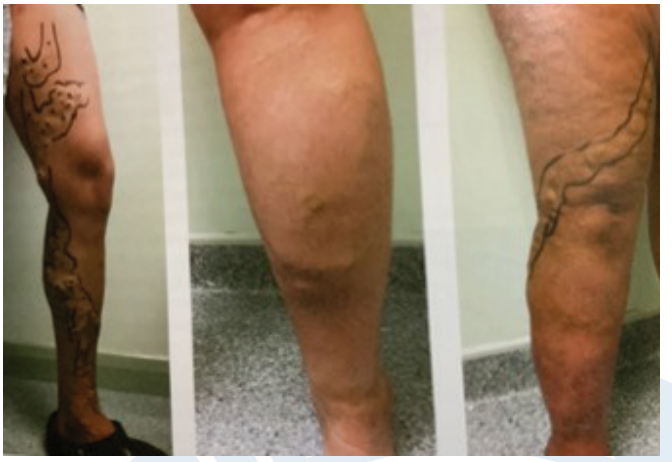


FIGURE- 5.9 Varicose Veins

CEAP Classification
Clinical, Etiology, Anatomy, Pathology
C_1 = Reticular veins (Telengiectasia)
C_2 = Varicose veins
C_3 = Oedema
C_4 = Skin changes
a = Eczema
b = Lipodermatosclerasis
c = Corona phlebectasia
C_5 = Venous ulcer → healed
C_6 = Venous ulcer → active

- RF : Prolonged standing
  (Traffic Policemen, Rickshaw Puller)
  - Pregnancy
  - Pelvic Tumour (Tubo ovarian mass)
  - Progesterone
  - Deep Vein Thrombosis
  - Genetic

- Ambulatory Venous Hypertension -
  In valvular incompetence → On walking
  → Blood move from Deep veins into Sup. veins
  → Pressure in Sup vein ↑ - AVH
- C/F : Aching discomfort in calf or heel, Bleeding

- Complications :
- Venous ulcer - MC - Around Medial. malleolus **Q July 25 IBQ**
  Gaiter's area (Area between 2 malleoli)

Active space

Marjolin's ulcer- skin ulcer in Burn Patients

↓

Can complicate into Marjolin's ulcer

↓

Can complicate into Squamous cell Ca

SCC:-

1. Slow growing
2. Can involve Lymph nodes
3. RxOC→Surgery Radiotheropy is not the RxOC

- Lipodermatosclerosis -

NEET PG 25

Brownish, Blackish, Leathery appearance in Lower Leg

- Champagne glass /bottle appearance

Q July 25 →

FIGURE- 5.10
Venous ulcer-medial malleolus

FIGURE- 5.11
Marjolin's ulcer- Will cause SCC

FIGURE- 5.12
Lipo dermato sclerosis

NEET PG 25 ←

- Clinical approach :
- Clinical Tests - Tourniquet Test - Brodie Trendelenburg Test (Saphenofemoral, Perforator valvular incompetence)
  - Fegan's Test Perforator incompetence

Active space

- Note - Perthe's Test - Deep Vein thrombosis
  Adson's Test - Cervical rib
- IOC for Varicose Veins - Duplex Imaging

↓

Micky Mouse Sign

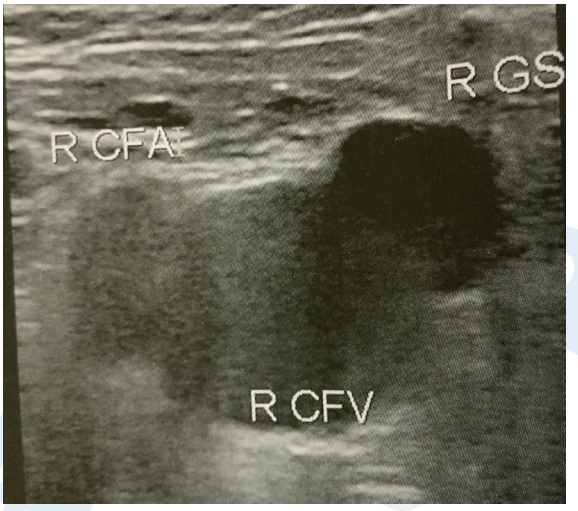


FIGURE- 5.13 Mickey mouse sign on Duplex Imaging in incompetent saphenofemoral valve in Varicose Veins

- Rx : Initially - Conservative Rx - Lifestyle modification

  Avoid Prolonged Standing

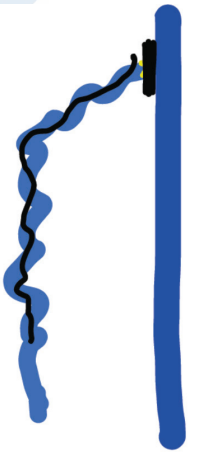
  Limb elevation

  Elastic compression stockings

- Gold Standard Rx-Endothermal Ablation (Laser/ Radiofrequency ablation) Q Jan 25 QQ.
- Sclerotherapy
- Sx - Trendelenburg opⁿ - Flush Ligation of LSV c̄ FV

  +

  Stripping of GSV (Mayo's vein stripper)



- MC Nerve Injured in removal of

  a) LSV below Knee => Saphenous nerve

  a) SSV below Knee => Sural nerve

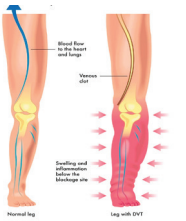
- In Perforator incompetence → SEPS

  Subfascial Endoscopic Perforator Surgery

VENOUS DISORDERS- DEEP VEIN THROMBOSIS

- MC Site - Calf vein (Soleal vein)
- MCC - Prolonged Immobilisation
  - ICU patient, Paraplegia, Multiple Sclerosis,
  - Bed ridden, Lower Limb fracture
  - Long haul flights

Virchow's Triad of Thrombosis
- Endothelial Injury
- Stasis (or Turbulence)
- Hypercoagulability



- Redness, Swelling, ↑ Temp., Pain in lower leg
- (Homan's sign - Resistance by calf ms to dorsiflexion of foot

FIGURE- 5.14 Homan's sign- Resistance on dorsiflexion of foot

- (Hamman's sign - Boerhave syndrome)
- Moses' sign - Calf pain on squeezing calf muscles
- These signs are Obsolete and should not be done
- IOC - Duplex Imaging Q
- Gold Standard Inv^{n} for veins but not done

↓

Ascending Venography

Active space

- Rx - Low mol wt Heparin
  ↓
  Oral Warfarin - for 3 months
- In Recurrent DVT - IVC Filter

LYMPHATIC DISORDERS- LYMPHOEDEMA

- Accumulation of Interstitial fluid
  In Lower Limbs - > 4.5 litres of fluid is collected
- MC - Secondary - Filariasis (Lower Limbs)
- MCC of Upper Limb Lymphoedema → Post mastectomy for breast Cancer (Post MRM)
- Primary - d/t hypoplasia of Lymphatic channels
  - L. congenita - < 2 yrs - Milroy's disease
  - L. praecox - 2 - 35 yrs - Meig's disease (MC)
  - L. tarda - > 35 yrs
- Secondary - diseases or infections - Filariasis, Cancers, Removal of LN by Sx, Rodiotherapy destruction of LN
- Chronic Filarial L. → Elephantiasis

FIGURE- 5.15 Primary Lymphoedema

FIGURE- 5.16 Secondary Lymphoedema

Active space

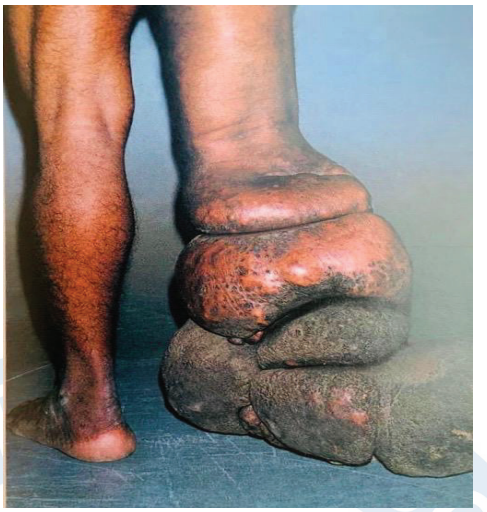


FIGURE- 4.17 Elephantiasis- Chronic Lymphoedema

- Rx : Manual Lymphatic drainage
- Massaging Pneumatic compression devices
  - Sx results poor
  - Nodo - venous shunt

Workbook Questions

1. A patient presented in the OPD complaining that on walking a certain distance, he feel pain in his calf region because of which he cannot walk any further and has to take rest. As per Boyd's grading it is classified as which of the following grades?
   A. I B. II
   C. III D. IV
2. A patient was admitted in the hospital for a long period of time following which she developed pain and swelling in her leg. Which of the following should be the investigation of choice to conclude the given pathology?
   A. DSA (Digital substraction angiography)
   B. Duplex imaging
   C. MRI
   D. Angiography

Answers: 1. (C), 2. (B)

Active space

SALIVARY GLANDS

SALIVARY GLAND CYSTS AND STONES

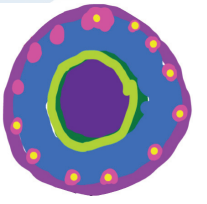
FIGURE- 7.1 Mucus Retention Cyst

Figure 54.2 Mucous retention cyst. A translucent swelling on the

Ranula

- Mucus extravasation cyst
- Floor of Mouth - Sublingnal gland
- Bluish, Translucent swelling
- Resemble Frog's belly
- Transillumination +
  (Brilliantly Transilluminant
  Swelling - Cystic Hygroma)
  - Other Transilluminat Swellings -
  - Hydrocele, Meningocele, rarely Branchial Cyst
- Some can resolve spontaneously
- If persist → Sx Excision of Cyst + Sublingual Gland
- MC Structure damaged in Ranula Sx - Submandibular duct
- " Nerve " " " - Lingual nerve QQ

FIGURE- 7.2 Ranula- Mucus Extravasation cyst- Sublingual Gland- Transilluminant

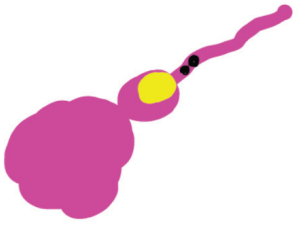


Active space

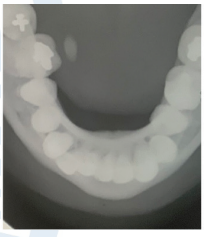
Plunging Ranula

- Mucus Retention cyst - in Sublingual gland
- Dumb bell shaped swelling

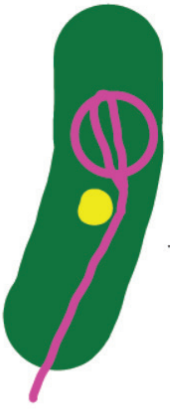
Submandibular Gland

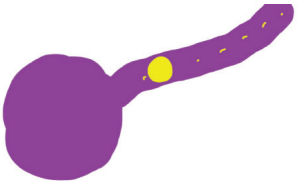


- SIALOLITHIASIS :
- 80% of Salivary stones → Subman gland/duct
- Dehydration - Smokers, Alcoholics
- Secretions - thick (viscid) - Submandibular duct is tortuous, against gravity → ↑ Stasis → Stone
- Composed of Phosphate/Oxalates
- Pain, Swelling - intermittent - at meal times
  at angle of mandible - MEAL TIME SYNDROME



- X - Ray - visible on X -Ray
  (Radiolucent stones - P G LUX - Lysine, Uric acid, Xanthine] Urinary stones
  Parotid duct ..... Gall bladder stones
- Rx - Plenty of fluids
  - < 4 mm stone → Sialendoscopy → Dormia Basket
  - ≥ 4 mm stone → ESWL/ISWL + Sialendoscopy and Dormia Basket
    (Extracorporeal shock wave Lithotripsy or Intracorporeal Shock wave Lithotripsy)





- If above fails

Sx → Incision at deeper surface of duct, remove stone, suture the duct
Sx → Subman. gland excision

- Nerves c̄ can be injured during Subman. gland excision
  - Marginal mandibular nv
  - Lingual nv
  - Hypoglossal nv

SALIVARY GLAND TUMOURS

- MC Site of Salivary Gland Tumours - Parotid
- MC Benign Tumour of Salivary Gland - Pleomorphic adenoma
- 2nd MC Benign " Parotid - Warthin's Tumour or Adenolymphoma
- MC Malignant Tumour in parotid - Mucoepidermoid Ca
- 2nd MC " " " " - Adenoid cystic Ca
- MC Malignancy in Subman, Subling, minor saliv-Adenoid cystic ca
- Salivary malignancy involving the nerve sheath - Adenoid cystic Carcinoma

Pleomorphic Adenoma

- Soft to firm, Smooth outline, no inv. of nerves, no inv of LN, not fixed - Swelling in Parotid region
- Involve Sup. Lobe of Parotid
- Rx : Sup. Parotidectomy (Conservative Parotidectomy) above Facial nerve
- Alternative - Extracapsular Parotidectomy Less Risk of Facial nerve injury

Active space

- Incision for Sup. Parotidectomy - "Lazy S Incision"

FIGURE- 7.3 Lazy S Incision for Superficial Parotidectomy for Pleomorphic adenoma of Parotid Gland

- Complications of Sx -
- Injury to nerves - Facial nerve - Temporary or Permanent weakness
- FREY'S Syndrome - Auriculotemporal syn - Gustatory sweating
- d/t regeneration of parasympathetic nerve fibre into skin of Parotid region, from Auriculotemporal nerve

Q July 25

- Sweating & erythema in Parotid region on smell or taste of food
- confirmed by Starch Iodide Test
- Prevention - Place a barrier between - Parotid bed & Skin
  - Sternomastoid muscle
  - Temporalis fascia
  - Artificial membranes
- Rx - Antiperspirants
  - Botulinum Toxin Injn

Active space

MISCELLANEOUS

- Classification of Surgical Wounds

I - Clean - No infecⁿ - No Ab- eg- Hernia, Thyroid, Breast opn

II - Clean - contaminated - Minimal Infection -

Prophylactic Ab - eg - Elective GI Sx

Cholecystectomy, Elective Appendicectomy

III - Contaminated - Significant Infection -

Both Prophylactic + Therapeutic Ab

eg - Emergency GI Sx -

Appendicectomy, Emergency Cholecystectomy

IV - Dirty - Gross infecn - Pus/ Abscess -

Both Prophylactic + Therapeutic Ab

Eg - Drainage of abscess, emergency Laparotomy for Perforation Peritonitis

- Best time of prophylactic antibiotic -

1 hour before surgical incision

- For Dense Hairy Skin -

Hair clippers are used just before operation

Active space

- Sebaceous or epidermoid cyst, or pilar cyst, is a benign, due to lockage of a sebaceous gland, leading to the accumulation of keratin and sebum inside a sac.

Locations- Face, neck, and upper back / Scalp / Chest or shoulders / Genital region

Central blackhead or opening / Foul-smelling, cheesy white or yellow discharge if ruptured

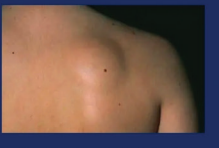
Treatment

Observation: If small and asymptomatic

Incision and drainage: Temporary relief for infected cysts

Complete excision: Removes entire cyst wall to prevent recurrence.

A 35-year-old male presents with a soft, non-tender swelling on the upper back. The lesion has been present for several months and occasionally drains a foul-smelling, cheesy material. What is the most appropriate management for this condition?



a. Incision and drainage alone.
b. Antibiotic therapy to treat the underlying infection.
c. Complete surgical excision of the cyst along with its capsule.
d. Wide Local excision.

Active space

- Lipoma : benign, slow-growing tumor of fat cells. most common type of soft tissue tumor just under the skin.

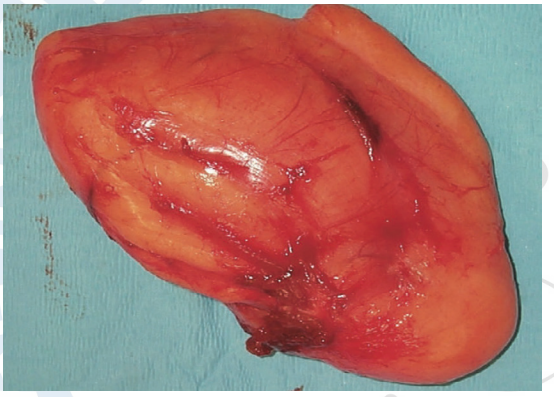
Locations- Neck, shoulders, and upper back / Arms and thighs /

Treatment :

Observation (if small and asymptomatic)

Surgical excision (if painful, growing, or cosmetically concerning)

Liposuction (for large lipomas, but may not remove all fat cells)



- Ganglion Cyst: benign, fluid-filled lump that develops near joints ortendons. It is filled with thick, jelly-like synovial fluid and commonly occurs in the hand or wrist.

Locations- Dorsal wrist (back of the wrist, most common) / Volar wrist (palm side of the wrist)

Transilluminates with light

Treatment

Observation as are harmless and disappear on their own

Aspiration (fluid removal with a needle, but recurrence is common)

Immobilization (wearing a brace to reduce movement and shrink the cyst)

Active space

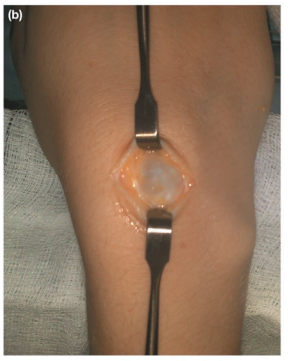
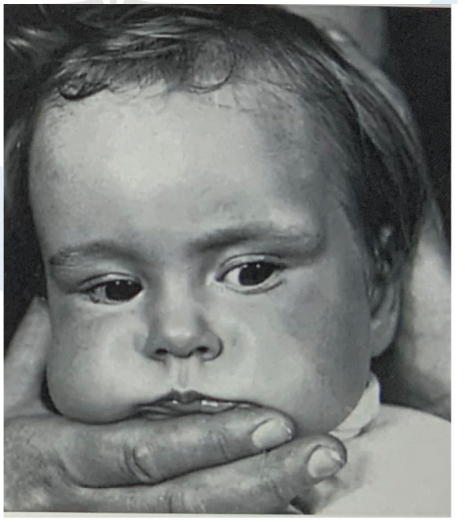


Figure 38.58 (a) Clinical and (b) surgical appearance of a dorsal wrist ganglion.

- **Angular Dermoid:**
  - Benign congenital choristoma of orbit
  - From Frontozygomatic suture line of bone during development
  - Can erode through bone into the duramater
  - IOC - CT Scan
  - Rx - Surgical Excision



Active space

Foreign Body Granuloma :

- Chronic inflammatory reaction as immune response
- To suture / mesh / splinter / glass / metal / Talc / hair / Tattoo
- Macrophages → Multinucleated giant cells (Frog Face like appearance on HPE)
- Fibroblast → Collagen
- Firm, Painless lump I with / without tenderness
- Rx : Observation / Ab Excision if Symp / complication

Basal Cell Ca :

- Site, white skinned / female
- UV Radiation
- MC Nodular / Nodulocystic
- Palisading
- Pearly, Rolled out edge
- Spread Locally
- Rarely metastasize
- Rx Mohs micrographic Sx

Figure 45.30 (a) A nodulocystic basal carcinoma (BCC). Note the characteristic pearly surface with telangiectasia. (b) An ulcerating BCC on the lower eyelid. (c) A recurrent morphoeic BCC. ((a, b) courtesy of Mr AR Greenbaum; (c) courtesy of St John's Institute for Dermatology, London, UK.)

Squamous Cell Ca :

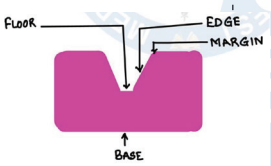
- White skinned / female
- Sun exposure
- Keratinisation Cytokeratin 1 to 10
- Raised Everted Edges
- Prognosis depends on -

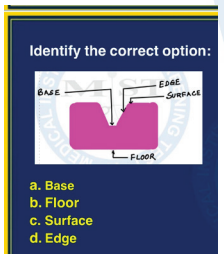
Active space

- Depth
- Size
- Nv / Vessel invasion
- Location / Aetiology / Immuno suppression

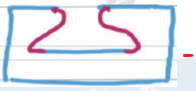
Rx WLE ± LN

- Ulcer - Edges





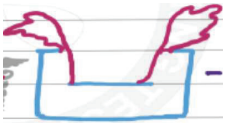
1. Sloping - - Venous ulcer, Healing ulcer, Inflammatory Ulcer

2. Undermined -  - Tuberculosis, Pyoderma Gangrenosum

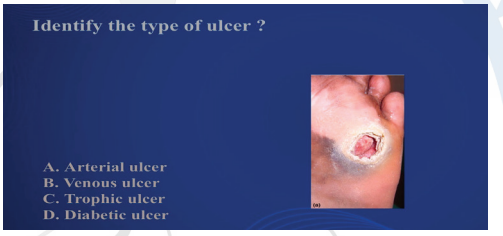
3. Punched out - Syphilis

Active space

4. Rolled out, Beaded - Basal cell Ca (Rodent ulcer)

5. Raised Irregular Everted -  - Squamous Cell Ca

- Ulcer at the tip of toes/dorsum of foot → Arterial ulcer
- Ulcer at the medial malleolus → Venous ulcer QQ **Q July 25**
- Ulcer at the Pressure points (Heel, Sole) → Trophic ulcer QQ



Pressure to cause a pressure sore = 30mm of Hg

- Bed Sore Grade

I. → Intact skin Redness

II. → Partial skin loss Ulcer

III. → Subcutaneous tissue exposed

IV. → Boneexposed

In bed sore - If bone is exposed → Grade 4

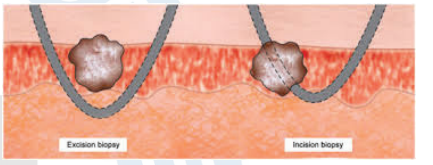
Active space

- Pressure in VAC Dressing = - 125mm of Hg

VAC Vacuum assisted closure or NPWT - Negative Pressure wound therapy

Figure 47.8 Negative-pressure wound therapy to promote wound healing in an open abdomen. The system consists of a non-adherent dressing overlaid by a sponge that is sealed with an airtight membrane and connected to a suction device.

- Biopsy :



- Hepatectomy :

Q. Left hemihepatectomy involve removal of which of the following segments of Liver?

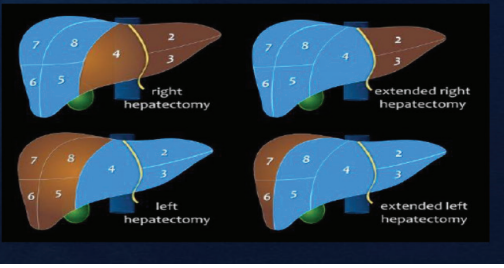
A. I,II,III and IV
B. V,VI,VII and VIII
C. II,III and IV ✓
D. II, III,IV,V and VIII

Right hepatectomy- V, VI, VII and VIII.

Extended Right hepatectomy- IV, V, VI, VII and VIII

Left hepatectomy- II, III and IV

Extended Left hepatectomy- II, III, IV, V and VIII



Active space

Pseudomyxoma Peritonei :

- Progressive Peritoneal tumor deposits + Mucinous ascites + Omental Cake + (Ovary involvement in females)
- Reason -

  MC → Perforation of mucinous adenoca of Appendix

  → Colorectal / Ovarian / Gastric Ca

  → Peritoneal Mesothelioma

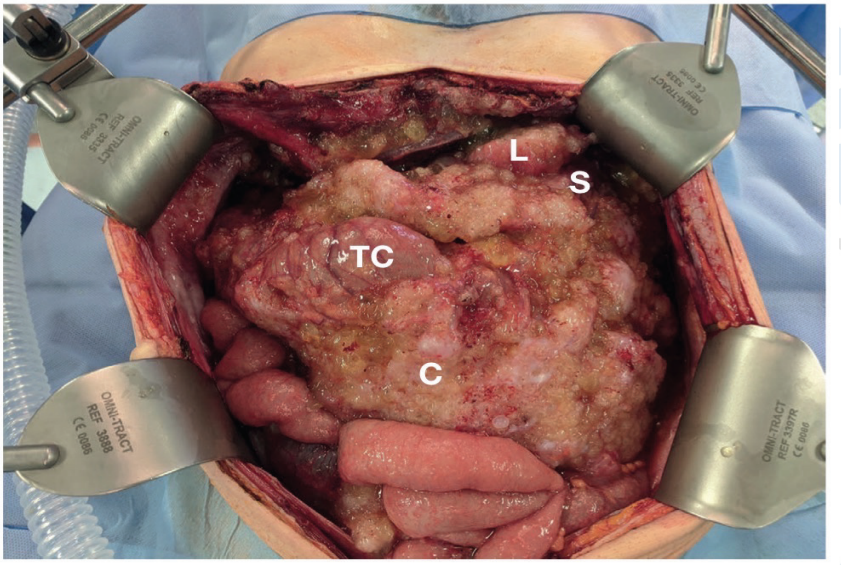


Figure 76.17 Pseudomyxoma peritonei with characteristic omental cake (C) encasing the transverse colon (TC) and extending to the greater curvature of the stomach (S). Tumour is seen to replace the lesser omentum (L).

- T marker -

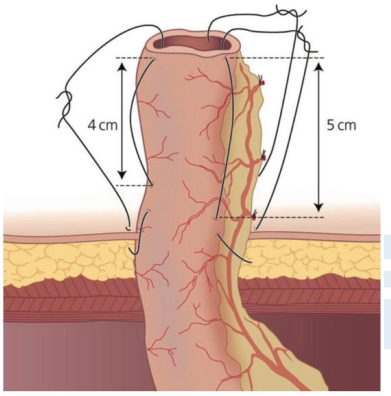
  CEA / CA - 125 /CA 19 - 9
- CT / MRI / Laparoscopy
- Rx - CRS + HIPEC

  → Cytoreductive Sx (Extensive Sx in abdomen to ↓ T burden)

  → Hyperthermic Intraperitoneal Chemotherapy before closing the abdomen

Active space

Ileostomy :



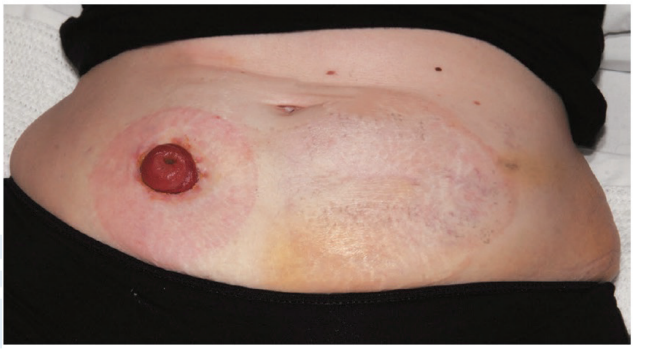


Figure 74.8 Spouted ileostomy in the right iliac fossa.

Colostomy :

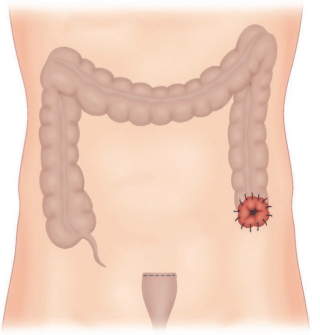
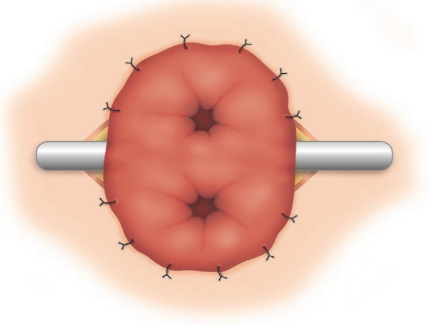


Figure 77.15 Hartmann's procedure with an oversewn rectal stump and an end left-sided colostomy following resection of the diseased segment of sigmoid colon.

Figure 77.19 A colostomy in the left iliac fossa.



77.18 Loop colostomy with a bridge.

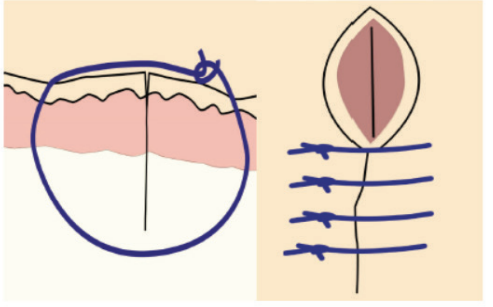
Active space

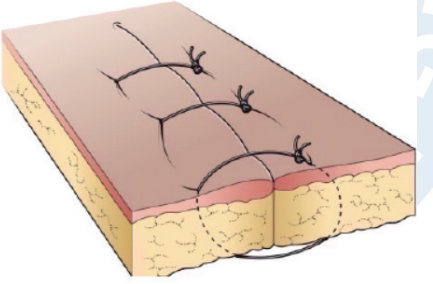
Xanthogranulomatous Pyelonephritis :

Figure 82.7 Xanthogranulomatous pyelonephritis with the 'bear's paw sign'.

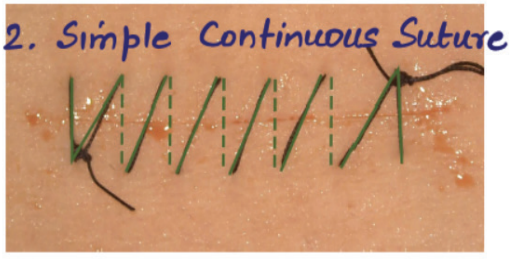
Suturing Techniques :

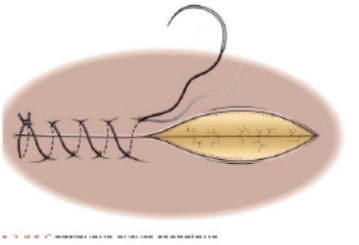
1 Simple Interrupted Suture **Q July 25**





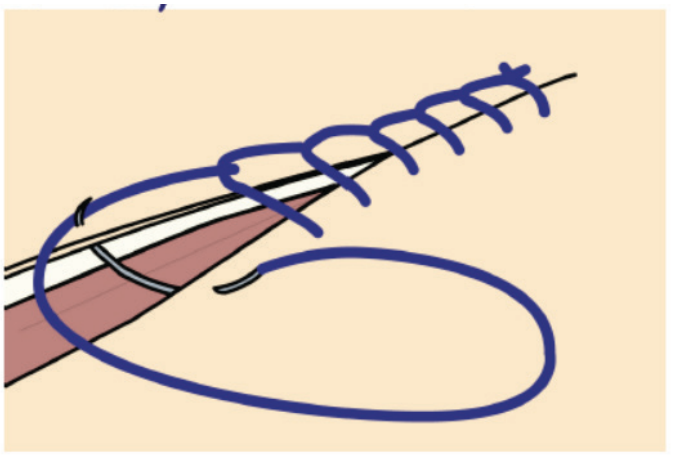
2 Simple Continuous Suture



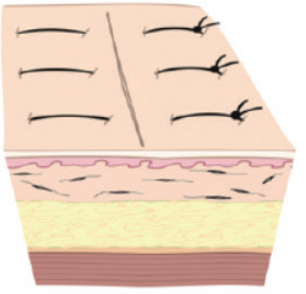


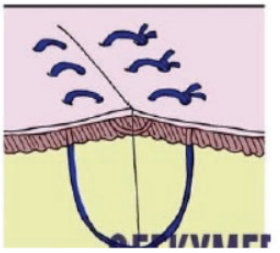
Active space

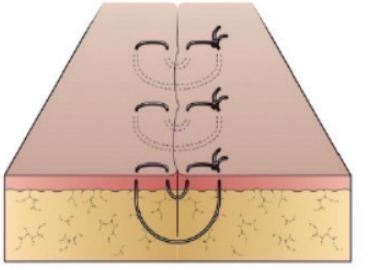
3 Simple Continuous Interlocking Suture



4 Vertical Mattress suture

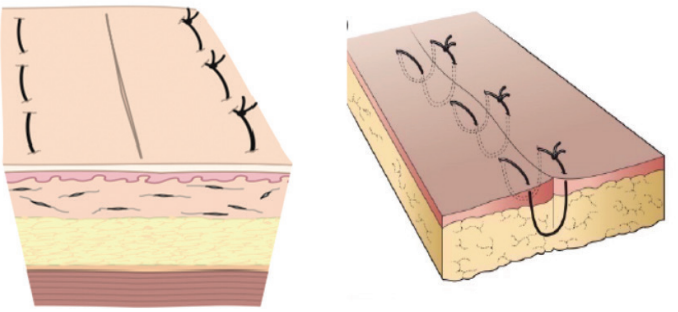


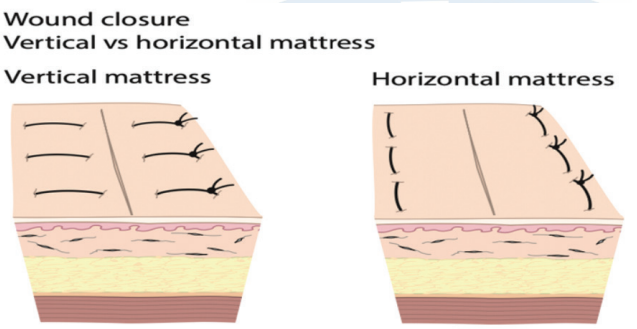




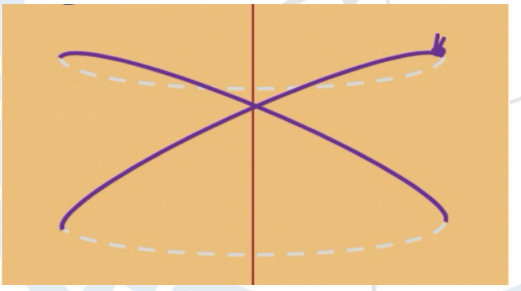
5 Horizontal Mattress suture

Active space

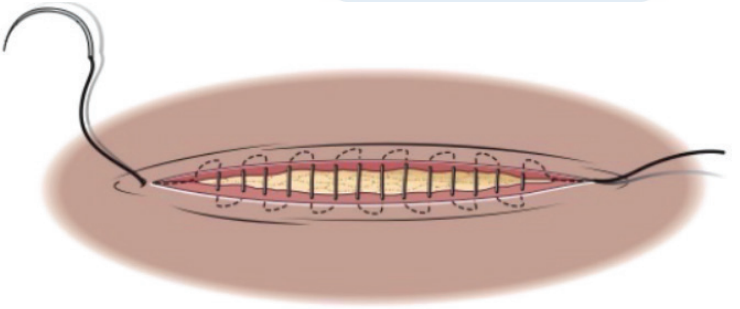




6 Figure of 8 suture



7 Sub Cuticular suture



Active space

Knots :

2 in ; 2 out -

Square (1 = 1) (Reef konot)

1in : 1 out

Not used

(Not Secure)

Granny Knot (1 x1)

Coil

Surgeon's knot (2=1)

Active space

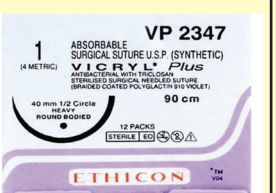
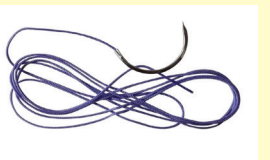
Sutures :

I. Non Absorbable :

	Silk	Nylon	Polyester	Polypropylene
Use	Skin	Skin	CVS, Ophtha, Plastic Sx	CVS, Ophtha, Plastic Sx
Absorption	Very slow (1-2 yrs)	15-20% per year	Nonabsor-bable	Nonabsor-bable
Tensile strenght	lost by 6 months	lost 15-20%per year	Infinite (>1yr)	Infinite (1yr)
Raw Material	Silkworm (Natural protein	Polyamide polymer	Polyester	Polymer of Prolene
Type	Braided Multi filament	Mono filament Braided Multi filament	Mono filament Braided Multi filament	Mono filament
Name	Silk	Ethilon, Dafilon	Ethibond	Prolene, Premilene

Active space

II. Absorbable :

	Catgut	Polyglactin	Polydioxanone (PDS)	Polydioxanone
Use	Stomas	Work horse Gut. Vascular	Same as Polyglatin	Subcuticular
Absorption	90 days (Phagocytosis)	60-90 days (Hydrolysis)	180 days ((Hydrolysis)	90-120 days
Tensile strenght	21-28 days	60% at 2 wks 30% at 2 wks	70% at 2wks 50% at 4wks 14% at 8wks	21 days
Raw Material	Collagen from Gut Mucasa of Sheep or Cattle	Copolymer of Lactide and Glycolide	Polyester Polyester	Copolymer of Glycolite and Caprolactone
Type	Chromic	Braided Multi filament	Mono filament NEET PG Q 25	Mono filament
Name	Chromic Catgut	Vicryl, Novosyn  	PDS	Monocryl, Monosyn

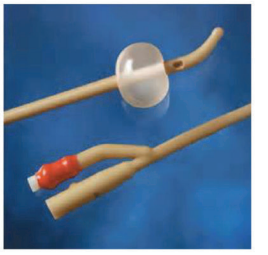
Van Nuys Prognostic classification for DCIS Also decide BCS vs Mastectomy

	1 Point	2 Point	3 Point
Tumor size	≤ 15mm	16-40mm	> 40mm
Margin width	≥ 10mm	1-9mm	< 1mm
Pathologic Classi	Grade I No Necrosis	Grade I / II + Necrosis or Grade II - Necrosis 40-59 yrs	Grade III or Grade II + Necrosis < 40yrs
Age	≥ 60yrs	40-59yrs	< 40yrs

4 - 6 = Excision
7 - 9 = Excision + Rth
10 - 12 = Mastectomy

Active space

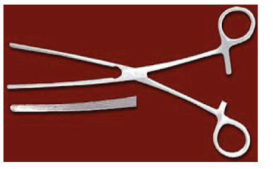
Instruments

Foley's Catheter	
Supracath (for Suprapubic Cystostomy)	
Vaccum Suction Drain (Romovac)	
Bard Parker Knife Handle (Scalpel) With blades	
Needle Holder	

Elderly patient has multiple clots in urine. Which catheter should be used in this patient which can facilitate irrigation of the bladder and aid in clot removal from the urinary bladder?

a. McKonly catheter
b. 22 French Foley's 2-way catheter
c. 22 French Foley's 3-way catheter
d. 22 French Foley's 4-way catheter

Active space

Plain Scissor Suture Cutting Scissor	
Tooth Forceps Plain Forceps	
Cheatle Forceps	
Intestinal Clamps	
Skin Stapler Staple Remover	
Linear Stapler Curvilinear Stapler Circular Stapler	

Active space

c. 22 French Foley's 3-way catheter

Explanation:
For a patient with multiple clots in urine, the most appropriate catheter to use is the: Foley's 3-way catheter (also called a triple-lumen catheter).
The third lumen in a 3-way catheter is specifically designed for continuous bladder irrigation (CBI). This helps:
Flush out blood clots from the bladder.
Prevent clot retention.
Maintain clear urine flow.

Why not the others?
1. McKonly catheter: Typically used in special circumstances but not ideal for managing clots.
2. Foley's 2-way catheter: Lacks the third lumen needed for irrigation.
3. Foley's 4-way catheter: Used for more complex cases, where we need to deliver medicines as well for intravesical therapies, but a 3-way catheter is simpler and sufficient for clot management in most cases.

Summary: Use Foley's 3-way catheter for patients with multiple clots in urine. Ensure proper irrigation to avoid bladder obstruction.

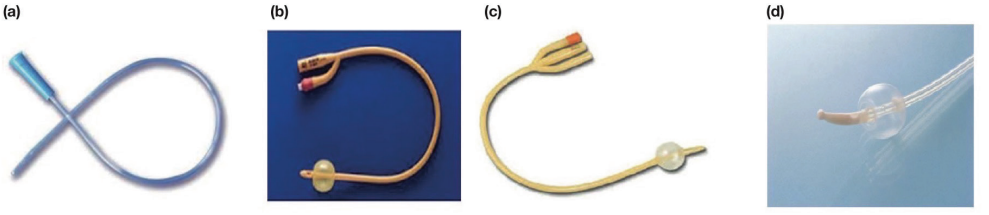


Figure 83.25 Types of urethral catheter. **(a)** Single-lumen catheter used for self-catheterisation; **(b)** standard urethral catheter with an inflatable balloon retention mechanism; **(c)** a three-way catheter with an extra channel for irrigation; **(d)** coudé tip catheter.

- 4 Way Foley's
  - B
  - U
  - I
  - Intravesical therapy

Active space